Carbon and nutrient dynamics in a temperate agroforestry system

Jyoti Sudhakar Narsude

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Declaration

I confirm that the work submitted is my own and that appropriate credit has been given where reference has been made to the work of others.

Dedication

This thesis is dedicated to Dr Bhimrao Ramji Ambedkar, the father and principal architect of the Indian Constitution. The Constitution he crafted grants me the right to be educated, to learn, and to grow. It gives me strength and continually inspires me through his broad and visionary perspective on humanity.

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"The more I learn, the more I realise how much I don't know." — Albert Einstein

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Abstract

Agroforestry is increasingly recognised as a restorative land-use strategy capable of enhancing climate change resilience, increasing carbon (C) sequestration, and delivering multiple ecosystem services. This thesis studied a mature (>30 years) silvoarable agroforestry system in northern England, where cropping in the alleys stopped in 2004. Of the poplar hybrids studied, Trichobel exhibited the highest aboveground biomass C stock (65.8 Mg C ha^{-1}) and greatest rates of C accumulation (mean annual C increment - MAIC: 2.1 Mg C ha^{-1} yr^{-1}; periodic annual C increment - PAIC: 2.3 Mg C ha^{-1} yr^{-1}), demonstrating substantial genotype specific differences in C sequestration potential. Carbon uptake by Trichobel was 52% greater than that of the least productive hybrid Gibecq. Upscaling these findings suggests that widespread adoption of poplar-based silvoarable systems could remove between 0.87 and 7.16 Tg CO_2 yr^{-1}, equivalent to offsetting between 2.1% and 17.5% of projected UK agricultural greenhouse gas (GHG) emissions by 2050. Higher summer temperatures reduced PAIC in Beaupre and Gibecq, while Trichobel responded positively to rainfall and wind speed, highlighting climate-driven differences in C sequestration. Agroforestry also increased soil organic carbon (SOC) stocks by 27% compared with adjacent conventional arable land, with a net SOC sequestration rate of 0.6 Mg C ha^{-1} yr^{-1}. Higher SOC concentrations in the topsoil (0-25 cm), together with positive relationships between soil nutrients availability, particularly nitrogen and phosphorus, and both live and dead root traits, indicate that root traits play an important role in regulating nutrient dynamics and SOC accumulation. In the hedgerow system, decomposition followed a biphasic pattern, with rapid initial mass loss (40-60%; 0-223 days) followed by slower decomposition (224-366 days) as readily decomposable material was depleted, while buried litter decomposed more rapidly than surface litter, reaching 99% mass loss within 12 months. Collectively, the findings demonstrate that C sequestration in agroforestry is not driven by tree presence alone, but also by the interactions among productive tree genotypes, favourable environmental conditions, enhanced SOC storage, root-mediated processes, and active nutrient cycling. Consequently, the effectiveness of agroforestry as a net-zero strategy depends not only on increasing tree cover, but also on managing system components in ways that optimise both aboveground and belowground C sequestration pathways. These results emphasise the importance of integrating tree selection, soil management, and ecological

functioning into future agroforestry design and policy to maximise climate-change mitigation benefits.

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Chapter 1. Carbon and nutrient dynamics in a temperate agroforestry system

1.1. Introduction

Agroforestry is widely regarded as a sustainable land management approach that intentionally integrates trees and shrubs with crops and/or livestock within the same space, generating both socioeconomic and ecological benefits (Castle et al., 2022; Raj et al., 2014; Torquebiau, 2000). Globally, agroforestry systems are estimated to cover nearly 1 billion ha (Cardinael et al., 2018c) and support the livelihoods of 1.2 billion people (Newton et al., 2022; World Bank, 2004). By integrating trees with crops in diverse spatial and temporal patterns, agroforestry systems provide a range of ecosystem services (Muhereza et al., 2020; Ntawuruhunga et al., 2023). These systems can also enhance human wellbeing through multiple benefits, such as enhancing food security (Waldron et al., 2017), improving water quality, combating climate change, and controlling the loss of biodiversity (Castle et al., 2022; Jose, 2009; Katila et al., 2015). As such, agroforestry is increasingly recognised as an important strategy for achieving the United Nations 2030 Sustainable Development Goals (SDGs) (Agroforestry Network, 2018).

Agroforestry takes several forms: silvoarable systems involve intercropping woody perennials with annual or perennial crops; while silvopasture systems combine trees, woodland, or shrubs with livestock grazing (Pagella & Whistance, 2019). Other agroforestry practices include hedgerows, which are rows of trees or shrubs planted along field boundaries (Holden et al., 2019), as well as shelterbelts and wooded or shrubby riparian buffers. These practices can improve water quality, reduce erosion, and support both crop-based (silvoarable) and livestock-based (silvopastoral) farming systems (Tufekcioglu et al., 1999). Forested environments can also support agroforestry through practices such as forest grazing and forest farming, the latter involving the cultivation of naturally occurring speciality crops such as ornamental or medicinal plants within forested ecosystems (Centre for Agroforestry, 2026). In urban and peri-

urban settings, home gardens combine trees or shrubs with vegetable production to maximise land productivity and biodiversity (Saha et al., 2009).

The growing impact of climate change on agroecosystems, coupled with global efforts to adapt and mitigate climate change (De Stefano & Jacobson, 2020; Masson-Delmotte et al., 2019), has renewed interest in agroforestry as a key strategy of sustainable development and a land-use practice with significant potential for wider adoption in temperate regions. This growing recognition is reflected in international climate policy, with approximately 40% of the 148 Nationally Determined Contributions (NDCs) submitted by non-Annex I countries to the United Nations Framework Convention on Climate Change (UNFCCC) under the Paris Agreement including agroforestry as a climate change mitigation tool (Rosenstock et al., 2019). In total, 41% of the National Communications (NCs) referenced agroforestry for climate change mitigation, 43% referenced it for adaptation, and 25% incorporated it as a tool for both mitigation and adaptation (Rosenstock et al., 2019). Despite this increasing integration of agroforestry into climate change policy and planning, the emphasis has focused on quantifiable mitigation outcomes, often at the expense of documenting and evaluating adaptation initiatives (Mulia et al., 2020; Rosenstock et al., 2019; Zahoor et al., 2021). In addition, limited data on the geographic distribution of agroforestry studies limits understanding of its global contribution to mitigating climate change (Quandt et al., 2023a).

Historical evidence suggests that agroforestry systems emerged alongside agricultural practices as early as 5,000 years ago in many parts of Asia (Mahmud et al., 2020). In temperate regions, however, the origins of agroforestry are closely associated with indigenous land management practices that integrated agricultural production with the use of forest resources (Bryant, 2023). Over time, agricultural practices in temperate zones evolved towards industrial agriculture, characterised by mechanisation, specialisation and the widespread adoption of monocultures. In contrast, agroforestry has remained more widely adopted in tropical regions, where favourable social and economic conditions, alongside environmental and ecological factors, support the growth of woody perennials within farming systems (Mahmud et al., 2020). Adoption in temperate regions has been limited over the past two centuries due to a range of biophysical, economic, and institutional constraints. These include slower tree growth, increasing climatic variability and establishment risk, historical legacies of agricultural

intensification, economic uncertainty and delayed returns, operational complexity at the scale of the farm, and the absence of enabling policy frameworks and advisory services (Marie-Laure, 2018; Jovanelly et al., 2025). As a result, despite growing recognition of its environmental and climate-related benefits, agroforestry remains less prevalent in temperate agricultural landscapes than in tropical regions. More recently, however, agroforestry adoption is increasing across temperate regions, including parts of the USA, Europe, Australia, New Zealand, Argentina, and East Asia, driven primarily by its potential to contribute to climate change mitigation and adaptation (Dmuchowski et al., 2024). However, temperate agroforestry systems generally store 10% less soil organic carbon (SOC) than in tropical and subtropical regions (Shi et al. 2018). In addition, Ma et al. (2020) reported that soil C stocks increased under agroforestry systems in both tropical and temperate regions; however, increases occurred more rapidly in the tropics, whereas temperate systems accumulated soil C more slowly and reached peak soil C levels later. These results suggest that time lags before soil C benefits become apparent may occur in both regions but are likely to be more pronounced in temperate agroforestry systems.

Nevertheless, agroforestry remains a potentially important nature-based solution for climate change mitigation due to its capacity to sequester C in both biomass and soils. In addition, agroforestry may provide opportunities for participation in C credit schemes, subject to robust monitoring, reporting, and verification requirements. The prospect of generating C revenues can incentivise farmers and other land managers to adopt agroforestry practices and invest in long-term landscape transformation (Marland et al., 2001; Pradhan et al., 2023).

Several studies have demonstrated that agroforestry systems substantially enhance C sequestration and are among the most effective nature-based solutions for climate mitigation. By integrating trees into agricultural landscapes, agroforestry increases C storage in both aboveground and belowground biomass and soil while creating favourable microclimatic conditions that improve resource efficiency potential and ecosystem functioning (Dhyani et al., 2021; Rehnus et al., 2013; van Noordwijk et al., 2021; Golicz et al., 2021; Smith et al., 2008; Aerts, 1997; Griscom et al., 2017; Ma et al., 2020; Ntawuruhunga et al., 2023). However, the C sequestration performance of agroforestry systems varies considerably according to system design and environmental

conditions. Factors such as tree density, species selection, previous land use, tree age, and regional climate affect the rate and magnitude of C accumulation, resulting in substantial variability in aboveground biomass carbon (AGB C) and SOC stocks (Nair et al., 2009ab; Takimoto et al., 2009; Chatterjee et al., 2018). These sources of variation introduce uncertainty into estimates of C storage and sequestration potential, limiting the ability to predict outcomes across different agroforestry systems and regions. Therefore, further research is needed to reduce these uncertainties, improve understanding of the factors controlling C dynamics, nutrient cycling and C sequestration and support evidence-based agroforestry management. Such knowledge is essential for informing policies and incentive mechanisms that promote long-term total C accumulation and contribute to greenhouse gas mitigation objectives.

In this context, this thesis aims to examine the key biotic and abiotic factors controlling C storage and sequestration in temperate agroforestry systems, with particular emphasis on the aboveground and belowground interactions that influence C and nutrient cycling. Given the increasing recognition of agroforestry as an important climate change mitigation strategy, a better understanding of the processes governing C accumulation and nutrient dynamics is needed to support evidence-based management, optimise system performance and support policy development in temperate regions. To address these knowledge gaps, this thesis explores the following research questions:

1. How do AGB C stocks and annual C accumulation rates vary among poplar hybrids over a 30-year period, and to what extent are these variations influenced by local climate conditions?
2. What effects does a 30-year-old poplar-based silvoarable system have on SOC storage and sequestration, and what potential contribution could the expansion of such systems make to achieving UK net-zero targets through enhanced C sequestration?
3. How do spatial variations in soil nutrient availability and root traits interact to regulate belowground C inputs, nutrient cycling, and SOC accumulation within a silvoarable agroforestry system?
4. How does leaf litter location (surface vs. buried) influence decomposition rates and the dynamics of C and nitrogen (N) content during decomposition?

The remainder of this introductory chapter is structured as follows. Section 1.2 provides an overview of the global distribution of agroforestry along with policy interventions that have supported its development. Section 1.3 focuses on agroforestry in the UK and its potential to help achieve national net-zero targets. Section 1.4 explores the multifunctional roles of agroforestry systems, while section 1.5 discusses the key challenges and barriers to their adoption. Section 1.6 reviews the C sequestration potential of agroforestry and examines the interactions between above and belowground components that influence C storage and nutrient cycling. Finally, section 1.7 presents the summary of research objectives followed by section 1.8 which outlines the research aim and objectives in relation to the identified knowledge gaps, along with an overview of the methodological approach and the structure of the thesis.

1.2. Worldwide agroforestry and its global and local policy drivers

Land plays a critical role in sustaining agrifood systems, supporting approximately 95% of global food production and underpinning a wide range of ecosystem services (FAO, 2025). However, land degradation driven by human activities such as deforestation, overgrazing, and unsustainable agricultural practices lead to soil erosion, reduce soil productivity, threatens soil productivity, food security, biodiversity, and exacerbates climate change (FAO, 2025). Globally, 90% of deforestation is caused by agricultural expansion, with 49.6% attributed to cropland expansion and 38.5% to livestock grazing (UN, 2024). Beyond its environmental impacts, the economic costs of reduced ecosystem services due to land degradation vary by region, from $16,746 to $29,623 per km² in Oceania to $124,191 to $224,434 per km² in Asia (Carswell et al., 2025).

Increasing global population and changing lifestyles are placing growing demands and pressure on food production systems. To achieve global food security, food production must increase by approximately 70% between 2005 and 2050 (Noel, 2015). These accelerating pressures can compromise soil resilience (Carswell et al., 2025) and contribute to the degradation of critical ecosystem services, including provisioning, regulating, and supporting functions, thereby exacerbating climate change (Kopittke et al., 2019). Conventional agricultural practices also contribute to soil degradation. For example, pesticides are harmful to non-target fungi, insects, and bacteria; synthetic N fertilisers can cause soil acidification and affect nitrification processes; deep tillage can

reduce fungal networks and invertebrates, leading to declines in soil organic matter and nutrient cycling; and plastic mulching may result in plastic residue accumulation in soil (Campanale et al., 2024; Goulding, 2016; Kopittke et al., 2019; D. Ma et al., 2024; Montràs-Janer et al., 2024; Pan et al., 2021; Schneider et al., 2017; Velásquez et al., 2018). As a result, there is increasing interest in agricultural practices that maintain soil health, reduce erosion, enhance C sequestration, and promote the sustainable management of cropland soils.

In this context, agroforestry is increasingly recognised as a nature-based land management strategy with the potential to deliver socioeconomic and ecological benefits to both humanity and the planet (Ellison and Wang-Erlandsson, 2019). Agroforestry involves the interaction among trees, shrubs, crops, and livestock, creating complex ecological relationships. These interactions encompass key ecological processes, including mutualism, commensalism, predation, and competition, which can have both positive and negative effects on biotic and abiotic components. For example, in temperate Europe, Vaccaro et al. (2023) observed potential yield losses for crops growing near mature trees, highlighting the importance of species selection in agroforestry systems. While the effects of trees on crop productivity are well documented, fewer studies have quantified how crop presence and management influence tree growth trajectories and the resulting long-term C sequestration potential of agroforestry systems. Consequently, uncertainties remain regarding the extent to which crop-tree interactions affect tree biomass accumulation and C dynamics over time. Since C4 grasses are generally less shade-tolerant (Taiz et al., 2015; Xie et al., 2020), yield reductions were expected, although some shade tolerant species do exist (Horton & Neufeld, 1998). Therefore, site management plays a crucial role in minimising resource competition, thereby directly influencing plant productivity and maximising ecological benefits (Powell & Bork, 2004**;** Aschehoug et al., 2016; Swieter et al., 2022; Báder et al., 2023). Effective management practices, including appropriate tree spacing, species selection, and grazing control, can help balance competitive and facilitative interactions, thereby enhancing the overall productivity and resilience of agroforestry systems.

However, large scale agroforestry systems have been practised globally for centuries. Agroforestry practices have existed in European countries since ancient times, and its

adoption is increasing due to the wide range of perceived socio-ecological benefits. These systems are mainly characterised by trees planted into pastures with livestock (silvopastoral) or integrated into arable land (silvoarable) (Eichhorn et al., 2006). Several traditional farming systems remain prevalent in Europe. For example, on Crete and the Aegean islands, fruit and nut trees are integrated into silvoarable systems, with almond trees occupying approximately 18,000 ha and fig trees covering approximately 10,200 ha (Eichhorn et al., 2006). In Central Europe, *Streubost,* a traditional method of cultivating fruit trees in croplands, meadows, and pastures, is widely recognised and can be classified as either silvoarable or silvopastoral. The Mediterranean region is characterised by the *Dehesa,* an agrosilvopastoral system that historically covered ~2,250,000 ha in southwest Spain, and the *Montados* in Portugal that occupied ~896,000 ha (Castro, 2008; McAdam et al., 2008; Moreno & Pulido, 2008). The Montados system typically involves cultivating low-density oak trees integrated with silvoarable or silvopastoral land. However, due to a decline in oak regeneration and plantation establishment, these systems are struggling to survive (Lopes-Fernandes et al., 2024).

The European Parliament has supported the adoption of agroforestry practices through the Common Agricultural Policy (CAP), mainly under rural development measures. Under this framework, farmers can receive per-hectare payments for establishing and maintaining agroforestry systems on their land (Nie et al., 2016). Data from the AgForward project estimated that the EU27 contains approximately 15.4 million ha of agroforestry, representing approximately 9% of the usable agricultural area and 3.6% of the total land area; when reindeer husbandry systems are included, this figure increases to 52.4 million ha (Augère-Granier, 2020). Of the 15.4 million ha classified as agroforestry, only 35,800 ha are categorised as silvoarable systems, whereas ~15.1 million ha consists of various forms of livestock-based agroforestry (Augère-Granier, 2020). This disparity highlights the dominance of silvopastoral and other livestock-integrated systems across Europe and suggests considerable potential for expanding silvoarable agroforestry. As concerns regarding climate change mitigation and adaptation continue to grow, the European Parliament has actively promoted agroforestry practices; however, significant untapped potential remains for expanding its adoption and integration across the region (McNie et al., 2016).

Agroforestry gained formal recognition as a climate change mitigation and adaptation policy following the Kyoto Protocol and has subsequently been incorporated into major international frameworks. These include the Reducing Emissions from Deforestation and Forest Degradation (REDD+) mechanisms, the UNFCCC Paris Agreement, the Convention on Biological Diversity (CBD), the United Nations Convention to Combat Desertification (UNCCD), and the United Nations SDGs. It has also gained support from international organisations such as the Food and Agriculture Organisation (FAO) and the World Bank (Dhyani et al., 2021; Meybeck et al., 2022; Rehnus et al., 2013; Meine van Noordwijk, 2019).

Growing international awareness of the role of agroforestry is reflected in recent climate assessments and policy reports. For example, Harrison and Gassner (2020) noted the role of trees on agricultural land received increased attention in the Intergovernmental Panel on Climate Change (IPCC) Working Group III report and earlier land-use assessments (Calvo Buendia et al., 2019). In addition, agroforestry is increasingly referenced in National Adaptation Plans (NAPs) and NDCs, further highlighting its perceived importance for C sequestration and climate change mitigation (Meybeck et al., 2020; Adaptation Fund, 2025).

National greenhouse gas inventories and global modelling studies estimate anthropogenic CO_2 emissions from land use at around 5.5 Gt CO_2 per year (IPCC Working Group III, Climate Change 2022). Yet discrepancies associated with measurement methods and reporting methodology introduce uncertainty into these estimates (IPCC Working Group III, Climate Change 2022). Agroforestry has therefore been promoted as an effective land-use strategy for mitigating these emissions by integrating trees into agricultural landscapes, thereby minimising the environmental impacts associated with conventional agriculture while also delivering ecological, socio-economic, and environmental benefits (Jovanelly et al., 2025; Smith et al., 2020).

Despite increasing policy support, important knowledge gaps remain regarding the magnitude and variability of C sequestration across different agroforestry systems. Currently available scientific evidence is often fragmented, system specific, and geographically constrained, limiting its usefulness for large-scale planning and policy design. Consequently, there is a need for more comprehensive quantitative and

qualitative assessments of C sequestration, alongside long-term monitoring across diverse environmental conditions. Enhancing the availability of comparative datasets would strengthen evidence-based decision making (Quandt et al., 2023).

1.3. Agroforestry in the United Kingdom

Agroforestry in the UK has transitioned from a traditional land-management practice to a research-led approach to sustainable agriculture and environmental restoration. Historically, integrated tree-livestock systems were widely practised through methods such as pannage, pollarding and woodland grazing, which provided a range of benefits including timber, fodder, fuel and food production. These practices developed in the early modern era into more structured systems, such as orchards, shelterbelts and parkland landscapes (Smith, 2010). However, throughout the 20th century, agricultural intensification significantly altered the British landscape. Efforts to increase crop yield resulted in the removal of trees from fields to provide machinery access, while the expansion of agricultural land following the Second World War and the introduction of the 1947 Agriculture Act contributed to substantial losses of semi-natural habitats (Hooftman and Bullock, 2012; Best and Coppack, 1992).

The simplification of farmland landscapes continued during the second half of the 20th century, when extensive hedgerow removal was undertaken to create larger fields (R. A. Robinson & Sutherland, 2002)**.** Since 1947, it has been estimated that approximately 50% of the UK's hedgerows have been lost (CPRE, 1999). Despite recent agri-environment schemes aimed at promoting hedgerow recreation and restoration (Defra, 2005), the Countryside Survey reported that the length of managed hedgerows has continued to decrease by 6.2% between 1998 and 2007 (Countryside Survey, 2008). These changes have contributed to biodiversity loss, habitat fragmentation, and reductions in ecosystem service provision, prompting renewed interest in agroforestry as a means of reintegrating trees into agricultural landscapes while supporting both productivity and environmental sustainability.

The limited adoption of agroforestry in the UK is primarily due to a combination of economic, policy and knowledge-based barriers. Farmers often report insufficient understanding of agroforestry practices, limited financial support for both adoption and

long-term maintenance, and a lack of awareness of the economic benefits and diverse income streams that agroforestry can provide (Farming Forum, 2022; Tranchina et al., 2024). A major barrier is the long-perceived lead time before trees generate an income. In addition, tenancy-related issues such as the need for landowner approval for tree planting and the reluctance of tenants to invest in long-term land improvements further reinforce negative perceptions among farmers, many of whom view agroforestry as offering low financial returns (Lawrence and Dandy, 2014; Valatin et al., 2016). Integrating trees into arable and livestock systems can also create practical management challenges, including interfering with machinery access and farming operations and constraints on future land-use opportunities (Felton et al., 2023).

Rising establishment and maintenance costs emphasise the need for financial incentives, through planting grants and ongoing management payments. Such support is likely to be essential if the UK is to meet its net-zero target, which relies on large-scale tree planting and increased tree cover across agricultural landscapes by 2050. Therefore, many of the barriers currently limiting agroforestry adoption also represent opportunities for policy interventions and the wider implementation of agroforestry systems as a climate mitigation strategy. (Kendon et al., 2023; Hanlon et al., 2021; Velásquez et al., 2018; Pan et al., 2021).

Climate change has further increased interest in agroforestry across the UK. The Met Office reports that the UK is experiencing wetter, sunnier, and warmer weather, while the frequency and severity of extreme events such as heatwaves, floods, and storms are increasing (Kendon et al., 2023). These changes are affecting the UK's agricultural sector (Hanlon et al., 2021). High rainfall negatively impacts agriculture by causing soil nutrient loss and creating waterlogged conditions, which can lead to water-related diseases (Velásquez et al., 2018; Pan et al., 2021). As a result, agroforestry is increasingly being recognised as a nature-based solution that can provide multiple benefits, such as flood mitigation, improved water regulation, reduced soil erosion, and increased ecosystem stability, highlighting its potential to address climate change adaptation and mitigation (Woodland Trust, 2020; Judson et al., 2026).

To maximise these benefits, a range of modern agroforestry systems have been developed to enhance land productivity while maintaining sustainable resource use. In

the UK, the main agroforestry systems include silvoarable, silvopastoral, hedgerows and home gardens (mixed tree-crop systems around homesteads), all of which aim to deliver both ecological and socioeconomic benefits (den Herder et al., 2017). Research and development in this field have been supported through several important initiatives, such as the establishment of the silvopastoral National Network Experiment in the 1980s, the National Silvoarable Network in 1992 (Hislop and Claridge, 2000), Martin Wolfe's organic silvoarable site (1994) and Stephen Briggs' large-scale apple-based silvoarable farm established in 2009 (Smith, 2010).

Tree species selection remains a critical component of successful agroforestry design. Broadleaf species such as cherry (*Prunus* spp.), oak (*Quercus* spp.), hazel (*Corylus avellana*), and field maple (*Acer campestre*) are among the most used because of their high nut and timber values (Forestry Commission, 2000). Poplar is also widely utilised for its rapid growth and economic benefits (Burgess et al., 2003). However, research has demonstrated that crop yields may decline in a competitive zone surrounding trees, with the magnitude of the effects varying according to crop type and tree species (Burgess et al., 2003., Vaccaro et al., 2023). Competition for light, water, and nutrients can also influence tree growth, particularly during establishment, with implications for long-term biomass accumulation and C sequestration. Hence, careful tree species selection, appropriate tree spacing, and effective site management are crucial for minimising crop competition and optimising overall system yields (Forestry Commission, 2002).

In 2020, the UK was the first major economy to legally commit to a net-zero emissions target to address climate change. However, current efforts remain insufficient to meet these targets (Climate Change Committee, 2022), particularly in the agrifood sector. To remain on track, the country must reduce greenhouse gas emissions by 81% from 1990 levels by 2035 (UK Climate Action Tracker, 2025). While the adoption of green energy and more efficient infrastructure is important, they are unlikely to deliver the required reductions alone. Large-scale implementation of nature-based solutions that increase C sequestration across agricultural landscapes will also be necessary.

Agroforestry has been identified as a key component of this transition. Achieving national net-zero objectives is expected to require that approximately 10% of UK farmland be managed under agroforestry systems (Burgess and Graves, 2022).

However, agroforestry currently occupies approximately 550,000 ha of the UK, representing about 3.3% of all agricultural land (Defra, 2025). Silvoarable agroforestry is rare, representing only 0.05% of the total arable area (den Herder et al., 2017). According to Burgess & Graves (2022), if 20% of the UK's arable land were converted to agroforestry, net-zero arable farming could be achieved by 2048. Additionally, converting 30% of pastures to silovopasture would be necessary to reach the overall targets by 2051 (Burgess & Graves, 2022). Such changes could reduce net agricultural emissions by 84% in arable systems and 80% in livestock systems (Burgess and Graves, 2022).

Despite growing evidence that agroforestry can enhance C sequestration within UK agricultural landscapes, important knowledge gaps remain regarding: (i) the quantification of soil and biomass C stocks and sequestration over time; (ii) nutrient cycling and belowground interactions; (iii) the long-term C sequestration potential of different agroforestry systems across environmental conditions; and (iv) trade-offs and synergies between C storage and agricultural productivity. Addressing these gaps will improve the evidence base for climate-smart land management.

1.4. Key functions of agroforestry

Agroforestry ecosystems provide a diverse range of ecological functions that collectively contribute to valuable ecosystem services (Fig. 1.1). Although agroforestry practices vary according to their main purpose and whether the trees are mixed with crops or livestock, all systems offer significant environmental benefits. As shown in Fig. 1.1, agroforestry systems facilitate provisioning, regulating, supporting, and cultural ecosystem services by enhancing environmental resilience. Trees can be planted in several differing configurations: (i) as windbreaks to reduce wind erosion and protect crops (Smith et al., 2021); (ii) along field boundaries as hedgerows; (iii) around mature trees to increase biomass accumulation; (iv) with shrubs and woodlands to increase habitat diversity (Brown et al., 2018); and (v) adjacent to water bodies as riparian buffers to improve water quality. Each of these agroforestry configurations provides various ecological and socioeconomic advantages.

1.4.1. Provisioning services

Agroforestry systems increase provisioning services by supplying a diverse range of products, such as food crops, fruits, fuelwood, timber, fibre, fodder, biofuels, and medicinal products (Awazi et al., 2025b). By integrating trees with crops and/or livestock, farmers can generate multiple revenue streams from annual crop production, fruit harvests, fodder, and timber production over short-, medium-, and long-term time horizons (Wiggering et al., 2006; Mukhlis et al., 2022). However, agroforestry systems generally produce a higher net present value than conventional monoculture systems, resulting in higher farm profitability and economic resilience (Báder et al., 2023; Atapattu et al., 2024). Agroforestry also offers additional income-generating opportunities through C markets and C credit schemes. However, significant knowledge gaps remain regarding the magnitude and variability of provisioning services delivered by agroforestry systems and the trade-offs between provisioning benefits and other ecosystem services across different agroforestry configurations (Báder et al., 2023).

1.4.2. Regulating services

Agroforestry systems play a pivotal role in the provision of regulating ecosystem services, in particular, climate regulation, water purification, pest control, and disease resistance (Fig. 1.1). There is significant evidence regarding the greater potential of C sequestration in agroforestry systems compared with monoculture practices due to higher AGB and BGB production (Tumwebaze & Byakagaba, 2016; Vigulu et al., 2019; Schneidewind et al., 2019; Guillot et al., 2021; Bibi et al., 2022). However, the magnitude and rate of C sequestration are strongly dependent on factors such as system design, species composition, soil type, climate, and temporal scale considered (Lorenz & Lal, 2014; Cardinael et al., 2018b).

Agroforestry can also contribute to pest regulation by interrupting pest life cycles and enhancing natural biological control, thereby reducing reliance on chemical pesticides (Muhammad Ahmad et al., 2024). However, such effects are often context specific. Trees may also provide alternative hosts for pests (Schroth et al., 2000), while increased shade, higher humidity, and reduced airflow may create favourable conditions for some

pests and plant diseases, possibly sustaining or increasing pesticide use (Staton et al., 2018).

Hydrological regulation is another key ecosystem service that agroforestry systems may provide. Trees can increase infiltration rates, enhance soil water storage, improve water quality, and buffer storm events by reducing runoff and nutrient loss (Idris & Mahrup, 2017; Welsh et al., 2018). However, the effectiveness of these outcomes depends strongly on climatic, geomorphological, spatial layout, and biological conditions (Seobi et al., 2005; Akdemir Eda et al., 2016; Chandler et al., 2018; Judson et al., 2026).

1.4.3. Supporting services

Trees and woodlands in agricultural landscapes promote habitat heterogeneity that supports key ecosystem functions, such as primary production, soil formation, and nutrient cycling. Agroforestry systems generally provide greater capacity for alpha-, beta-, and gamma-level biodiversity than simplified agricultural systems (Santos et al., 2022). However, biodiversity responses are uneven across agroecosystems and depend heavily on species composition, landscape context, and management intensity (Daleo et al., 2023; Vanneste et al., 2024).

The higher structural complexity characteristic of agroforestry systems may improve both biomass production and soil health. Appropriate combinations of tree and crop species can achieve 20–80% greater total biomass by optimising resource use, such as enhanced light capture and the maintenance of favourable microclimatic conditions (Rijal et al., 2025). Nevertheless, increased biomass accumulation may also result in trade-offs. Competition between trees and crops for light, nutrients, and water may reduce crop yields, while elevated soil respiration may lower net C storage (Swieter et al., 2022). In conclusion, the benefits of agroforestry systems for supporting ecosystem services are not universal and their effectiveness remains highly context-dependent and management-dependent.

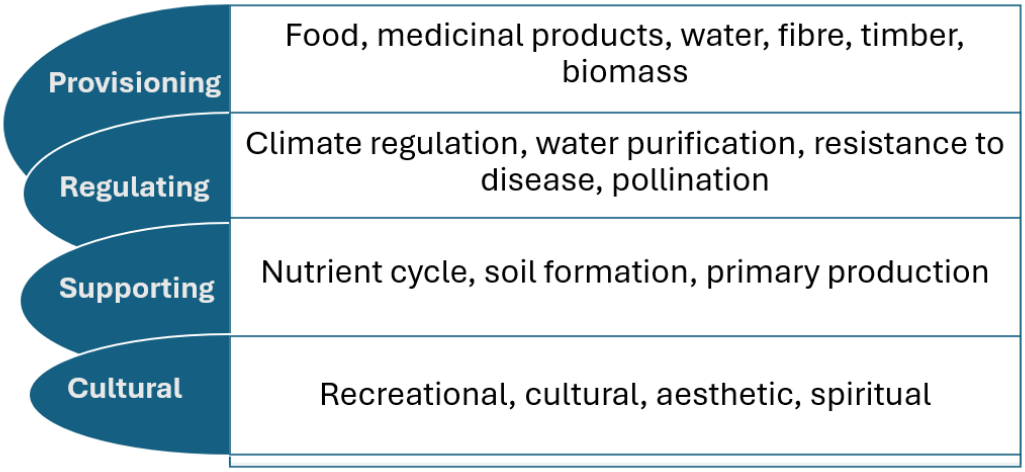


Figure 1.1 Classification of agroforestry ecosystem services based on the Millennium Ecosystem Assessment framework showing the contribution of agroforestry systems to providing provisioning, regulating, supporting, and cultural services (an original figure adapted from ideas in MEA, 2025; Jose, 2009; Mlambo et al., 2025; Mathieu et al., 2025).

1.4.4. Cultural services

Agroforestry landscapes contribute to cultural services by supporting aesthetic values, cultural identity and traditional knowledge systems of land stewardship, with gender-differentiated roles and practices (Jose, 2009). Such multifunctional landscapes can enhance visual appeal, provide recreational opportunities, encourage tree planting and sustain the culture of interactions between people and agroecosystems (Fagerholm et al., 2019; Torralba et al., 2016). Agroforestry systems may also facilitate gender equality through increasing women's access to productive resources, participation in decision-making opportunities, and engagement with culturally valued knowledge related to tree-based systems, especially in smallholder and subsistence farming contexts (Doss et al., 2018; Meinzen-Dick et al., 2014).

1.5. Challenges and research needed in temperate agroforestry

Agroforestry is increasingly acknowledged as a nature-based solution that can contribute to climate change mitigation and adaptation, biodiversity conservation and resilient agricultural production. Many countries have integrated agroforestry into climate and land-use strategies, while the success of agroforestry systems has still been

highly dependent on ecological, economic and policy factors (Fig. 1.2) (Tranchina et al., 2024; Felton et al., 2024).

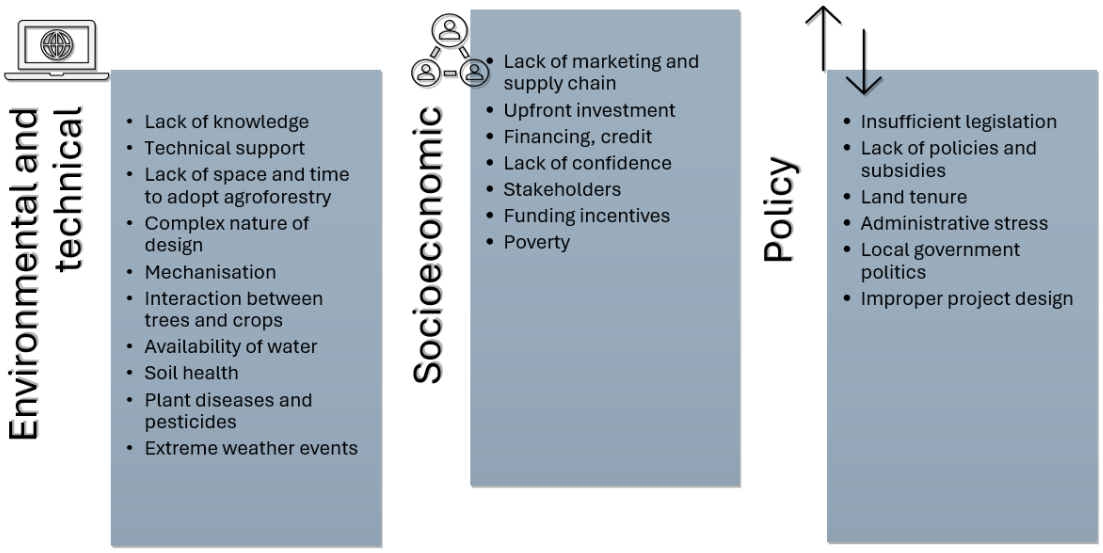


Figure 1.2 Barriers to establishment of agroforestry systems, categorised into three broad groups: environmental and technical, socioeconomic, and policy-related factors (an original figure based on concepts presented in Tranchina et al., 2024)

Climate variation in temperate regions, including changing precipitation patterns and higher temperatures, may affect tree growth and crop productivity, making the long-term performance of agroforestry systems uncertain (McRae, 2024; Jovanelly et al., 2025). Species selection and system design are important, as poorly designed systems may create competition for light, water and nutrients between trees and crops. However, appropriate tree spacing and alley widths can reduce these trade-offs while maintaining ecosystem benefits (Felton et al., 2024).

Economic and policy factors also affect adoption. While agroforestry can provide additional income streams through timber and non-timber products, financial returns are often realised over longer time periods for farmers in the process of establishing the system (Felton et al., 2023). In Europe, implementation may also be hampered by the administrative complexity of agri-environment schemes and CAP regulations (Augère-Granier, 2020; Sollen-Norrlin et al., 2020).

Despite increasing policy support and recognition of agroforestry's climate mitigation potential, significant knowledge gaps remain regarding the extent and distribution of C stocks in temperate agroforestry systems, particularly belowground. Uncertainties in soil and root C pools, C sequestration rates and nutrient cycling processes make it challenging to accurately assess climate impacts and inform policy decisions (Jha, 2018; Biffi et al., 2022; Judson et al., 2024). Thus, addressing these knowledge gaps is crucial for better accounting of C and for the evidence-based adoption of agroforestry.

Thus, this thesis will focus on above- and belowground C stocks, C sequestration rates, nutrient cycling, and root characteristics, including root biomass density, root length density, root C stocks, and root C and nitrogen concentrations, in temperate agroforestry systems. These components are examined to better understand the mechanisms by which agroforestry contributes to climate change mitigation and long-term ecosystem sustainability.

1.6. Carbon stocks, sequestration and nutrient cycling in agroforestry systems

1.6.1. Carbon stocks and sequestration rate

In recent years, there has been a growing interest in the climate change mitigation potential of introducing woody species into agricultural landscapes via agroforestry (Judson et al., 2023). Trees capture carbon dioxide (CO_2) through photosynthesis and store it as C in AGB, belowground biomass (BGB) and SOC pools (Báder et al., 2023). Agroforestry systems can therefore contribute to C sequestration through various pathways. Carbon sequestration refers to the transfer of atmospheric C into long-term terrestrial reservoirs (USGS, 2023; Nair et al., 2021bc), and in agroforestry, this occurs through the accumulation of woody biomass, litter inputs, root turnover, and rhizodeposition occurs through the accumulation of woody biomass, such as AGB (trees' main stem, branches, leaves) or BGB (leaf litter inputs, deposition, complex root turnover and rhizodeposition. These processes enhance SOC through the decomposition of litter and woody residues and incorporation of root-derived C into the soil (Awazi et al., 2025b). The detailed mechanisms of C sequestration are discussed in sections 1.6.2–1.6.4.

The soil C pool is a critical edaphic component that regulates ecosystem structure and function, containing approximately 2300 Pg of C in the top 1 m of soil, including 1500 Pg (10^{15} g) of organic and 750 Pg of inorganic C (Batjes, 1996; Munoz-Rojas et al., 2012). This reservoir exceeds both the atmospheric C pool (770 Pg) and the vegetation C pool (610 Pg) (Nair et al., 2021b; Canadel & Monteiro, 2021). However, land-use change is a key driver of variation in terrestrial C stocks (Yu & Song, 2023), while soil CO_2 emissions from both human and natural activities make soils the second-largest source of CO_2 globally (Le Quéré et al., 2009; Van Der Werf et al., 2009). Agroforestry practices can mitigate greenhouse gas emissions by increasing C storage in biomass and soils, thereby reducing atmospheric CO_2 levels. Where sequestration exceeds system emissions, agroforestry may function as a net C sink (Baah-Acheamfour et al., 2017).

The global removal potential of agroforestry has been estimated at approximately 0.1–5.7 Gt CO_2 yr^{-1} (Mlambo et al., 2025). Furthermore, increased tree cover on agricultural land increased global biomass C stocks by >2 Pg C between 2000 and 2010, while further expansion could add an additional ~4–6 Pg C (incremental adoption) to ~12–19 Pg C (systemic adoption) (Zomer et al.,2016; Zomer et al., 2022). Although temperate regions generally contain higher AGB C stocks than tropical regions, C sequestration rates are typically greater in tropical regions due to faster tree growth (Mazumder et al., 2025). In addition to biomass C storage, agroforestry can substantially increase SOC stocks. Global meta-analyses of SOC stocks by Shi et al. (2018) and Lorenz and Lal (2014) indicated that agroforestry systems could store an additional ~5.3 Pg C in SOC worldwide. Reported SOC sequestration rates typically range from ~0.2–0.4 Mg C ha^{-1} yr^{-1}, depending on climate, agroforestry system type, stand age, and baseline SOC levels. In temperate regions, Mayer et al. (2022) reviewed 63 observations and found that hedgerows and silvoarable systems exhibited the highest SOC sequestration rates, particularly in topsoil, ranging from 0.2–0.3 Mg C ha^{-1} yr^{-1}. Overall, these findings highlight the importance of considering both aboveground and belowground pathways in determining the overall C sequestration potential of agroforestry systems.

Table 1.1 Total biomass C (above ground and below ground, including tree biomass) on agricultural land in 2000 and in 2010 (adapted from Zomer et al., 2016)

Region	Total C stored by biomass in agricultural land (Pg C)			Total agricultural area (km^2)
	2000	2010	Increased total percentage of C due to an increase in tree cover	
Australia/Pacific	2.11	2.28	8.06	790,658
Central America	1.42	1.52	6.45	239.235
Central Asia	0.48	0.47	-1.04	839.949
East Asia	2.37	2.53	6.95	1,795,893
Eastern and Southern Africa	2.31	2.3	-0.17	1,573,527
Europe	2.13	2.15	0.96	2,299,766
North Africa	0.11	0.11	-0.01	155,948
North America	3.31	3.4	2.68	2,073,033
Russia	1.07	1.07	0.02	1,669,166
South America	11.34	12.13	6.95	3,888,792
South Asia	2.3	2.48	7.85	1,827,025
Southeast Asia	10.03	10.69	6.59	1,648,268
West and Central Africa	5.57	5.45	-2.18	2,390,980
Western Asia	0.75	0.79	4.72	955,689
Global C = (agri + tree integration)	45.3	47.37	4.57	
Baseline	11.08	11.08		
Contribution by trees	34.22	36.29	4.57	

Table 1.1 shows regional variations in total biomass C stocks (Pg C) on agricultural land in 2000 and 2010, based on observed changes in tree cover derived from remote sensing. Baseline C stock (11.08 Pg C) represents the estimated biomass C stored in agricultural land excluding additional C stored by tree cover. Tropical regions exhibited the largest absolute increases in biomass C stocks between 2000 and 2010, reflecting their greater potential for rapid biomass accumulation. In contrast, temperate areas, including large parts of Europe, showed smaller absolute changes in total biomass C stocks. However, when total biomass C stocks are expressed on an area basis (Mg C ha^{-1}), temperate agricultural areas often contain relatively high biomass C densities. This indicates that temperate agroforestry systems may already contain substantial woody biomass stocks, with limited scope for rapid increases in C stocks compared with regions with lower baseline C stocks.

Sequestration rates generally decrease with stand age and may approach saturation over time (Nair et al., 2010). Soil C accumulation is also very heterogeneous, commonly localised in surface horizons and strongly influenced by soil properties, agroforestry design, climate, and previous land use, making large-scale upscaling using limited data a challenge and potentially associated with large uncertainties (Cardinael et al., 2018c). In temperate regions, these challenges are exacerbated by the limited availability of long-term datasets and inconsistent reporting of belowground C pools and fluxes, such as fine-root turnover and rhizodeposition, which are important components of the overall C balance (Cardinael et al., 2018b; Meybeck et al., 2022).

As such, while agroforestry represents a promising nature-based solution for combating climate change, uncertainties in C accounting, long-term permanence, saturation effects, and landscape-scale implementation highlight the need for cautious interpretation of sequestration estimates and their integration into broader emissions-reduction and land-management strategies. Despite the high net C removal potential of agroforestry systems, considerable uncertainty persists regarding the scalability, permanence, and reliability of these estimates. Current projections of CO_2 removal (0.1–5.7 Gt $CO_2 yr^{-1}$) are predicated on widespread adoption, sustained long-term management, and stable climatic conditions, all of which may be inapplicable (Osei & Oelbermann, 2025; Shi et al., 2018; Zomer et al., 2022). Additionally, C accumulation in both biomass and soil is not indefinite.

1.6.2. Factors influencing carbon sequestration potential in agroforestry

Fig. 1.3 illustrates the key biological and management-related factors that regulate C sequestration in agroforestry systems, as well as the major pathways through which atmospheric C is incorporated into vegetation and soil C pools. The subsequent sections focus on how agroforestry system design and management (Section 1.6.2.1) and climatic conditions (Section 1.6.2.2) affect the accumulation and long-term storage of C in vegetation and soils.

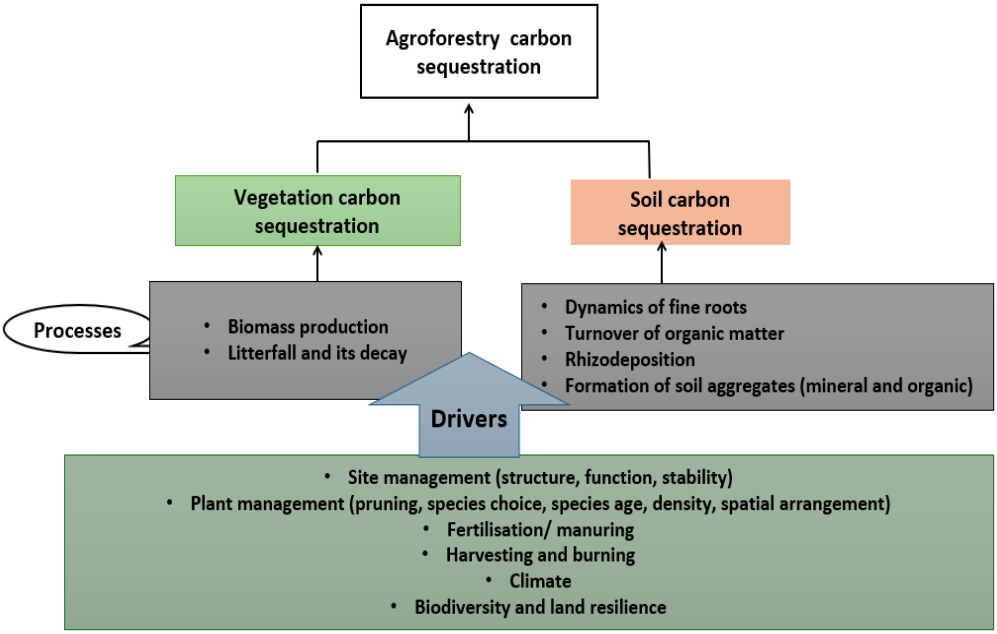


Figure 1.3 Conceptual framework illustrating the key factors affecting C sequestration in agroforestry systems

1.6.2.1. Configuration and management of agroforestry systems

Tree density and the spatial arrangement of trees within agricultural landscapes also influence C sequestration potential by affecting biomass production, belowground C allocation, and resource competition (Alexander et al., 2025). Under conditions of limited soil resources, trees modify the biomass, morphology, and chemical composition of fine roots to enhance nutrient acquisition. This response may be particularly pronounced in systems with lower tree densities, where individual trees have greater access to available resources (Bo et al., 2018). Increasing tree density can increase stand-level biomass C storage; however, it may also reduce understory crop productivity due to intensified competition for light, water, and nutrients. For example, poplar clones grown in wide alleys within temperate silvoarable systems in Italy exhibited faster growth rates than those in densely planted poplar plantations, where inter-tree competition constrained biomass accumulation (Piotto et al., 2024). Collectively, these findings demonstrate that both spatial configuration and structural management are key determinants of biomass dynamics and, consequently, the C sequestration capacity of agroforestry systems.

Management practices have a profound impact on C sequestration by regulating biomass production, SOC inputs, and decomposition processes (Fig. 1.3). Species selection, pruning intensity, and tree density influence AGB and BGB accumulation and consequently the amount of C captured and stored in vegetation. Pruning can promote compensatory growth and improve biomass turnover; however, repeated or intensive pruning can reduce woody biomass and limit long-term C storage (Fig.1.3). For example, Axe et al. (2017) report that untrimmed hedges stored significantly more AGB C (42.0 ± 3.78 t C ha^{-1}) than trimmed hedges, indicating the trade-off between structural management and C sequestration.

Enhanced productivity and greater biomass inputs to soil can also result indirectly from management practices, such as manure or fertilisation, especially within alley cropping systems, where crop yields are often improved. However, over-application of fertiliser may negatively impact soil health through acidification or salinisation, contributing to land degradation and potentially offsetting long-term gains in C sequestration (Fig. 1.3; Hijbeek et al., 2019; Liu et al., 2023). Litterfall and pruning residues represent major pathways of C transfer to the soil. Their contribution to SOC depends on the amount, its quality and chemical composition (e.g., lignin and nitrogen content), as well as environmental and management conditions that regulate the rate of decomposition (Gartner and Cardon, 2004). Moreover, rhizodeposition, fine root turnover, and soil aggregate formation all contribute substantially to SOC accumulation (Reid and Goss, 1982; Nguyen, 2009). However, increased organic matter inputs can also stimulate microbial activity and accelerate CO_2 release through priming effects, partially counteracting SOC gains.

1.6.2.2. Climatic factors

Climatic factors, particularly temperature and precipitation, strongly influence SOC sequestration rates by regulating primary productivity, litter inputs, and decomposition processes. A meta-analysis conducted in China found that shelterbelts and silvoarable systems enhanced soil C storage (both topsoil and subsoil), with the magnitude of the effects primarily influenced by annual mean temperature, precipitation, land use, and soil class; subtropical zones showed the highest SOC storage capacity (Hübner et al. 2021). Similar findings were reported in other meta-analyses by Chatterjee et al. (2018),

Feliciano et al. (2018), and Shi et al. (2018) for temperate and tropical regions, which showed that favourable moisture availability and thermal conditions, combined with greater system complexity, generally enhanced SOC stocks relative to less diverse land use systems. However, uncertainties persist due to variations in land management practices, soil types, plant species, and climatic conditions, which affect the C sequestration potential of agroforestry. De Stefano and Jacobson (2018) further highlighted methodological challenges, including the lack of standardised sampling protocols, variable sampling procedures, inconsistent study designs, and insufficient reporting of variance estimators and explanatory variables. These limitations hinder precise quantification of climate-driven changes in SOC.

Biomass allocation among leaves, stems, and roots reflects the balance between aboveground and belowground resource acquisition, enabling vegetation to adapt to varying environmental conditions (Zhang et al., 2023). The distribution of biomass is influenced by the availability of essential resources such as light, atmospheric CO_2, water, and soil nutrients, which collectively determine plant growth strategies under different climatic regimes. Climatic variables including mean annual temperature, annual precipitation, soil temperature, sunshine duration, growing degree days, and the frequency of ground and air frost significantly affect biomass accumulation, leaf area development, DBH, canopy cover, and overall tree growth. Temperature and precipitation are recognised as major drivers of AGB production because they regulate biogeochemical processes such as photosynthesis, respiration, and water availability (Ma et al., 2023; Yao et al., 2022). The presence of multiple tree and crop components may create diverse microclimates, improve resource-use efficiency, enhance soil moisture retention, and buffer the impacts of environmental extremes. Consequently, agroforestry systems could potentially better sustain tree growth and biomass production under adverse climatic conditions than monocultures (Quandt et al., 2023b; Ntawuruhunga et al., 2023).

1.6.3. Interactions between aboveground and belowground carbon sources

Aboveground and belowground C pools in agroforestry systems are closely linked through continuous C transfer processes that regulate ecosystem productivity and long-term C storage. Carbon fixed through photosynthesis is initially allocated to AGB,

including stems, branches, and foliage, where approximately 50% of the dry biomass is represented as C (Nair et al., 2012; Nair et al., 2021abc; Shepherd & Montagnini, 2001). The amount of C accumulated in AGB varies depending on species composition, stand age, site characteristics, and management practices (Nair, 2009a).

The interaction between aboveground and belowground C pools occurs through several pathways including, litterfall, pruning residues, root turnover, and dissolved organic C movement through the soil profile. These fluxes regulate the transfer of C and nutrients from vegetation to soil and are therefore key determinants of soil organic carbon (SOC) accumulation in agroforestry systems (Ramos et al., 2018). Litter quantity and quality strongly influence decomposition rates, microbial processing and subsequent C stabilisation. In particular, litter chemistry, including concentrations of lignin, nitrogen, tannins, and polyphenols, influences microbial activity and the balance between nutrient mineralisation and immobilisation, with implications for long-term C stabilisation (Cusack et al., 2009).

Carbon storage within agroforestry systems is also influenced by species-specific growth and allocation strategies. Fast-growing tree species can accumulate substantial amount of C is stored in AGB, although C losses also occur simultaneously through soil respiration and root-associated microbial activities (Bae et al., 2013). Variations in wood C concentration and biomass allocation across species and developmental stages further affect overall C storage. Climatic conditions also influence these processes; for example, the rapid turnover of soil organic matter under warm, humid tropical conditions may limit soil C accumulation compared to temperate systems (Shepherd and Montagnini, 2001; Nair et al., 2009b). Collectively, these factors highlight the complex, species- and climate-dependent relationship between AGB production and belowground C storage in agroforestry systems.

Balanced nutrient supply is essential for sustaining ecosystem services, and plant species selection plays a central role in regulating nutrient cycling rates in agroforestry systems (Kopittke et al., 2019). Compared with conventional agriculture systems, agroforestry can enhance nutrient turnover through increased biomass production and decomposition, which replenish C and nutrient soil pools (Sanchez, 1995; Rhoades, 1997; Khanna, 1997; Young, 1997; Nair et al., 2021bc). Integrating functionally diverse

species, particularly nitrogen (N)-fixing trees can further improve nutrient availability. For example, Sharma et al. (1994) demonstrated that combing N-fixing *Alnus* with non-N-fixing trees and cardamom enhanced nutrient cycling in the Eastern Himalaya. Tree species also differ in their contribution to nutrient return via litter decomposition, with species such as *Dalbergia sissoo* and *Azadirachta indica* promoting the replenishment of nitrogen phosphorus and potassium (Bisht et al., 2003; Hasauzzaman & Hossain, 2015; Hossain et al., 2011). Poplar-based agroforestry systems have similarly been identified as sustainable options for maintaining soil nutrient status (Das & Chaturvedi, 2005; Tzec-Gamboa et al., 2023).

Nutrient cycling in agroforestry remains inherently complex due to interactions among trees, crops, soil biota and environmental conditions. In temperate regions, nutrient dynamics and crop responses are influenced by shading intensity and local environmental conditions (Chirko et al., 1996; Dufour et al., 2013; Artru et al., 2017), while management practices, such as tree spacing and species composition strongly influence nutrient cycling (Nair et al., 1995). Agroforestry systems can modify soil biological communities increasing bacterial and earthworm abundance and modifying fungal assemblages (Vaupel et al., 2023). Furthermore, soil biodiversity and understory vegetation diversity contribute to ecosystem multifunctionality and stability, reinforcing the ecological benefits of agroforestry relative to monocultures (Jiao et al., 2022; Meyer et al., 2018; Wang et al., 2023a).

Belowground processes play a critical role in regulating nutrient cycling, nutrient retention, and C sequestration in agroforestry systems. Trees access nutrients from deeper soil horizons through extensive root systems, while nutrient release from organic residues is often accelerated during periods of high moisture availability (Hailu & Estifanos, 2023). Longer-term nutrient retention depends on root–soil interactions, mineral weathering and geochemical cycling, particularly in deeper soil layers where nutrient losses are reduced (Issac and Borden, 2019). Two mechanisms are particularly important in agroforestry systems: biological nutrient pumping, whereby trees translocate nutrients from deeper soil layers to surface horizons via litterfall, and the biological safety net effect, whereby tree roots capture nutrients that would otherwise be lost through leaching (Nair et al., 2021b). Nitrogen dynamics are particularly influential in shaping nutrient cycling and productivity in temperate agroforestry systems. Tree N

uptake is regulated by site conditions, competition, biomass production, and edge effects, all of which affect nutrient acquisition and crop performance (Udawatta & Jose, 2012; Gamble et al., 2016). Plants exhibit adaptive nutrient capture strategies in response to spatial and temporal variability in soil resources (Lambers et al., 2008). These responses are strongly modulated by soil microbial communities, which mediate nutrient transformations and retention. For example, Beule et al. (2020) reported enhanced abundances of microbial groups (Acidobacteria, Actinobacteria, alpha and gamma proteobacteria, Firmicutes, and Verrucomicrobia) associated with nutrient cycling and denitrification in poplar-based agroforestry systems than in monocultures, suggesting enhanced potential for N retention and reduced nutrient losses.

The interaction between aboveground and belowground C pools can be further enhanced by landscape features such as hedgerows, grass strips, and tree strips. These components increase AGB production while simultaneously promoting the transfer of organic C to the soil through litterfall, root turnover, and rhizodeposition, thereby contributing to nutrient retention and C sequestration in agricultural landscapes (Falloon et al., 2004). Carbon allocated belowground supports root growth, nodule development, and rhizodeposition, which contribute organic matter and biologically fixed N to the soil. However, important knowledge gaps remain regarding the pathways through which C and nutrients are transferred, transformed, and retained within agroforestry systems, limiting the ability to optimise system design and management (Nygren et al., 2012).

Roots represent the largest component of BGB, typically accounting for 20–40% of a tree's total biomass, and are increasingly recognised as a major contributor to long-term C storage. Root derived C can be stabilised through interactions with soil minerals, microbial communities and soil aggregates, often exhibiting longer residence times than aboveground litter inputs (Lorenz & Lal, 2014). Studies in agroforestry systems have shown that root-derived C inputs can equal or exceed those that are derived from aboveground biomass, highlighting the importance of root turnover and rhizodeposition as drivers of soil C sequestration (Jackson et al., 1997; Dhillon & Van Rees, 2016). Consequently, understanding root dynamics is essential for evaluating the long-term capacity of agroforestry systems to sequester C, maintain soil fertility, and enhance ecosystem resilience (Nair et al., 2009ab).

Fig. 1.4 illustrates the pathways through which above- and belowground C inputs contribute to soil C sequestration. Aboveground inputs include litter derived from leaves, flowers, branches, fruits, and woody biomass, which, upon decomposition, enrich soil organic matter and indirectly affect microbial activity in the rhizosphere. In contrast, belowground inputs originate primarily from roots in the form of exudates, sloughed-off cells, and other by-products, directly stimulate microbial activity in the rhizosphere, altering the decomposition of soil organic matter and potentially influencing the priming effect through enhanced mineralisation (Chen et al., 2019).

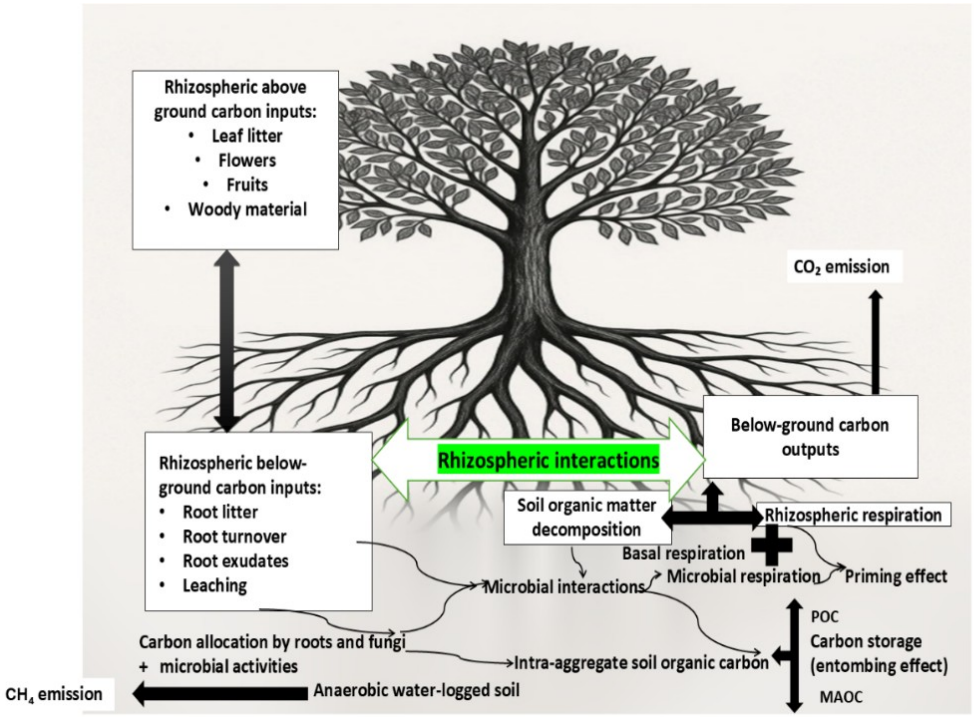


Figure 1.4 How rhizospheric components link to the formation of particulate organic C and mineral-associated organic C. (author's own figure developed based on concepts described in Dhyani et al., 2019).

Soil microbial activity can be quantified through respiration measurements, including basal respiration, which reflect microbial decomposition of soil organic matter independent of live roots (autotrophic respiration), and rhizosphere respiration, which includes both root respiration and microbial respiration stimulated by root activity (Hogan et al., 2023). Together, these above and belowground pathways regulate the balance between C storage and C loss from soils and therefore play a fundamental role in the long-term sequestration potential of agroforestry systems.

1.6.4. Root traits as drivers of carbon sequestration and nutrient dynamics

Root acquisition of water and nutrients depends on multiple root traits. Architectural traits (e.g., depth and length) define root system spatial configuration; morphological traits (e.g., diameter, size) determine absorptive surface area; physiological traits include respiration, nutrient uptake kinetics, and exudation; and biological traits involve microbial associations (Malamy, 2005; Osmont et al., 2007; Sousa-Nunes & Somers, 2010; Contreras-Cornejo et al., 2009; Zamioudis et al., 2013). Root exudates also modify root morphology specifically root length, branching density, and biomass distribution while mediating plant microbe interactions (Kong et al., 2018; Semchenko et al., 2014). In agroforestry systems, complementary root traits between trees and crops can reduce direct competition for resources and improve nutrient-acquisition efficiency compared with monoculture systems (Isaac & Borden, 2019).

In agroforestry, root distribution strongly influences SOC dynamics. For example, Upson and Burgess (2013) reported deeper coarse roots in poplar intercropped systems, which reduced SOC stock below 60 cm depth beneath tree rows, possibly due to the priming effect of newly introduced C. Dead roots contribute C inputs comparable to leaf litter, highlighting the importance of their turnover in soil C cycling (Defrenet et al., 2016). While studies using soil cores and allometric equations suggest minimal effects of root competition on fine root biomass, competitive interactions can still alter net primary production (NPP) in shaded plants (Charbonnier et al., 2017; Bai et al., 2023b). NPP, the balance of gross primary productivity and respiration, is central to quantifying ecosystem C cycling (Song et al., 2023). Collectively, these findings emphasise the central role of root traits, turnover, and species interactions in SOC sequestration and agroforestry productivity. Soil bulk density and pore structure largely determine root penetration capacity (Hughes & Wilde, 1988). Increased root branching under low microporosity conditions enhances root density and exploration. Root length density and biomass strongly influence soil volume exploration and nutrient availability (Reich et al., 1998; Kembel et al., 2008; Mommer et al., 2011; Ravenek et al., 2016). Within the root trait spectrum, low tissue density roots are associated with high N uptake, high turnover, and elevated respiration, contributing significantly to soil C cycling (Reich et al., 1998). However, elevated atmospheric CO_2 can substantially alter root morphology, biomass, and stoichiometry. Nie et al. (2013) observed that elevated CO_2 increased root

length by 26%, diameter by 8.4%, and total biomass by 28.8%. However, it simultaneously reduced root N concentration by 7.1% and increased the C:N ratio by 7.8%, without affecting root C concentration. These changes may enhance soil C sequestration but could also alter respiration, rhizodeposition, and fungal colonisation.

Root turnover, defined as the production and mortality of roots, is a fundamental component of ecosystem C and nutrient cycling (Aber et al., 1985). Turnover rates vary by root size: coarse roots have low turnover and high maintenance respiration, while fine roots exhibit rapid turnover (up to 100% annually), contributing roughly one-third of NPP (Jackson et al., 1997; Clark et al., 2001; Jourdan et al., 2008). The transformation of live and dead roots through turnover and decomposition drives soil C mineralisation and accumulation, while simultaneously regulating N dynamics through mineralisation and immobilisation. Root C: N ratios are negatively correlated with lifespan, linking nutrient stoichiometry to C cycling processes (De Deyn et al., 2008). On average, 40–50% of total soil CO_2 efflux originates from root respiration, though contributions vary by species, nutrient availability, and mycorrhizal associations (Bond-Lamberty et al., 2004; De Deyn et al., 2008; Martin & Stutz, 2004). These findings reinforce the critical role of root traits in sustaining soil fertility, microbial activity, and regulating C and nutrient cycle.

Studies have found that fluctuations in SOC over time are primarily influenced by the balance between C inputs from plant litter (aboveground components) and root turnover and exudates (belowground components), which are decomposed by microbes (Ong et al., 1991; McCormack et al., 2015; Oliveira et al., 2018; Mayrinck et al., 2019; Panchal et al., 2022). Increased biomass input in agroforestry systems, from both above and belowground sources, can lead to SOC losses through rhizomicrobial respiration, a phenomenon known as the priming effect (Fontaine et al., 2007; Sayer et al., 2011). However, studies have shown that greater tree diversity and density generally enhance SOC by increasing C inputs from biomass (Lange et al., 2015; Saha et al., 2009). Microorganisms with a fungal network associated with roots can alter mineralisation, leading to a priming effect (e.g., via root cell respiration, with emissions of CO_2 and CH_4). This labile form of C, derived from root and aboveground decomposition products, and a recalcitrant form of C from roots and from microbial interactions with soil minerals, promote the formation of soil aggregates (Tisdall, 2020). In contrast,

some aggregates lead to anaerobic, waterlogged soil due to C allocation by roots and fungal networks (Reid and Goss, 1982). This effect, though, is not uniform throughout the soil profile; it varies with depth depending on root length density, which can contribute more organic matter to deeper soil layers (Cardinael et al., 2018a). Thevathasan & Gordon (2004) found in alley cropping that about 35% of SOC is present at a depth of 0-15 cm within a 2 m distance from poplar trees in Southern Ontario.

Beule et al. (2021) reported that temperate poplar-based agroforestry showed an abundance of Basidiomycota fungi in alley cropping compared with crop monoculture. The ectomycorrhizal fungi, such as *Cortinarius*, *Geopora*, and *Inocybe*, associated with poplar enhance nutrient acquisition. Soil biota and soil aggregates exhibit a reciprocal relationship: soil organisms facilitate aggregate formation, while aggregates provide habitats for biota (Oades & Waters, 1991; Oades, 1984; Blanco-Canqui & Lal, 2004). Macroaggregates are primarily formed and stabilised by soil fauna such as earthworms and termites, whereas fungi and bacteria contribute to microaggregate formation (Tisdall, 2020). Microaggregates often contain higher microbial biomass and humic substances, with plant roots both living and dead supporting microbial communities. Beuschel et al. (2019) found that a poplar-based agroforestry system had greater variations in ratios of soil nutrient-fungal and soil bacterial C in tree rows compared to crop trees because of an increase in the input of lignin and residues of a suberin-rich tree. Such shifts in soil microbial and fungal communities influence the C stabilisation in soil aggregates. This effect may especially be significant in mature systems and deeper soil layers, providing insights into the long-term effectiveness of poplar in alley cropping systems.

In agroforestry systems, the chemical composition of plant residues plays a major role in altering decomposition dynamics and long-term soil C sequestration. The C:N ratio of plant residues strongly influences decomposition and C sequestration. Low C: N residues (e.g., soybean) decompose rapidly, while high C: N residues (e.g., wheat) decompose slowly, promoting long term SOC storage (Blanco-Canqui & Lal, 2004). Lignin and other recalcitrant compounds (aromatic, phenolic, lipids) enhance aggregate stability and prolong SOC residence time (Melillo et al., 1982; Monreal et al., 1997).

Lignin rich materials, due to their slow decomposition and binding properties, play a critical role in stabilising macroaggregates and protecting SOC.

Aggregate-associated SOC protection is a key mechanism for long-term C sequestration, particularly in systems like agroforestry that enhance labile organic matter. Biochemical recalcitrance and mineral interactions further stabilise SOC within aggregates, forming a passive C pool (Gale et al., 2001; Nair et al., 2021b). For example, Gunina et al. (2015) found that a no-tillage practice increased the abundance of macroaggregates with 70-80% C content in poplar agroforestry. Hence, they conclude that C content in the mineral fraction was 1.3-1.5 times greater in poplar-based agroforestry and short rotation coppice than in standard cropland management. Microclimatic conditions under trees influence the function of microfauna, mesofauna, and macrofauna, substantially altering the physicochemical and biological properties of soil and further affecting SOC storage (Chauhan et al., 2012; Jinger et al., 2023).

In summary, root traits influence C sequestration and nutrient cycling through multiple interconnected pathways, including resource acquisition, root turnover, rhizodeposition, microbial interactions, and aggregate formation. Variations in root architecture, morphology, physiology, and symbiotic associations determine both the quantity and quality of C and nutrient inputs to soil, ultimately affecting SOC accumulation and stability. While increased biomass inputs can enhance long-term C sequestration, they may also stimulate microbial activity and accelerate SOC losses through priming effects. Furthermore, interactions among roots, microbial communities, and soil aggregates play a central role in regulating C retention within soil profiles. Despite significant advances in understanding these processes, substantial uncertainties remain regarding the relative contributions of living roots, dead roots, and rhizodeposition to C and nutrient fluxes across agroforestry systems. Addressing these knowledge gaps is essential for improving predictions of SOC dynamics, optimising nutrient-use efficiency, and enhancing the long-term sustainability and climate-change mitigation potential of agroforestry systems.

1.7. Summary

Although numerous studies have demonstrated the potential of agroforestry systems to enhance AGB, SOC stocks, and C sequestration (Cardinael et al., 2018ab; Kuyah et al., 2019; Cardinael et al., 2018c), important knowledge gaps remain regarding the mechanisms controlling long-term C storage and nutrient dynamics in temperate agroforestry systems. In particular, the relative contributions and interactions of aboveground inputs, such as leaf litter, and belowground inputs, including live and dead roots, to SOC sequestration and nutrient cycling remain poorly understood. While litter- and root-derived inputs are recognised as major drivers of soil C accumulation and belowground ecosystem functioning (Gill and Jackson, 2000; Jobbágy and Jackson, 2001; Lorenz and Lal, 2014; Feng et al., 2022), limited information is available on how litter quality, root traits, litter decomposition, soil nutrient availability, and environmental conditions interact to regulate C storage and nutrient availability within temperate agroforestry systems.

Furthermore, long-term studies examining root dynamics and root-soil interactions across spatial and temporal scales remain scarce (Udawatta and Jose, 2011; Bayala and Prieto, 2020). Similarly, the role of litter inputs and decomposition in supporting nutrient cycling and sustaining belowground C sequestration processes in temperate agroecosystems has received little attention. Consequently, the mechanisms linking aboveground productivity, belowground ecological processes, and long-term C sequestration remain poorly resolved.

Addressing these knowledge gaps is essential for improving understanding of the biological and environmental controls on C sequestration and for optimising agroforestry systems as climate-resilient land management strategies. Therefore, this thesis investigates the relationships among tree productivity, leaf litter decomposition, root traits, soil nutrient dynamics, and C sequestration within a mature temperate agroforestry system, providing an integrated assessment of the aboveground and belowground processes that regulate long-term C storage.

1.8. Research approach

The University of Leeds Research Farm provides a unique opportunity to investigate the long-term impacts of agroforestry in a temperate agricultural landscape. The farm encompasses a range of land-use types, including a mature silvoarable system planted in the 1990s and established hedgerows adjacent to conventionally managed arable land (Fig. 1.5).

The mature silvoarable system at the University of Leeds farm consists of four poplar hybrids (Beaupre, Robusta, Trichobel, and Gibecq) planted at 10 m intervals in replicated experimental blocks covering 2.3 ha (Burgess et al., 2005). These hybrids differ in parentage, growth performance, and disease susceptibility, characteristics that may influence biomass production and C sequestration. Beaupre (*Populus trichocarpa* × *P. deltoides*) is a fast-growing interamerican hybrid with high biomass production potential, but latterly it has become increasingly susceptible to leaf rust. Robusta (*P. nigra* × *P. deltoides*) and Gibecq (*P. deltoides* × *P. nigra*) are widely used euramerican hybrids known for their productivity and adaptability, while Trichobel (*P. trichocarpa*) is an intraspecific selection of black cottonwood (Burgess et al., 2005). Investigating these contrasting genotypes provides an opportunity to assess how genetic variation influences tree growth, AGB accumulation, and C sequestration within temperate agroforestry systems.



Figure 1.5 Silvoarable and hedgerow study site: (a) Alley-cropping treatments between tree rows during the early establishment phase of the plantation (2004) – photo by Chris Wright, who managed the Field Research Unit; (b) silvoarable system in 2023, showing the development of unmanaged vegetation between tree rows following the cessation of cropping treatments in 2004; (c) placement of leaf litterbags within hedgerows during winter (January 2024); and (d) dense summer hedgerow vegetation, illustrating the challenges associated with litterbag retrieval.

Diameter at breast height has been collected for over 30 years, providing a unique long-term dataset for examining trends in AGB C stocks and annual C sequestration among poplar hybrids. The presence of an on-site Met Office weather station further enables the assessment of genotype-specific responses to climatic variability and the influence of environmental conditions on the long-term growth and C sequestration. Furthermore, quantifying AGB C alongside BGB C (using an established root: shoot ratio) and SOC stocks in both tree rows and alleys, allows a whole-system assessment of C storage and sequestration. This information is essential for evaluating the climate change mitigation potential of silvoarable systems, developing future land-use scenarios and net-zero target strategies.

In addition to the silvoarable system, the University of Leeds farm contains established hedgerows comprising native trees and shrub species, providing a complementary

system for investigating litter decomposition and nutrient cycling dynamics. Although hedgerows are well recognised for their contribution to biodiversity conservation, soil conservation and water quality protection (Baudry et al., 2000), their role in driving C and nutrient cycling within agricultural landscapes remained comparatively understudied. Investigating the decomposition of leaf litter from two common native shrub species, hawthorn (*Crataegus monogyna*) and hazel (*Corylus avellana*), using litterbags placed on the soil surface and at depth allows the assessment of species-specific decomposition rates and nutrient release patterns under contrasting environmental conditions.

Evaluating these processes over seasonal timescales will provide insights into the mechanisms that control organic matter turnover, nutrient availability, and C cycling within hedgerow systems. Together with the silvoarable experiment, this approach facilitates an integrated assessment of the aboveground and belowground processes that govern C sequestration and ecosystem functioning in temperate agroforestry systems. Such knowledge is essential for improving our understanding of how different agroforestry components contribute to climate change mitigation, soil health, and the long-term sustainability of agricultural landscapes.

1.8.1. Aim and objectives

The overall aim of this thesis is to evaluate the aboveground and belowground processes governing C storage and accumulation in temperate agroforestry systems and to examine the biotic and abiotic factors influencing C and nutrient cycling. Specifically, the thesis evaluated the effects of agroforestry configuration, tree genotype, time, climate variability, soil nutrient availability, root dynamics, and litter decomposition on aboveground and belowground C pools and nutrient dynamics.

This aim will be addressed through the following specific research objectives:

1. To quantify AGB C stocks and sequestration rates in a 30-year-old poplar based silvoarable system and assess the influence of weather variables on annual C accumulation (Chapter 2).
2. To quantify SOC stocks and sequestration rates within a poplar-based silvoarable system and evaluate its whole-system C sequestration potential

(Chapter 3). This objective integrates AGB C, BGB C, and SOC sequestration rates to estimate the net CO_2 uptake potential of four poplar hybrids. Using three scenarios for expanding silvoarable agroforestry across UK cropland and arable land, the study evaluates the potential contribution of silvoarable agroforestry to achieving the UK Climate Change Committee's targets over 25 years (to 2050).

3. To investigate the relationships among root traits, soil nutrient availability and SOC storage in a 30-year-old poplar silvoarable system, thereby improving understanding of the role of belowground processes in C sequestration and nutrient cycling (Chapter 4).
4. To evaluate the contribution of hedgerows to C and nutrient cycling by assessing leaf litter decomposition rates and C: N dynamics in relation to litterbag placement (surface versus buried) and seasonal variation (Chapter 5).

Collectively, these objectives provide an integrated assessment of the aboveground and belowground mechanisms regulating C sequestration in temperate agroforestry systems. By linking tree productivity, SOC storage, root dynamics, nutrient availability, and litter decomposition processes, this thesis advances understanding of the factors controlling long-term C accumulation and nutrient cycling, while providing evidence to support the design and management of agroforestry systems for climate change mitigation, soil health enhancement, and sustainable land use.

1.8.2. Research methodology

Following the identification of key research gaps in the literature review, Fig. 1.6 presents a conceptual framework developed for this study that illustrates the relationships among the principal drivers and outcomes investigated. The framework incorporates the key factors discussed throughout Section 1.6, including tree species and genotype selection, tree age, site management, agroforestry system type (silvoarable/hedgerows), and weather variables. These drivers are hypothesised to influence three interrelated outcome categories: C and nutrient cycling, soil fertility, and C sequestration. By conceptualising the interactions among these factors within poplar-based silvoarable systems and hedgerows, the framework provides the theoretical foundation for the research questions, objectives, and analyses undertaken in the present study. It also highlights the pathways through which aboveground and belowground

processes interact to regulate long-term C storage, nutrient dynamics, and ecosystem functioning in temperate agroforestry systems.

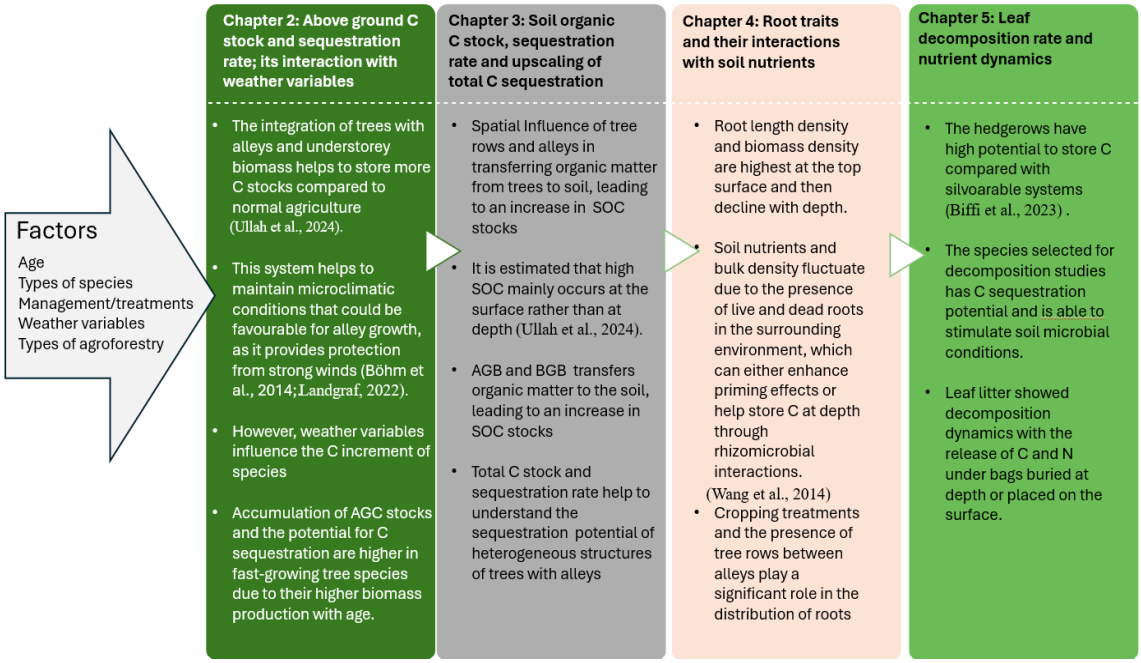


Figure 1.6 Conceptual framework of interactions between principal drivers and outcomes investigated for agroforestry systems

Table 1.2 summarises the alignment between the thesis objectives, the key factors investigated, and the resulting outcomes. Objective 1 was addressed using an observational approach (Chapter 2). Annual DBH measurements collected over a 30-year period were used to calculate AGB, AGB C stocks, and C sequestration rates for four poplar hybrids. Statistical analyses were applied to evaluate the influence of weather variables on the periodic annual C sequestration rates across the different hybrids and to assess differences in growth and C accumulation among hybrids.

Objective 2 was achieved through a combination of field and laboratory methods (Chapter 3). Soil core samples were collected from tree rows (n=24), alleys between tree rows (n=24), and an adjacent arable field (n=48) to quantify bulk density and SOC content at depths of 0–15 cm and 15–30 cm. These data were used to quantify SOC stocks and sequestration rates within the silvoarable system. The resulting estimates were integrated with AGB C and BGB C stocks to quantify whole-system C

sequestration and upscaled to evaluate the potential net CO_2 uptake of four poplar hybrids using three different agroforestry expansion scenarios.

Table 1.2 The overall structure of the thesis

Thesis Chapter	Objective	Factors (fctr) and Outcomes
Chapter 1: Introduction and review of the literature	Establish background, research gaps, aim/objectives of study	Systematic review of literature on above- and belowground C and nutrient cycling in agroforestry
Chapter 2: Aboveground biomass C stocks and variability in annual sequestration rates for a 30-year-old temperate poplar silvoarable system	Quantify AGB, C stock and sequestration rate, and influence with weather variables	Four poplar hybrids (fctr) – AGB C stock, mean annual AGB C increment, periodic annual AGB C increment, influence of weather variables
Chapter 3: SOC stock and upscaling the total C increment of the poplar hybrids to achieve net zero targets	Quantify SOC stock, increment rate, total C increment = AGB C + BGB C+ SOC and upscaling to quantify CO2 uptake under agroforestry expansion scenarios and contribution to net-zero targets till 2050	Four poplar hybrids (fctr), three treatments (fctr) (tree rows, alleys, and arable field), age of poplar hybrids (fctr), – SOC stock and increment per treatment, total C increment per hybrid
Chapter 4: Interactions of soil nutrients with root traits	Soil and root sampling (soil coring, washing, live and dead roots classification, scanning, image interpretation, and lab analysis); Root trait measurement (length density, biomass density, diameter, C, N, C stock, C:N ratio); Soil nutrient analysis (SOC, SOC stock, total N, available N, P, bulk density, C/N)	Four poplar hybrids (fctr), three treatments (fctr), (tree row, alley, and arable field) – soil nutrients and root C, N concentration with root traits
Chapter 5: Leaf decomposition and nutrient release under hedgerows in a temperate agricultural landscape	Preparation of leaf litter bags, collection of leaf litter, placement of leaf litter bags (surface and at depth), leaf litter mass loss and nutrient analysis of leaf litter remaining after retrieving bags.	Shrub species (fctr) - Leaf litter mass loss%, decomposition dynamics, residual concentration of C, N and C/N with release of C and N
Chapter 6: Conclusion & Recommendations	Integrate findings across objectives; Identify implications for ecosystem service delivery in temperate agroforestry	Comparative synthesis of results; Integration with existing literature; Identification of management policy implications

Objective 3 was achieved using a field-based approach (Chapter 4). Soil samples were collected from four faces of three soil pits excavated within a tree row, an alley, and the adjacent arable field. Root samples were obtained using the soil coring method and processed alongside soil samples in the laboratory to determine nutrient concentrations. Prior to chemical analysis, root samples were scanned to quantify key morphological traits, enabling assessment of the relationships between root characteristics, soil nutrient availability, and SOC within the silvoarable system.

Objective 4 was achieved through a field and laboratory investigation of leaf litter decomposition dynamics within a hedgerow adjacent to agricultural land (Chapter 5). Leaf litter from two native hedgerow species was collected and placed in litterbags either on the soil surface or buried at a depth of 5 cm. Litterbags were retrieved periodically over 366 days and analysed to determine mass loss, decomposition rates, and nutrient dynamics, including changes in C: N ratios during decomposition.

Lastly, Chapter 6 synthesises the findings from the individual research chapters, evaluates their implications for C sequestration and nutrient cycling in temperate agroforestry systems, and identifies the study's principal limitations, along with priorities for future research.

Chapter 2. Long-term dynamics of aboveground biomass carbon stocks and sequestration rates in a 30-year-old temperate poplar silvoarable system

Abstract

Poplar (*Populus spp.*) is well known for its ecological and economic importance and its considerable potential for carbon (C) sequestration in agroforestry systems. However, the long-term dynamics of aboveground biomass (AGB) C stock and periodic annual carbon increment (PAIC) in mature temperate silvoarable systems remain poorly understood. This study quantified AGB C stocks, mean annual carbon increment (MAIC) and PAIC for four poplar hybrids (Beaupre, Robusta, Gibecq, and Trichobel) growing in a silvoarable system in northern England over a 30-year period to 2023. The AGB C stocks were calculated using a poplar-specific allometric equation based on diameter at breast height (DBH). Significant differences in AGB C stock, MAIC, and PAIC were observed among hybrids. By 2023, Trichobel achieved the largest mean AGB C stock (65.8 Mg C ha^{-1}), MAIC (2.1 Mg C $ha^{-1}yr^{-1}$), and PAIC (2.3 Mg C $ha^{-1}yr^{-1}$), representing almost double the values recorded for Beaupre, Gibecq, and Robusta. Across all hybrids, PAIC increased during the establishment phase (1994 -1998), decreased between 2000 and 2005, and subsequently stabilised, whereas MAIC increased rapidly during the early years before reaching a relatively stable trajectory. Trichobel consistently maintained the highest MAIC and exhibited sustained productivity throughout the study period, indicating a later maturity and greater long-term growth potential than the other hybrids. Cropping management strongly influenced AGB C accumulation during the active silvoarable phase (1993-2004). The continuous fallow treatment had the highest PAIC and MAIC, exceeding continuous cropping by 59.5% and 45.1%, respectively, indicating that reduced crop competition promoted greater tree growth and C uptake. In contrast, crops growing adjacent to tree rows constrained tree growth and C accumulation. Following the cessation of cropping, differences among treatments gradually diminished, highlighting the importance of crop-tree competition in determining C sequestration in silvoarable systems. Climate-growth relationships differed among hybrids and were driven primarily by summer weather conditions. Higher summer and summer-half-year soil temperature were

associated with lower PAIC in Beaupre (-36%, $p = 0.02$) and Gibecq (-38%, $p = 0.04$), whereas Trichobel showed positive responses with summer rainfall (+29%, $p = 0.01$) and seasonal wind speed (+15-39%, $p < 0.05$). These findings demonstrate marked genotype-specific differences in growth responses to climatic variability and suggest that hybrid selection will be increasingly important under future climate change. Overall, the results demonstrate the value of PAIC as an indicator of long-term C sequestration dynamics and identify Trichobel as the hybrid with the greatest potential for AGB C accumulation in temperate silvoarable systems. These findings provide important evidence to support hybrid selection, agroforestry design, and climate change mitigation strategies aimed at maximising landscape-scale C sequestration.

2.1. Introduction

Climate change is increasingly reshaping land-use systems worldwide, making it vital to adopt land-based strategies that can simultaneously support climate mitigation, adaptation and food security. The Intergovernmental Panel on Climate Change (IPCC) has identified land-use change, particularly deforestation and conversion of land for agriculture, as a major contributor to global greenhouse gas (GHG) emissions, accounting for approximately 5.2 Gt CO_2 per year (IPCC, 2019). At the same time, terrestrial ecosystems remove an estimated 11.2 Gt CO_2 yr^{-1} from the atmosphere, resulting in a net C sink of approximately 6.0 ± 3.7 Gt CO_2 yr^{-1} from the agriculture, forestry and land use (AFOLU) sector. Thus, AFOLU is both a driver of climate change and a key opportunity for mitigation through improved land management practices (Jia et al., 2022).

Agroforestry, the intentional integration of trees and shrubs into farming systems, has emerged as a promising nature-based solution capable of sequestering C in woody biomass and soils while delivering co-benefits such as improved soil fertility, enhanced biodiversity, and flood mitigation (Cardinael et al., 2015, 2018b; Mayer et al., 2022; Terasaki Hart et al., 2023; Stanton et al., 2022; Judson et al., 2026). Carbon sequestration in agroforestry systems occurs through the capture of atmospheric carbon dioxide (CO_2) by trees via photosynthesis and its subsequent storage in biomass and soils. Global assessments estimate that agroforestry practices could mitigate 10–15 Gt CO_2eq per year by 2050, contributing 20–30% of the mitigation required to meet the

Paris Climate Agreement targets (Griscom et al., 2017; UNEP, 2018). However, this potential is unevenly distributed across regions. Western Europe is estimated to contribute only 0.18 Gt CO_2eq yr^{-1}, substantially less than tropical regions in Africa and Asia, where faster tree growth and longer growing seasons prevail (Roe et al., 2021). However, this regional gap also reflects substantial uncertainty regarding the mitigation potential of temperate agroforestry systems, which remain poorly understood. Carbon sequestration rates are highly sensitive to variability in soil type and climate, variation in system design (including tree species, planting density, alley width and crop rotations) and interactions between trees and crops, such as competition for light, water, and nutrients (Quandt et al., 2023; Laub et al., 2025). Although numerous studies have quantified total C stocks in agroforestry systems, relatively few have examined how C sequestration rates vary throughout system development. In Europe, for instance, silvoarable systems (cropping in alleys between rows of trees) have been reported to sequester between 0.77 and 3 Mg C ha^{-1} yr^{-1} (Palma et al., 2007; Reisner et al., 2007). However, these estimates mainly come from modelled C dynamics dominated by biomass accumulation (trees and crops) and only indirectly reflect soil C changes. Differences in system ages, designs and management further limit direct comparisons among studies and contribute to uncertainty in C sequestration estimates.

In the UK context, uncertainty in C sequestration presents a significant challenge, particularly for agroforestry systems. Approximately 70% of UK land is agricultural, and the UK Climate Change Committee has recommended expanding agroforestry from the current 3.3% of farmland to at least 10% by 2050 to support net-zero emissions targets (CCC, 2021; Tosh, 2021). While there is increasing momentum behind tree planting, including national targets of planting at least 30,000 ha of new woodland annually (Westaway et al., 2023), the evidence base on quantifying the contribution of agroforestry to climate change mitigation remains limited.

Most agroforestry studies in temperate regions focus on relatively young systems, which are poorly suited to estimating long-term C sequestration due to tree growth patterns and the time required to reach maturity (> 20 years; Dupraz et al., 2019). Furthermore, while total C stocks and average sequestration rates are commonly reported, the temporal dynamics of C accumulation are not well characterised. Existing long-term studies typically express sequestration as a mean annual C increment (MAIC)

over the study period (e.g., Wotherspoon et al., 2014). Although MAIC is useful as a summary metric, it obscures how the annual C increment varies through time - periodic annual C increment (PAIC) - in response to changing tree growth patterns and environmental conditions. This lack of temporal resolution limits our understanding of when agroforestry systems deliver their greatest climate benefits and how stable those benefits remain over multiple decades.

Poplar (*Populus* spp.) is frequently identified as a high-potential agroforestry species due to its fast growth, high biomass production and substantial C sequestration capacity. Across temperate agroforestry systems, average total C stock has been estimated at approximately 63 Mg C ha^{-1} (Montagnini and Nair, 2004), although poplar-based systems frequently exceed this average due to their exceptional biomass productivity. Wotherspoon et al. (2014) found that 25-year-old poplar intercropping systems (111 trees ha^{-1}) stored more C in AGB than comparable systems with other tree species. However, C sequestration efficiency is thought to vary substantially among poplar subspecies and hybrids, driven by differences in wood density, growth form and environmental tolerance (Jha, 2018). Despite their recognised potential, few studies have quantified annual changes in AGB C within mature poplar systems under temperate climates (Montagnini & Nair, 2004; Nair, 2012; Udawatta & Jose, 2012), and none have examined long-term PAIC in dynamics in detail.

Climate variability further complicates the role of agroforestry in climate change mitigation. Recent studies show that poplar growth rates are highly sensitive to long-term climate change (e.g., Kong et al., 2024), and projected warmer, drier summers in northwest Europe (IPCC, 2021), are likely to alter growth patterns and rates of C uptake (Kumar et al., 2018). Despite this, there is a lack of specific evidence on how climate variability affects the growth and sequestration performance of fast-growing poplar hybrids in silvoarable systems (Seserman et al., 2018; Trouillier et al., 2020), limiting confidence in future mitigation projections.

Long-term assessments of the C sequestration potential of temperate agroforestry systems are thus needed to improve climate change mitigation modelling and support evidence-based policy development. This study addresses this knowledge gap through the analysis of a 30-year-old silvoarable experiment in Yorkshire, UK, comprising four

poplar hybrids and a unique dataset of annually recorded tree growth. Specifically, the study aims to: (i) quantify the AGB C stock of four poplar hybrids, (ii) characterise long-term and inter-annual variation in C accumulation through analysis of PAIC over a 30-year period, and (iii) evaluate the impact of local climate variability on PAIC. By providing a rare, long-term assessment of C dynamics, hybrid performance, and climate sensitivity in a mature silvoarable system, this study advances the understanding of the potential contribution of temperate poplar-based agroforestry to climate mitigation and achievement of national net-zero targets.

2.2. Methods

2.2.1. Site description and experimental design

The experimental site is located at the University of Leeds research farm, in Yorkshire (53°53' N, 1°15' W), northern England, at an altitude of 42 to 70 metres above sea-level (asl). The soil is a well-drained, loamy, calcareous brown earth derived from the Aberford series of Calcaric Endoleptic Cambisols, which are characteristic of arable farming in this region (Holden et al., 2019; Pun et al., 2024) and underlain by dolomitic limestone of the Cadeby formation (British Geological Survey, 2020).

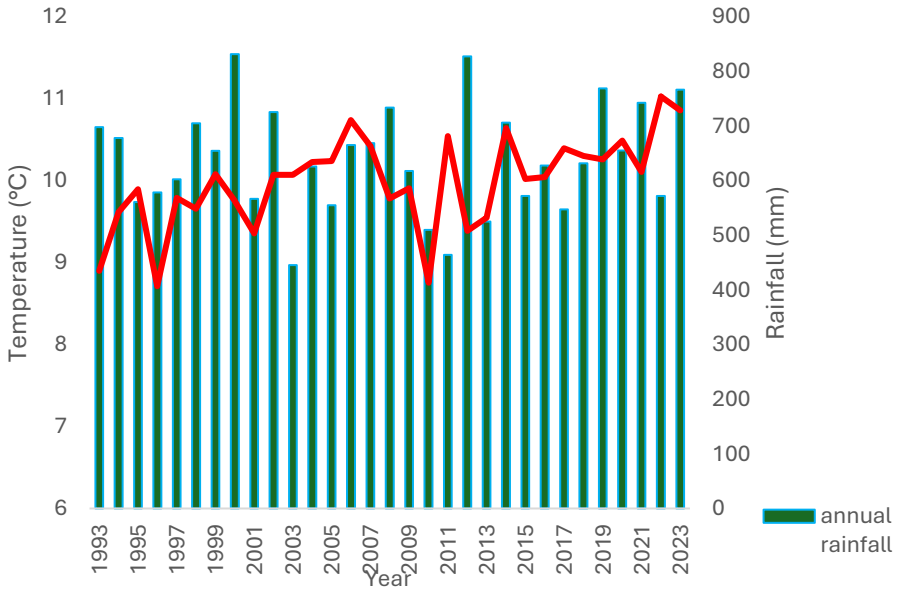


Figure 2.1 Mean annual air temperature and total annual precipitation 1993-2023 at the study site with data from Bramham and Church Fenton weather stations

The long-term (1993-2023) mean annual temperature was 10.9°C, and mean annual rainfall was 767 mm (Met Office, 2023). However, Fig. 2.1 shows that there has been considerable inter-annual variability over this period. Mean annual temperature was 8.9 °C in 1993 and 10.9 °C in 2023, with a maximum of 11.0 °C in 2022 and it displays a general upward trend over the period. Highest rainfall occurred in 2000 (831.5 mm) and 2012 (827.8 mm), and lowest in 2003 (445.4 mm) and 2011 (464.3 mm).

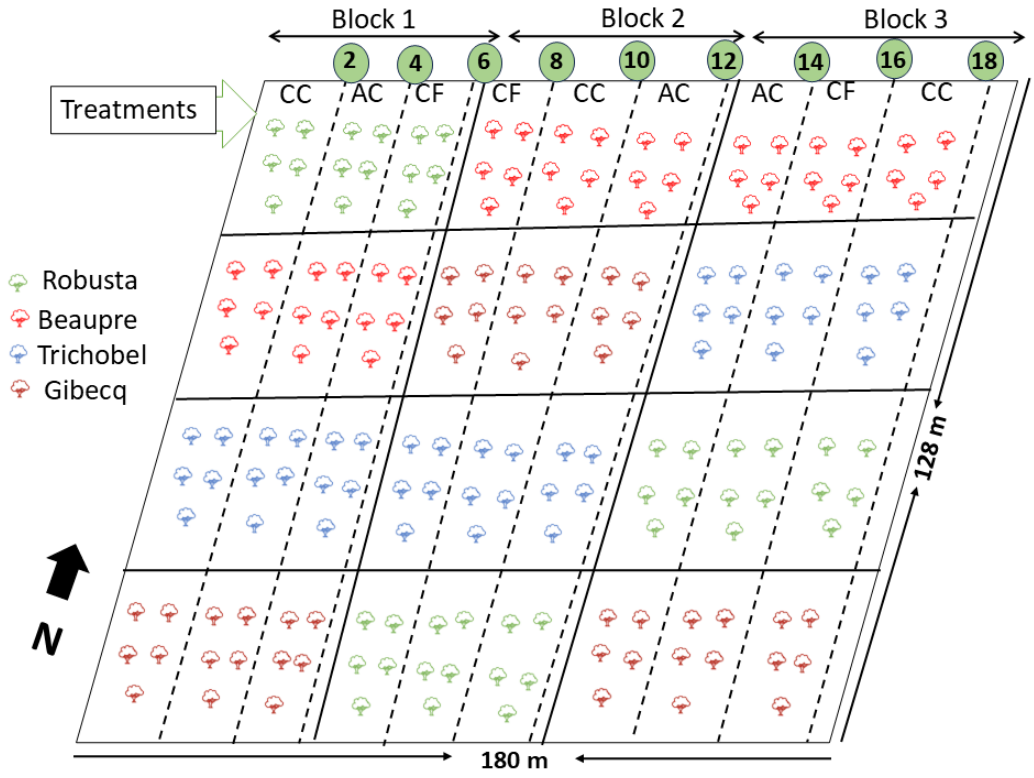


Figure 2.2 Layout of the poplar agroforestry experimental site. Each dashed vertical line represents the even numbered row of trees, which were used for diameter at breast height (DBH) measurements. Each row contains five trees of four poplar hybrids (Robusta, Beaupre, Trichobel and Gibecq). In the alleys between the tree rows, three cropping systems were used between 1992 and 2004: continuous cropping (CC), alternate cropping (AC), and continuous fallow (CF). These treatments are arranged across three experimental blocks (Block 1, Block 2, and Block 3)

The poplar silvoarable experimental system was established in 1992 and comprises four fast-growing hybrids, Trichobel, Gibecq, Beaupre, and Robusta, which represented the major groups of poplars being grown in Britain at the time (Burgess et al., 2005). Between 1992 and August 2004, the alleys between the tree rows were assigned one of three cropping systems: continuous fallow (CF), continuous cropping (CC), or alternate cropping (AC). After the 2004 harvest, cropping in the alleys stopped due to reduced

yields resulting from competition for resources such as sunlight, nutrients, and water. Each combination of four poplar hybrids and three cropping systems was included within three replicated blocks in a randomised block design (n=3), to account for variation within the field and edging effects of the experiment (Fig. 2.2). The site consists of 19 tree rows-oriented north-south, with trees planted at intervals of 6.4 metres (m) apart along each row, and tree rows spaced 10 m apart. Within each tree row, the hybrids were planted in groups of five consecutive trees, with a single guard tree at each end of the row. The planting design results in a tree density of 166 trees per ha. For further information on the establishment and management of trees, see Burgess et al. (2005).

Each autumn, from 1992 to 2003, the 10 m-wide alleys between tree rows were ploughed in 8 m strips, resulting in damage to tree roots at a depth of 20 cm in the arable area and leaving the tree row as an uncultivated 2-m-wide strip (Burgess et al., 2005). In the CF, two adjacent alleys never had crops planted in them, but they were regularly cultivated so that the measurement row of poplars between them would have minimal competition from crops or weeds. The CC included a rotation of cereal and non-cereal crops on either side of a measurement tree row, while the AC involved CC on one side of the measurement row and the fallow treatment on the other side of the tree row (Burgess et al., 2003). All crops were managed as in commercial systems, receiving standard fertiliser and pesticide applications as appropriate. Since 2004, the alleys have been unmanaged, resulting in the development of a perennial understorey of grasses within the alleys, which are not managed.

2.2.2. Estimating AGB C stock and increment rates

Tree DBH was recorded every winter (between November-March) from 1993 until 2023 for all 20 trees in the even numbered rows (2, 4, 6, 8, 10, 12, 14, 16 and 18) only, resulting in DBH measurement for 180 trees (45 for each hybrid) each year. However, DBH data were missing for the following years: 1999, 2000, 2003, and 2004. Tree diameters were measured using a girth tape at a height of 1.3 m above the ground. Over the measurement period (1993-2023), the mean DBH across all trees was 28.6 ± 12.4 cm.

Based on this I estimated tree biomass using the allometric equation developed by Fortier et al. (2017). This equation was selected because it was derived using destructive harvest data from a poplar hybrid (*Populus canadensis* × *Populus maximowiczii* - clone DN × M-915508), is applicable for temperate conditions similar to those of the study site, relies primarily on DBH measurements, and performs well across a broad range of tree sizes (DBH range 7.5-30.8 cm), which corresponded with the DBH range in my study. These characteristics made it particularly suitable for the long-term dataset, in which DBH was the only consistently measured tree attribute throughout the monitoring period. The equation is based on allometric scaling theory, which uses a power function of tree growth to estimate biomass, with DBH as the predictor variable:

$$\ln(AGB\ per\ year) = \ln(a) + b\ (\ln DBH) \quad \text{(Equation 2.1)}$$

where a (-2.14) and b (2.26) are allometric coefficients that denote the slope and intercept of the line given in Fortier et al. (2017). Equation 2.1 was used to estimate AGB for all 180 trees in each year with available DBH data.

To calculate the AGB C content (kg C) of each tree, the AGB (kg) was multiplied by 0.5, as a default biomass-to-C standard conversion factor to assess C content of temperate forests (Meragiaw et al., 2021; Piper et al., 2003; Salas Macías et al., 2017; Fang et al., 2007; Yasin et al., 2021).

The AGB C stock (Mg C ha^{-1}) was calculated for each hybrid and measurement year (t) by scaling tree-level AGB C content (kg C) to a per unit area basis and assuming a tree planting density of 166 trees per hectare as:

$$AGB\ C\ stock_t = AGB\ C\ content_t\ x\ \frac{166\ (number\ of\ trees\ per\ ha)}{1000} \quad \text{(Equation 2.2)}$$

Annual AGB C sequestration was expressed in two ways:

(1) MAIC (Mg C $ha^{-1}yr^{-1}$) represents the cumulative average annual rate of AGB C sequestration from planting to a given tree age. For each hybrid and measurement year, it was calculated as:

$$MAIC_t = \frac{AGB\ C\ stock_t}{number\ of\ years\ since\ planting} \quad \text{(Equation 2.3)}$$

(2) PAIC (Mg C $ha^{-1}yr^{-1}$) represents the annual increase in AGB C stock between two consecutive measurement years (t_1 and t_2). PAIC for a given interval was calculated as:

$$PAIC_{t1-t2} = (AGB_{t2} - AGB_{t1})/(t_2 - t_1) \qquad \text{(Equation 2.4)}$$

2.2.3. Climate station data

Daily meteorological data for the study period were obtained from the UK Met Office CEDA Archive Bramham automatic weather station (WMO station 00534) located at the University of Leeds farm (54 m a.s.l; 53° 52′ 1.2″,1° 19′ 58.8″ W) (Met Office, 2023). To fill occasional gaps in the Bramham data, linear regressions were developed for each meteorological variable using overlapping data from nearby UK Met Office automatic weather stations at Cawood (WMO station 00535) (6 m a.s.l; 53° 49' 38.64" N, 1° 08' 48.16" W), located 19.2 km away from Bramham, and Church Fenton (WMO station 03355) (8 m asl; 53°49′58.8″N, 1°12′0″W), located 14.2 km away from Bramham. Minimum grass temperature data were gap-filled using only the Church Fenton station because it was the only station with corresponding data available. After gap-filling, soil temperature at a depth of 10 cm remained unavailable for 21 days in January 1996 and for 12 days in April 2000. Wind speed data were missing for the years 1993, 2002, 2003, 2006, and 2007. The annual number of windy days was calculated as the number of days with wind speeds greater than 17 knots (Beaufort scale 5).

Growing Degree Days (GDD) were calculated using a reference temperature of 5.5 °C consistent with temperate plant phenology (Piña-Rey et al., 2021). For each day, GDD was calculated as the difference between the mean daily temperature and the base temperature when the mean daily temperature was above 5.5 °C; otherwise, the daily GDD contribution was set to zero. For example, a mean daily temperature of 7.0 °C, contributed 1.5 °C GDD units (7.0-5.5 °C). Days with mean temperatures above 5.5 °C were classified as growing days. Air frost days and ground frost days were defined as days on which the minimum air temperature and minimum grass temperature, respectively, fell below 0 °C.

To ensure that weather variables corresponded to the period during which growth occurred, climatic data were aligned with the biological growth year rather than the

calendar year. Thus, weather conditions associated with a DBH measurement in year t were assigned to the previous growth year ($t - 1$). Hence, to evaluate the influence of broader seasonal climatic conditions on PAIC of the four poplar hybrids, monthly weather data were aggregated into biologically relevant seasonal windows. The seasonal classifications were defined as follows: Winter: December of the year preceding the growth year plus January-February of the growth year, Spring: March-May of the growth year, Summer: June-August of the growth year, Autumn: September-November of the growth year, Winter half-year: October-December of the year preceding the growth year plus January-March of the growth year, Summer half-year: April-September of the growth year.

2.2.4. Statistical analysis

Statistical analyses were conducted in R version 4.3.3 (R Core Team, 2024). The influence of hybrid, cropping treatment, and block on AGB C stock, MAIC, and PAIC was assessed across three distinct time periods: The first period (1993-2004) represented the active cropping phase, during which cropping treatments were maintained between tree rows, showing direct evaluation of their effects on tree growth and C sequestration. The second period (2005-2023) represented the post-cropping phase following the cessation of cropping in 2005 and was analysed to investigate any residual or carryover effects of the earlier treatments. A final analysis covering the entire study period (1993-2023) was conducted to assess long-term C accumulation patterns and the cumulative impacts of cropping treatments.

As the AGB data were not normally distributed, according to the Shapiro-Wilk test, non-parametric Kruskal-Wallis tests were used to determine differences in AGB C stocks, MAIC, and PAIC among poplar hybrids, experimental blocks, and agricultural management cropping treatments. For each hybrid and measurement year, median AGB C stocks, MAIC and PAIC were calculated. Median values with interquartile range were used to reduce the effects of outliers and account for variability within the dataset. The interquartile range (IQR), defined as the difference between the third quartile (Q3) and the first quartile (Q1), was used to describe the variability of the middle 50% of the data. Carbon sequestration dynamics were evaluated by comparing temporal patterns in MAIC and PAIC among the four poplar hybrids.

The relationships between weather variables and PAIC were investigated using generalised linear models (GLM). As PAIC data were non-normal, continuous, positive, and right-skewed, gamma regression models with a log link were applied using the glm() function. Separate models were developed for each poplar hybrid to examine:

1. Relationships between PAIC and annual weather variables

2. Relationships between PAIC and seasonal weather variables (spring, summer, autumn, and winter), and

3. Relationships between PAIC and weather variables aggregated over the summer half-year and winter half-year.

The weather predictors considered in the analysis included mean annual air temperature, soil temperature at 10 cm depth, sunshine hours, growing degree days (GDD), number of growing days, ground frost days, air frost days, mean wind speed, number of windy days, and total rainfall.

To reduce multicollinearity among predictors, absolute pairwise correlation analyses were conducted prior to model fitting to assess relationships among weather variables. Variables exhibiting strong intercorrelation ($|\rho| \geq 0.7$) were considered collinear and were not included simultaneously in the same model. Variance inflation factors (VIF) were also calculated, and variables with VIF > 5 were removed to reduce redundancy and multicollinearity. All predictor variables were standardised prior to analysis to facilitate comparison of effect sizes. Statistical significance was assessed at $\alpha = 0.05$ to identify robust and meaningful relationships.

For each hybrid, the full model was compared with an intercept-only null model using likelihood ratio tests. Model fit was assessed using analysis of deviance (*anova*()) with a likelihood ratio test. Model performance was evaluated using McFadden's pseudo-R^2, calculated as $R^2 = 1 - D_{model}/D_{null}$, where D_{model} and D_{null} represent the residual deviance of the fitted and null models, respectively. This metric quantifies the improvement of the fitted model compared to the null model. Model comparisons were performed using the Akaike Information Criterion (AIC), with lower AIC values indicating greater support for a model after accounting for both goodness of fit and model complexity

2.3. Results

2.3.1. Aboveground biomass carbon stock of the poplar hybrids

There were no significant differences in AGB C stock between replicate treatment blocks. The AGB C stocks increased steadily throughout the 30-year monitoring period in all poplar hybrids (Fig. 2.3). However, the magnitude and rate of C accumulation varied significantly among hybrids. The Kruskal-Wallis test showed significant differences in C stock among hybrids during all three periods: 1993-2004 ($\chi^2 = 53.873$, df = 3, $p < 0.001$), 2005-2023 ($\chi^2 = 878.65$, df = 3, $p < 0.001$), and 1993-2023 ($\chi^2 = 332.84$, df = 3, $p < 0.001$). Hence, hybrid identity was a major determinant of C stock across both the active cropping and post-cropping phases, as well as over the entire 30-year study period. At the beginning of the study, in 1993, Beaupre had the highest median C stock (0.16 Mg Cha^{-1}**,** IQR = 0.06 Mg C ha^{-1}), followed by Trichobel (0.08 Mg C ha^{-1}, IQR = 0.04 Mg C ha^{-1}), Robusta (0.02 Mg C ha^{-1}, IQR = 0.03 Mg C ha^{-1}), and Gibecq (0.01 Mg C ha^{-1}, IQR = 0.03 Mg C ha^{-1}). By 2002, all hybrids showed substantial increases in AGB C stock, with Beaupre still having the highest stock (12.73 Mg C ha^{-1}, IQR = 3.39 Mg C ha^{-1}), followed by Trichobel (11.79 Mg C ha^{-1}, IQR = 2.98 Mg C ha^{-1}), Robusta (9.30 Mg C ha^{-1}, IQR = 2.71 Mg C ha^{-1}), and Gibecq (8.80 Mg C ha^{-1}, IQR = 3.03 Mg C ha^{-1}) (Fig. 2.3).

Agricultural cropping treatments (CF, CC, and AC) showed a significant effect on AGB C stocks during both the early assessment period (1993–2004), as indicated by the Kruskal–Wallis test ($\chi^2 = 26.29$, df = 2, $p < 0.001$), and the later period after cropping activity (2005-2023) ($\chi^2 = 6.479$, df = 2, $p = 0.039$). During the active (1993-2004) and post-cropping (2005-2023) periods, AGB C stocks were greatest under the CF, exceeding CC by 53.7% and AC by 2.8%, respectively. However, when the analysis was extended to encompass the entire study period (1993–2023), the effect of management treatments was substantially reduced and became marginally non-significant ($\chi^2 = 5.91$, df = 2, $p = 0.052$).

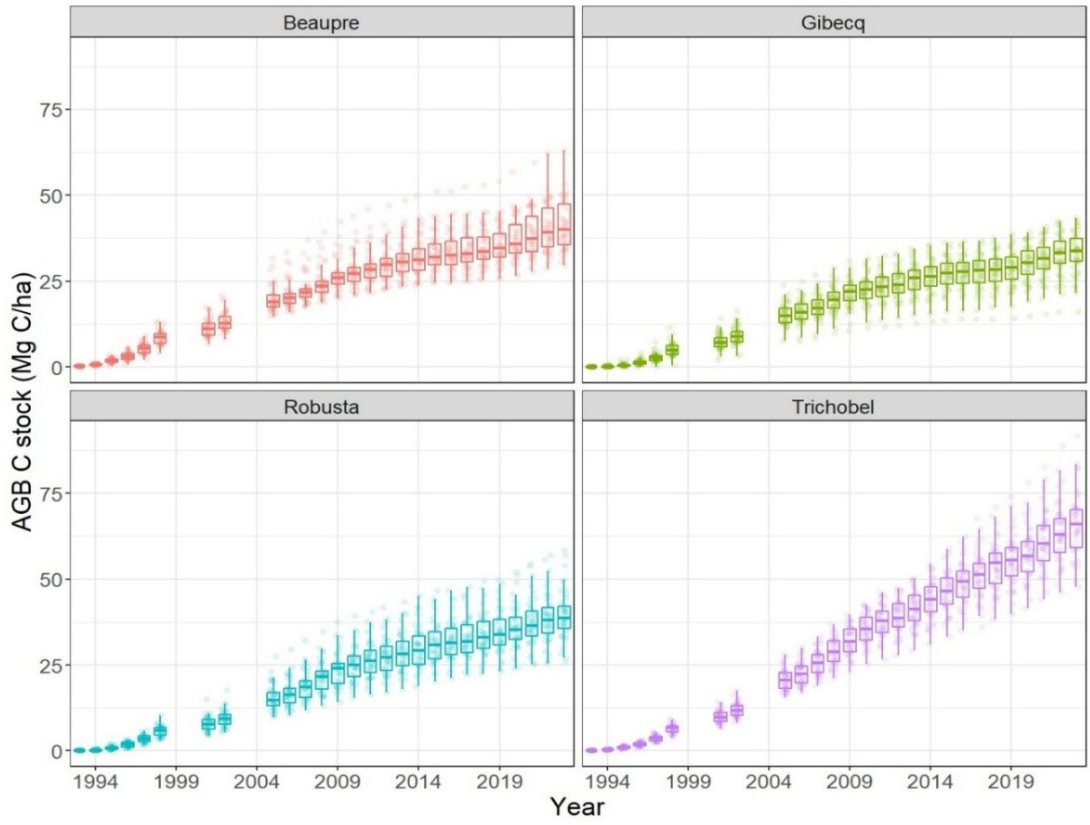


*Figure 2.3 Box and whisker plots showing the median and the 25th and 75th percentiles of the AGB C stock (*Mg C ha^{-1}*) over the period 1993 to 2023 for each poplar hybrid (n=45 for each hybrid in each year)*

From 2005 onward, Trichobel surpassed Beaupre in AGB C stock and maintained its lead for the remainder of the study period. By 2023, more than 30 years after establishment, Trichobel had accumulated the largest median C stock (65.99 Mg C ha^{-1}, IQR = 11.02 Mg C ha^{-1}), representing the highest long-term C storage among the hybrids. In contrast, Gibecq had the lowest AGB C stock (33.77 Mg C ha^{-1}, IQR = 6.66 Mg C ha^{-1}), approximately half that of Trichobel. Beaupre and Robusta exhibited intermediate AGB C stocks of 40.08 Mg C ha^{-1} and 38.28 Mg C ha^{-1}, respectively (Fig. 2.3).

Overall, while Beaupre accumulated C most rapidly during the early stages of stand development, Trichobel showed superior long-term performance and ultimately achieved the greatest AGB C storage. Gibecq consistently exhibited the lowest C accumulation throughout the study period, while Robusta showed moderate growth and C sequestration compared with the other hybrids.

2.3.2. Mean annual carbon increment and periodic annual carbon increment over 30 years

The Kruskal-Wallis test showed significant differences in MAIC among hybrids during 1993-2004 ($\chi^2 = 146.03$, df = 3, $p < 0.001$), 2005-2023 ($\chi^2 = 1839.2$, df = 3, $p < 0.001$), and across the entire study period, 1993-2023 ($\chi^2 = 948.96$, df = 3, $p < 0.001$). The substantially higher χ^2 value across whole study period indicates greater divergence in MAIC among hybrids over time. Between 1993 and 1998, the median MAIC increased rapidly across all hybrids as stands developed (Fig. 2.4). For example, Beaupre's median MAIC increased from 0.16 Mg C $ha^{-1}yr^{-1}$ (IQR = 0.10) in 1993 to 1.43 Mg C $ha^{-1}yr^{-1}$ (IQR = 0.47) in 1998 (Fig. 2.4). Over this early growth phase, Beaupre consistently exhibited the highest MAIC, followed by Trichobel, Robusta, and Gibecq.

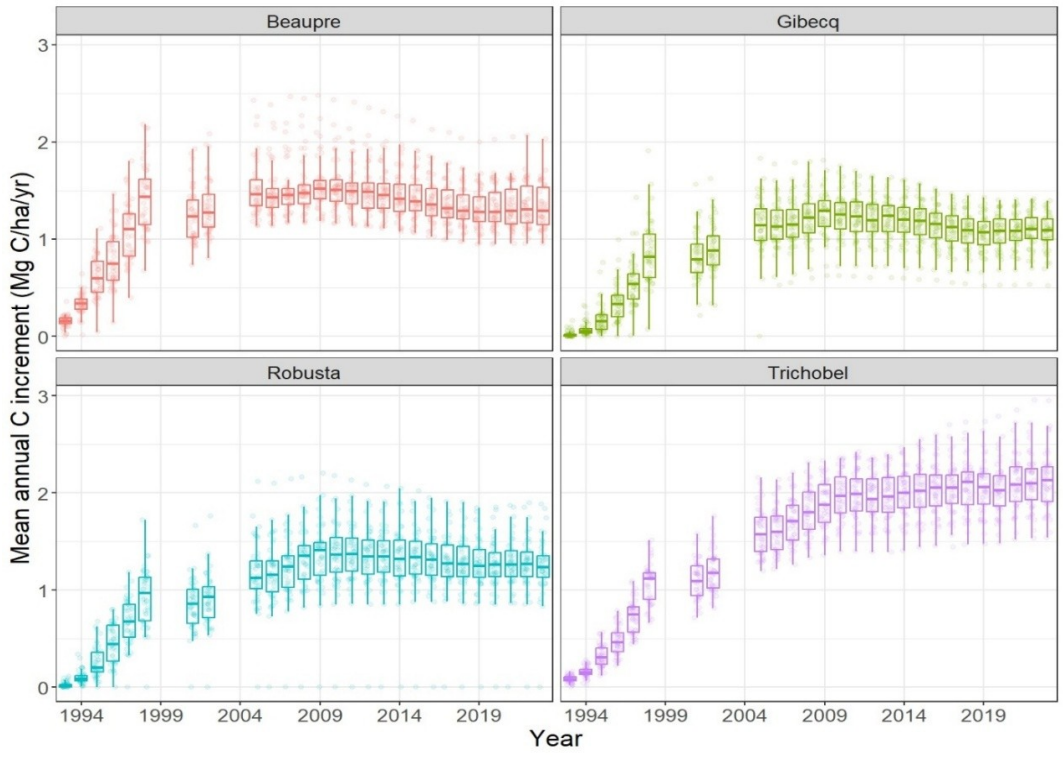


Figure 2.4 Box and whisker plots showing the median and the 25th and 75th percentiles of the mean annual C increment (MAIC Mg C $ha^{-1}yr^{-1}$*) over the period 1993 to 2023 for each poplar hybrid (n=45 for each hybrid in each year)*

From 2001 to 2014, MAIC continued to increase gradually across all hybrids (Fig. 2.4). MAIC for Beaupre increased from 1.23 Mg C $ha^{-1}yr^{-1}$(IQR = 0.37) to 1.42 Mg C $ha^{-1}yr^{-1}$ (IQR = 0.27), Gibecq from 0.79 Mg C $ha^{-1}yr^{-1}$(IQR = 0.29) to 1.20 Mg C $ha^{-1}yr^{-1}$(IQR = 0.25), Robusta from 0.86 Mg C $ha^{-1}yr^{-1}$(IQR = 0.35) to 1.32 Mg C $ha^{-1}yr^{-1}$(IQR = 0.37), and Trichobel from 1.09 Mg C $ha^{-1}yr^{-1}$(IQR = 0.30) to 2.00 Mg C $ha^{-1}yr^{-1}$(IQR = 0.32). Block effects were not significant during the establishment period (1993-2004; $\chi^2 = 1.45$, df = 2, $p = 0.485$) or across the entire study period ($\chi^2 = 1.45$, df = 2, $p = 0.48$). However, significant differences among blocks were detected during the later stage of stand development (2005-2023; $\chi^2 = 6.43$, df = 2, $p = 0.040$).

From 2005 onward, Trichobel exhibited the greatest increase in median value of the MAIC, rising from 1.58 Mg C $ha^{-1}yr^{-1}$(IQR = 0.34) in 2005 to 2.13 Mg C $ha^{-1}yr^{-1}$(IQR = 0.36) by 2023, outperforming all other hybrids (Fig. 2.4). MAIC generally increased until approximately 2011, after which values stabilised or declined for Beaupre, Robusta, and Gibecq. In contrast, Trichobel maintained elevated MAIC throughout the remainder of the monitoring period.

The Kruskal-Wallis test showed significant differences in PAIC among hybrids during 1993–2004 ($\chi^2 = 63.25$, df = 3, $p < 0.001$), 2005–2023 ($\chi^2 = 1036.9$, df = 3, $p < 2.2 \times 10^{-16}$), and across the entire study period (1993–2023) ($\chi^2 = 803.78$, df = 3, $p < 0.001$) As shown in Fig. 2.5, all hybrids exhibited a rapid increase in PAIC during the early establishment phase (1994-1998), reflecting vigorous juvenile growth. During this period, Beaupre recorded the highest PAIC values, with the median increasing from 0.51 Mg C $ha^{-1}yr^{-1}$ (IQR = 0.19) in 1994 to 3.02 Mg C ha-1yr-1 (IQR = 0.69) in 1998.

Following this initial phase, PAIC trajectories diverged among hybrids. By 2002, Trichobel had achieved the highest median PAIC (1.93 Mg C $ha^{-1}yr^{-1}$; IQR = 0.60), exceeding Beaupre (1.84 Mg C $ha^{-1}yr^{-1}$), Gibecq (1.67 Mg C $ha^{-1}yr^{-1}$), and Robusta (1.39 Mg C $ha^{-1}yr^{-1}$) (Fig. 2.5). This trend continued, and by 2005, 12 years post-establishment, Trichobel exhibited the highest PAIC at 2.90 Mg C $ha^{-1}yr^{-1}$(IQR = 0.59), compared with Beaupre (2.12 Mg C $ha^{-1}yr^{-1}$), Gibecq (2.03 Mg C $ha^{-1}yr^{-1}$), and Robusta (2.01 Mg C $ha^{-1}yr^{-1}$). From 2001 onwards, PAIC generally ranged from a maximum of 1.14 Mg C $ha^{-1}yr^{-1}$(IQR = 0.37) for Trichobel to a minimum of 0.73 Mg C $ha^{-1}yr^{-1}$(IQR

= 0.24) for Gibecq, illustrating the increasing divergence in growth performance among hybrids.

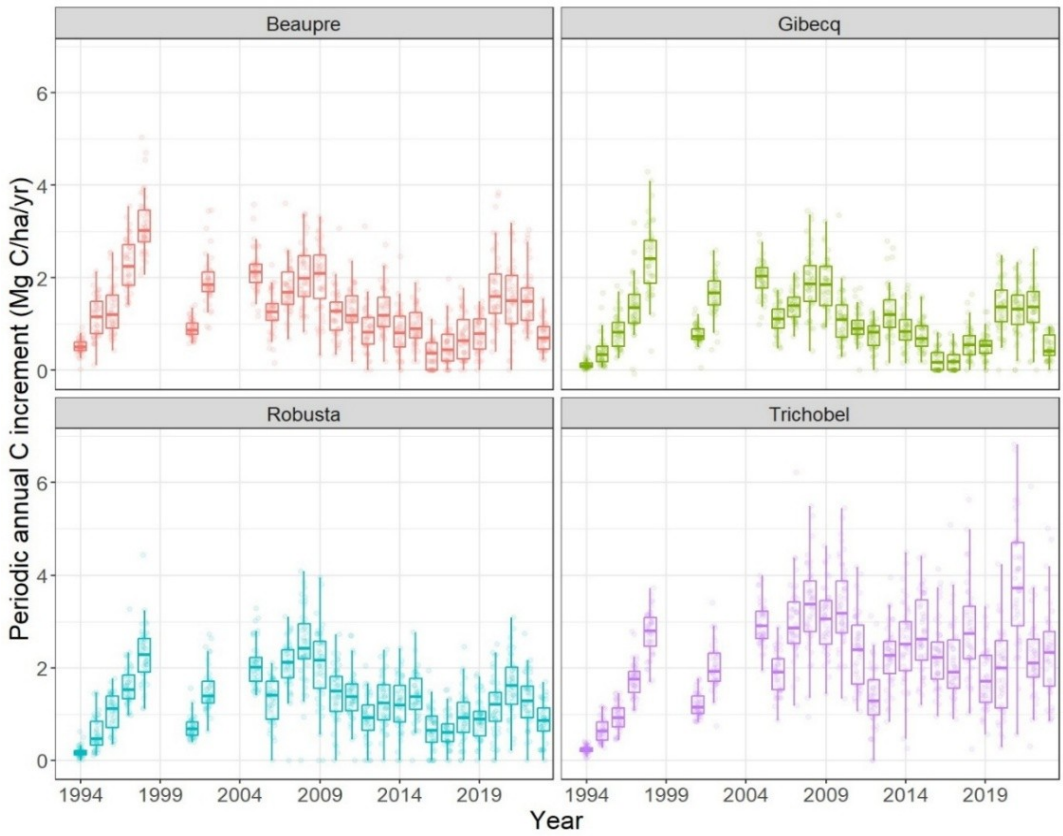


Figure 2.5 Box and whisker plots showing the median and the 25th and 75th percentiles of periodic annual carbon increment (PAIC Mg C $ha^{-1}yr^{-1}$*) over the period 1993 to 2023 for each poplar hybrid (n=45 for each hybrid in each year)*

Trichobel displayed a distinct growth trajectory characterised by sustained productivity during stand maturation. In contrast to the other hybrids, which showed progressive declines in PAIC with increasing age, Trichobel maintained relatively high PAIC values throughout the study period and particularly after 2009. Although a general decline in PAIC was observed for all hybrids after 2015, Trichobel consistently retained the highest PAIC (Fig. 2.5). Peak PAIC for Trichobel occurred in 2021, reaching 3.72 Mg C $ha^{-1}yr^{-1}$ (IQR = 1.80), representing the highest PAIC across the 30-year study period. By 2023, PAIC had declined across all hybrids, yet remained highest for Trichobel at 2.33 Mg C $ha^{-1}yr^{-1}$(IQR = 1.17), significantly exceeding the values of Beaupre (0.7 Mg C $ha^{-1}yr^{-1}$), Gibecq (0.42 Mg C $ha^{-1}yr^{-1}$), and Robusta (0.86 Mg C $ha^{-1}yr^{-1}$). In contrast, no significant effect of block found on PAIC during 1993–2004 ($\chi^2 = 0.909$, df = 2, $p = 0.635$), 2005–2023 ($\chi^2 = 2.451$, df = 2, $p = 0.294$), or across the entire study

period (1993–2023) ($\chi^2 = 0.901$, df = 2, $p = 0.637$). This analysis indicates greater differentiation in PAIC among hybrids over the longer study period, while the effects of cropping treatments and block remained non-significant.

Cropping management significantly affected both MAIC and PAIC during the active cropping phase (1993-2004; MAIC: $\chi^2 = 47.4$, df =2, $p < 0.001$; PAIC: $\chi^2 = 52.6$, df=2, $p < 0.001$). Following the cessation of cropping, no significant treatment effect was detected for MAIC during the post-cropping phase ($\chi^2 = 1.15$, df = 2, $p = 0.561$), whereas PAIC remained significantly affected ($\chi^2 = 9.12$, df = 2, $p = 0.010$). Across the entire study period (1993-2023), treatment effects were significant for MAIC ($\chi^2 = 6.38$, df = 2, $p = 0.041$) but not for PAIC ($\chi^2 = 1.13$, df = 2, $p= 0.568$).

Continuous fallow treatment exhibited the highest MAIC and PAIC during the active cropping period (1993-2004), exceeding CC by 45.1% and 59.5%, respectively. During the post-cropping period (2005-2023), CC showed the highest MAIC (10.3-17.6% greater than AC and CF) but no significant effect, whereas CF had the highest PAIC (3.7-4.4% greater). Across the entire study period (1993-2023), CC maintained the highest MAIC and PAIC, surpassing AC by 5.9% and 2.9%, respectively.

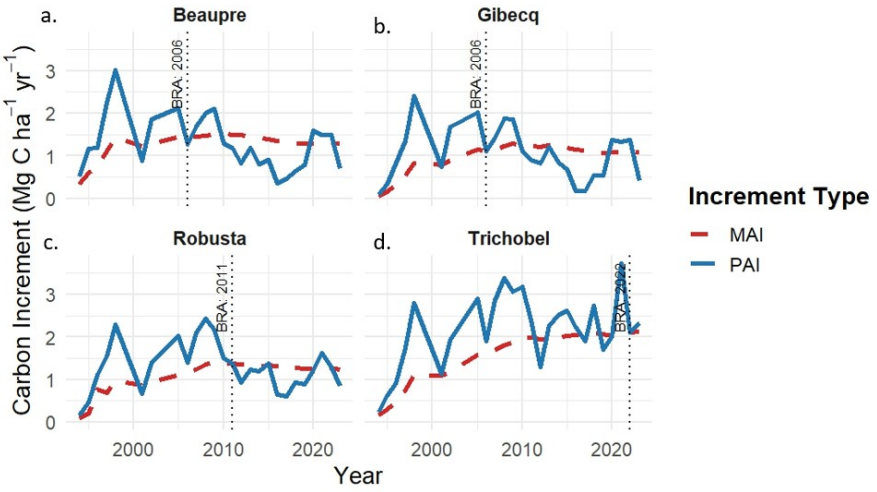


Figure 2.6 Comparison of PAIC and MAIC over the experimental period (1994-2023). Panels show: (a) Beaupre, (b) Gibecq, (c) Robusta, and (d) Trichobel. The biological rotation age (BRA) is defined as the age at which PAIC and MAIC are equal, indicating the rotation age that maximises mean productivity.

Comparison of MAIC and PAIC trajectories revealed contrasting patterns of stand development among hybrids (Fig. 2.6). For Beaupre, Robusta, and Gibecq, MAIC was higher than PAIC during the early establishment, with the two measures converging during the late 1990s. Thereafter, PAIC declined progressively, particularly after 2011, indicating age-related reduction in growth rates. The hybrids shared a similar developmental pattern, characterised by high levels of PAIC during the early years (1994–2000), followed by declining productivity as the age of the stand increased.

Trichobel, however, showed a markedly different growth pattern. Periodic annual carbon increment remained comparatively high throughout the study period and closely tracked MAIC even in later years, suggesting sustained productivity and a weaker age-related decline in growth. For example, in 2022, both PAIC and MAIC were approximately 2.10 Mg C $ha^{-1}yr^{-1}$. By contrast, other hybrids exhibited larger differences between the two measures. Beaupre recorded PAIC and MAIC values of 1.27 and 1.43 Mg C $ha^{-1}yr^{-1}$, respectively, in 2006, while Gibecq recorded values of 1.11 and 1.13 Mg C $ha^{-1}yr^{-1}$during the same year. Robusta saw PAIC and MAIC converge in 2011, with values of 1.38 and 1.37 Mg C $ha^{-1}yr^{-1}$, respectively. Overall, the close correspondence between PAIC and MAIC in Trichobel during stand maturity indicates sustained C sequestration capacity and continued productivity well beyond the period when growth rates began to decline in the other hybrids.

2.3.3. Influence of weather variables on periodic annual carbon increment

Weather variables were not significant predictors of PAIC in the annual, winter half-year, or most seasonal models. The only exception was for the summer model, in which significant relationships were observed (Suppl. Table A.1). In particular, the summer model explained a substantial proportion of PAIC variation in Beaupre (R^2 = 49%, p = 0.01) and Trichobel (R^2 = 47%, $p < 0.01$), while the relationships for Gibecq (R^2 = 43%, p = 0.04) and Robusta (R^2 = 59%, p = 0.02) were significant. As shown in Table 2.1, significant individual predictor effects in the summer model were observed only for Beaupre and Trichobel. In contrast, none of the individual summer weather variables were significant predictors for Gibecq and Robusta. However, summer-half soil temperature was significantly associated with Gibecq growth (Table 2.1). This suggests

that, for Gibecq, the contribution of summer climate variables to model performance arose from their combined effect rather than from any single weather predictor.

Table 2.1 Significant associations between seasonal weather variables and PAIC for the studied poplar hybrids

Hybrid	Seasonal period	Weather variable	Exp(β)	95% CI	% Change in PAIC	p-value
Beaupre	Spring	Wind speed	0.7	0.50-0.98	-29.6	0.04
Beaupre	Summer	Soil temperature	0.78	0.63-0.96	-22	0.03
Beaupre	Summer half-year	Soil temperature	0.64	0.44-0.93	-36.3	0.02
Gibecq	Summer half-year	Soil temperature	0.62	0.38-1.00	-38.2	0.04
Trichobel	Autumn	Wind speed	1.39	1.13-1.70	39.3	<0.001
Trichobel	Autumn	Rainfall	0.8	0.67-0.96	-19.8	0.04
Trichobel	Summer	Rainfall	1.29	1.11-1.49	28.5	0.01
Trichobel	Summer	Wind speed	1.15	1.02-1.30	15.4	0.04
Trichobel	Winter	Wind speed	1.23	1.02-1.48	23.2	0.04

Note: Exp(β) values > 1 indicate a positive influence on PAIC, whereas values < 1 show a negative influence. Percentage changes were calculated as $(\mathrm{Exp}(\beta) - 1) \times 100$. Because predictors were standardised, all effects correspond to a one-standard-deviation increase in the weather variable.

Several significant associations between seasonal weather variables and PAIC were identified (Table 2.1). Negative influences were observed for Beaupre and Gibecq, where higher soil temperatures, particularly during the summer half-year, were associated with lower PAIC. In Beaupre, increased spring wind speed was also associated with reduced PAIC. In contrast, Trichobel exhibited predominantly positive associations with climatic variables: higher wind speeds and summer rainfall were associated with increased PAIC, whereas autumn rainfall was negatively associated with PAIC. Overall, soil temperature emerged as the primary climatic factor associated with reduced PAIC in Beaupre and Gibecq, while rainfall and wind speed were the main variables associated with PAIC in Trichobel. No individual weather variables were significantly associated with PAIC in Robusta.

2.4. Discussion

2.4.1. Aboveground biomass carbon stock and carbon sequestration rate

Few studies have quantified AGB C stocks in mature (~30-year-old) temperate silvoarable systems, and meaningful comparisons between studies are challenging due to a wide variation in tree species, management practices, climatic conditions, soil types, and planting densities (Abedi et al., 2024; Oliveira et al., 2018; Ramos et al., 2018). To date, no study has reported PAIC for poplar hybrids in mature silvoarable systems, as MAIC is usually presented. Comparisons are further constrained because many published studies do not report both system age or tree density alongside AGB C stocks and MAIC, limiting the opportunities to benchmark long-term C sequestration potential across systems.

The present study shows that both hybrid type and cropping treatments influenced AGB C accumulation within the poplar silvoarable agroforestry system although their importance varied throughout stand development (Fig. 2.3). Hybrid effects became increasing evident over time, suggesting that genetic characteristics were a major determinant of long-term C sequestration potential. In contrast, the lack of significant block effect for both AGB C stock and MAIC indicates limited spatial heterogeneity across the experimental area. This suggests that differences in productivity were primarily associated with hybrid performance and cropping management rather than underlying site variation.

Differences among hybrids are likely associated with variation in growth rate, biomass allocation patterns, resource-use efficiency and physiological performance, resulting in distinct C accumulation trajectories through time (Truax et al., 2014). However, the temporal growth patterns observed among hybrids demonstrate that early growth performance is not necessarily a reliable indicator of long-term C sequestration potential (Fig. 2.3). While MAIC provides an integrated measure of average productivity, PAIC revealed substantial temporal variation in growth rates and showed that early productivity rankings did not persist throughout stand development. Although Beaupre exhibited the highest AGB C stock and MAIC during the first six years after establishment, this advantage was not maintained throughout the study period (Fig. 2.3

and 2.4). From 1998 onwards, Trichobel showed consistently greater increases in PAIC and ultimately accumulated the highest AGB C stock by 2023 (Fig. 2.3). These contrasting growth trajectories illustrate how hybrid rankings can change substantially through time as trees mature. Consequently, estimates of C sequestration potential derived from measurements made during the early years after establishment may not accurately represent long-term productivity. Such temporal variability may reflect differences in photosynthetic assimilation, biomass allocation strategies, resource-use efficiency, and responses to changing competitive conditions among hybrids (Wang et al., 2023b; Hilmers et al., 2025). Importantly, these findings suggest that short-term monitoring alone may provide an unreliable basis for predicting the long-term climate-change mitigation potential of silvoarable systems.

Cropping management strongly influenced C uptake into AGB during the active cropping period (1993-2004), likely through effects on tree growth and increased competition for light, water, and nutrients where crops were grown on either side of tree rows (Burgess et al., 2003; Jose et al., 2004. The lower rates of C accumulation observed under more intensive cropping are consistent with evidence that competition between crops and trees can constrain tree growth and C uptake in agroforestry systems (Jose, 2009; Udawatta & Jose, 2012; Nair, 2012; Cardinael et al., 2018c; Forrester et al., 2021). During the active cropping period, the CC treatment exhibited the lowest C accumulation, with MAIC 45.1% lower and AGB C stocks 53.7% lower than CF, while AC showed intermediate values between CC and CF. These results indicate that higher cropping intensity constrained tree growth and AGB C sequestration. Although cropping treatment differences in total AGB C stocks were not significant over the full study period, the early growth suppression observed in CC demonstrates the influence that crop-tree competition can exert on biomass accumulation. Had these competitive effects persisted for longer, greater divergence in long-term C sequestration may have emerged. Similar observations have been reported in other silvoarable agroforestry systems (Vaccaro et al., 2023).

Following the cessation of cropping, differences among treatments gradually diminished over time. Although trees in the previously cropped treatments continued to exhibit lower MAIC and AGB C stocks than those in the CF treatment, these differences were no longer statistically significant during the post-cropping period. In

contrast, PAIC remained significantly affected by treatment, indicating that differences in periodic C accumulation persisted after cropping had ceased. While the convergence of MAIC and AGB C stocks suggests a degree of recovery in the previously cropped treatments, the continued treatment effect on PAIC indicates that growth responses remained distinct among treatments during the post-cropping phase.

Reduced competition for water and nutrients may have enabled trees to increase growth rates and partially recover biomass accumulation lost during the cropping phase. Although growth was initially supressed under the CC treatment, subsequent increases in productivity enabled trees to compensate for these early limitations, resulting in comparable and, in some cases slightly greater long-term AGB C stocks and MAIC. Therefore, where aboveground C sequestration is the primary management objective, establishing trees in pasture systems or minimising competition from adjacent crops may be more effective than maintaining intensive alley-cropping arrangements close to tree rows.

Comparisons with previous studies (Table 2.2) further emphasise the importance of tree density and system design in determining AGB C stock and MAIC. Jha et al. (2018) reported lower AGB C stocks in a poplar agroforestry system in southern France than those observed for Trichobel in the present study. Although direct comparison is limited because stand age was not reported, the lower planting density of 139 trees ha^{-1} compared with 166 trees ha^{-1} in the present study may have contributed to reduced biomass accumulation. Similarly, Peichl et al. (2006) reported lower AGB C stocks and MAIC values in a 13-year-old Canadian silvoarable system established at a density of 111 trees ha^{-1}, while Wotherspoon et al. (2014) subsequently reported similarly lower values from the same system after 25 years of growth. Although differences in climate, genotype, management history, and methodological approaches complicate direct comparisons, stand age, tree density, and site management practices are also likely to contribute to differences in stand-level C accumulation across these agroforestry systems (Table 2.2).

Further evidence for the importance of stand structure comes from comparisons with short-rotation forestry systems. Oliveira et al. (2018) reported substantially higher MAIC values in short-rotation poplar plantations across several Spanish climatic

regions. However, these systems were established at extremely high densities ranging from 5,555 to 13,333 cuttings ha^{-1} and therefore differed fundamentally from the low-density silvoarable design employed in the present study. The higher productivity observed by Oliveira et al. (2018) likely reflects the intensive management and high stocking densities characteristic of short-rotation biomass production systems. Direct comparisons should therefore be interpreted cautiously because propagule density at establishment does not necessarily reflect surviving tree density during stand development, and silvoarable agroforestry systems are designed to balance tree production and agricultural outputs rather than maximise biomass production alone.

The influence of spatial configuration has also been highlighted in recent agroforestry research. Judson et al. (2025) found that combined soil and tree-biomass C stocks increased by approximately 0.7 Mg C ha^{-1} yr^{-1} relative to an adjacent arable control in a mature temperate silvoarable system with 12 m alley spacing. In contrast, no significant C benefit was observed where alley widths were 24 m. These findings suggest that the influence of tree rows declines with increasing distance into the alley and that wider alley configurations may reduce stand-level biomass accumulation. Together with the density-related comparisons discussed above, this evidence indicates that both tree stocking density and spatial arrangement are key determinants of C sequestration in agroforestry systems.

Overall, the findings of the present study demonstrate that long-term C sequestration in poplar agroforestry systems is governed by interactions among hybrid performance, stand structure, and management practices. Hybrid selection influenced AGB C stock and MAIC throughout stand development, although the magnitude and direction of this effect changed through time. For example, Beaupre showed higher C accumulation during the early years, whereas Trichobel ultimately achieved the greater long-term AGB C stock, demonstrating that early productivity does not necessarily translate into superior long-term sequestration. The long-term dataset highlights the importance of extended monitoring for accurately characterising productivity trajectories and C sequestration rates.

Table 2.2 Comparison of AGB C stocks and mean annual C sequestration rates between studies

Hybrid/Species	Region	Koppen's scale	Age	AGB C stock (Mg C ha^{-1})	Mean annual C increment (Mg C ha^{-1} yr^{-1})	Tree density (ha^{-1})	References
Beaupre	Yorkshire, England	Cf	25	33.96	1.36	166	Present study
Gibecq				27.38	1.10		
Robusta				32.66	1.31		
Trichobel				50.99	2.04		
Populus deltoides* nigra DN -177	Canada	Cf, E	25	20.70	0.83	111	Wotherspoon et al. (2014)
Populus deltoides* Populus nigra clone DN -177	Canada	Cf, E	13	15.25	1.17	111	Peichl et al. (2006)
Beaupre	Yorkshire, England	Cf	13	19.61	1.51	166	Present study
Gibecq				15.23	1.17		
Robusta				15.52	1.19		
Trichobel				20.80	1.60		
Populus euramericana	Southern France	Cs	n/a	75.76	n/a	139	Jha, (2018b)
Populus*canadensis Monch + P.*generosa Henry*P. nigra L.	Spain	Cs, Cw, and Cf	3	1.95-3.24	3.89 - 6.48	5,555 stem cuttings to 13,333 cuttings	Oliveira et al. (2018)
Beaupre	Yorkshire, England	Cf	3	1.84	0.61	166	Present study
Gibecq				0.55	0.19		Present study
Robusta				0.78	0.26		Present study
Trichobel				0.96	0.32		Present study

(Koppen's classification: C temperate climate: (-3°C < Tmin < 18°C) - Cs: temperate climate with dry summer, Cw : temperate climate with dry winter, Cf: humid temperate climate, E: polar climate (Tmax <+ 10°C) Note: only three studies reporting AGB C data were available for temperate regions)

Despite increasing policy interest in agroforestry as a climate-change mitigation strategy, empirical evidence quantifying AGB C stock in mature low-density silvoarable systems remains scarce, creating uncertainty in estimates of their climate-change mitigation potential. These findings have important implications for C accounting frameworks and agroforestry incentive schemes, which frequently rely on sequestration rates derived from relatively young systems. The observed shifts in hybrid performance through time indicate that sequestration estimates based solely on short-term measurements may overestimate or underestimate long-term C storage potential. Continued long-term monitoring across a wider range of species, genotypes, planting densities and management regimes will therefore be essential for improving future C accounting, refining climate change mitigation models and informing evidence-based agroforestry policy and practice.

2.4.2. Variability in AGB C accumulation over 30 years

Among the four hybrids studied, Trichobel demonstrated the greatest long-term AGB C sequestration potential, achieving the highest AGB C stock and maintaining productivity throughout the 30-year period. To our knowledge, no previous study has examined the relationship between MAIC and PAIC in mature poplar agroforestry systems. The inclusion of PAIC provided additional insight into temporal C dynamics that would not have been apparent from MAIC alone, enabling changes in C uptake through time to be evaluated. These findings highlight the importance of long-term monitoring when assessing the C sequestration potential of agroforestry systems.

The findings are consistent with previous research demonstrating substantial interspecific variation in C accumulation within temperate agroforestry systems. For example, Wotherspoon et al. (2014) reported considerable differences in total C stocks (AGB + BGB + SOC; 91-113 Mg C ha^{-1}) and MAIC (0.8-2.1 Mg C ha^{-1} yr^{-1}) among tree species in a 25-year-old tree-based intercropping system in Canada, illustrating the importance of species-specific growth characteristics for long-term C storage. Similarly, Cardinael et al. (2018ab) observed that C accumulation rates in agroforestry systems tend to decline with stand age, reflecting the transition from rapid biomass production during establishment to slower growth as trees mature. This pattern is widely observed in forest ecosystems, where C uptake is generally greatest during periods of rapid

growth and declines as physiological and structural constraints increase with age (Ryan et al., 1997).

The temporal patterns observed among the poplar hybrids reflected these broader stand development processes. By year 15, Beaupre (67%) and Gibecq (64%) had already accumulated 67% and 64% of their final 2023 AGB C stock, respectively (Fig. 2.7), indicating relatively rapid early growth and C accumulation. Their growth trajectories suggest an earlier transition towards maturity and a greater reduction in biomass accumulation rates in later years. In contrast, Robusta and particularly Trichobel exhibited more sustained C accumulation throughout the study period. Although age-related declines in productivity were evident across all hybrids, the magnitude of this decline differed substantially among genotypes.

Importantly, whereas Beaupre and Gibecq accumulated a large proportion of their C during the establishment phase, Trichobel continued to accumulate C at relatively high rates during the middle and later stages of stand development. Consequently, its competitive advantage emerged gradually and was only fully apparent after several decades of growth. This finding demonstrates that early growth performance is not necessarily a reliable predictor of long-term AGB C sequestration potential and suggests that assessments based solely on short monitoring periods may misrepresent the long-term C sequestration potential of individual hybrids.

Trichobel consistently exhibited higher PAIC values than the other hybrids throughout the study period and experienced only small decreases as the stand matured (Fig. 2.6 & 2.7), indicating a prolonged period of productivity. This pattern was especially evident in 2022, when Trichobel recorded a PAIC value that closely matched its MAIC (Fig. 2.6). Because MAIC typically declines with increasing stand age, the convergence of PAIC and MAIC at a relatively high level suggests that annual C uptake remained close to the long-term average rate despite the stand approaching 30 years of age. In contrast, when PAIC falls substantially below MAIC, current C accumulation is lower than the historical average and cumulative AGB C stock reflects C accumulated during earlier developmental stages. This sustained productivity exhibited by Trichobel therefore contributed to its superior long-term C sequestration performance and ultimately resulted in the highest AGB C stock among the hybrids studied.

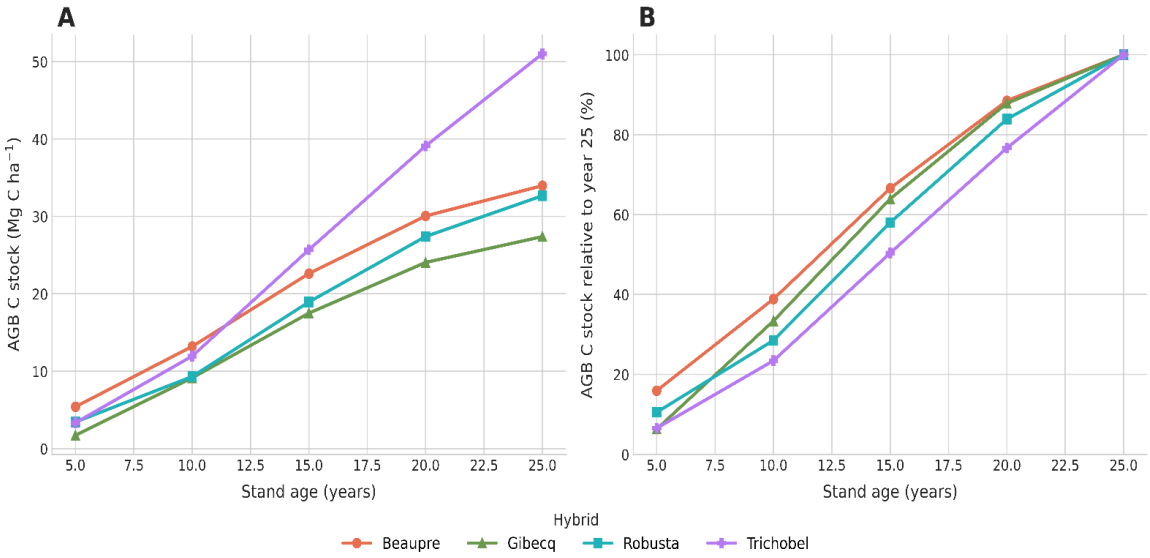


Figure 2.7 Mean aboveground biomass (AGB) carbon stock of four poplar hybrids: (A) AGB C stock (Mg C ha⁻¹ yr⁻¹) at 5,15, 20, and 25 years; (B) Percentage of AGB C stock at 5,10,15 and 20 years relative to 25 years

Over the study period, the progressive reduction in cropping-treatment differences after cropping ceased suggests that alley-cropping management largely influenced C accumulation in AGB during periods of active crop-tree interaction. Once crops were no longer present, trees across treatments were exposed to more similar growing conditions, which may have facilitated compensatory growth in previously suppressed trees and reduced differences in C accumulation over time. The convergence of AGB C stocks and sequestration rates among treatments indicates that early management did not impose a permanent limitation on long-term stand-level C sequestration. This finding suggests that growth reductions associated with crop-tree competition during establishment can be at least partially offset when competition is alleviated. From a management perspective, however, the lower rates of C accumulation observed during the cropping phase indicate that reducing crop-tree competition may be beneficial when maximising long-term tree growth and AGB C sequestration is the primary objective. Consequently, less intensive silvoarable designs, wider uncropped buffer zones adjacent to tree rows, or silvopastoral systems may provide greater opportunities for long-term biomass accumulation and C storage.

2.4.3. Effect of weather variables on PAIC

The results indicate that weather variables influenced PAIC primarily through summer conditions highlighting the importance of the peak growing season in determining annual AGB C accumulation. Summer weather variables explained a substantial proportion of PAIC variation, particularly for Beaupre and Trichobel, suggesting that growth during peak physiological activity is highly responsive to environmental conditions (Table 2.1). This seasonal sensitivity is likely linked to physiological and biochemical processes that regulate C uptake and biomass production (Muhammad et al., 2021). However, the effects of seasonal weather variables on PAIC differed markedly among the poplar hybrids, indicating substantial variation in climatic sensitivity (Table 2.1).

For Beaupre, PAIC was negatively associated with both summer soil temperature and summer-half-year soil temperature, while Gibecq showed a similar negative association with summer-half-year soil temperature (Table 2.1). These findings suggest that elevated soil temperatures were associated with conditions likely to increase evaporative demand and moisture stress, thereby limiting growth. Drought and heat stress are widely recognised as major constraints on poplar productivity because they reduce stomatal conductance, photosynthetic activity, biomass accumulation, and hydraulic functioning (Rosso et al., 2023; Wang et al., 2025). The stronger response to summer-half-year period compared with summer alone suggests that cumulative thermal conditions may exert a greater influence on growth than short-term seasonal fluctuations. This interpretation is supported by findings from temperate European poplar plantations, where prolonged climatic water deficit and heat stress have been associated with reduced biomass productivity (Bianchi et al., 2023). In addition to the effects of soil temperature, Beaupre exhibited a significant negative association between PAIC and spring wind speed (Table 2.1). Because spring corresponds to the onset of cambial reactivation and canopy development, increased wind speed during this period may enhance evapotranspiration and atmospheric water demand, thereby reducing water availability for growth. Wind has been shown to influence plant hydraulic functioning and transpiration rates, particularly during periods of active growth (Schymanski & Or, 2015; Schneider et al., 2025). Together with the negative effects of summer and summer-half-year soil temperatures, this pattern suggests that Beaupre and Gibecq may

be particularly sensitive to weather conditions that increase water demand and physiological stress.

The susceptibility of Beaupre to climatic stress may also be linked to disease dynamics, as Beaupre is known to be particularly susceptible to leaf rust (*Melampsora larici-populina*) (Lonsdale & Tabbush, 2002). Severe rust infections can lead to premature defoliation, reduced photosynthetic activity, and lower biomass production. Previous studies have shown that early-season rust infections result in greater reductions in growth because photosynthetically active leaf area is lost during the period of maximum C assimilation (Pinon & Frey, 2005). Consequently, the lower PAIC during warmer years may reflect the combined effects of thermal stress and increased disease susceptibility. Although increased summer rainfall may favour fungal development (Elad et al., 2016), no significant direct relationship between rainfall and PAIC of Beaupre was detected in the present study.

The observed negative effect of soil temperature may also reflect local microclimatic conditions. Kovács and Vityi (2022) reported significantly lower soil temperatures in poplar agroforestry systems compared with conventional plantations and demonstrated a strong relationship between soil and air temperature. Their findings suggest that moderated soil thermal conditions may help maintain favourable growing environments and reduce temperature-related stress, supporting the interpretation that elevated soil temperatures can constrain poplar growth, and thus C uptake.

In contrast to Beaupre and Gibecq, Trichobel exhibited several positive climatic responses. PAIC increased with summer rainfall, indicating that water availability during the main growing season was an important driver of growth (Table 2.1). This finding is consistent with studies across temperate Europe showing that precipitation and water availability are among the strongest determinants of hybrid poplar productivity, while moisture deficits generally reduce biomass accumulation (Bianchi et al., 2023; Rosso et al., 2023). Adequate moisture availability during the active growing season supports photosynthesis, nutrient uptake, canopy expansion, and cambial activity, thereby promoting growth and C uptake (Wang et al., 2025).

Wind speed showed contrasting effects among hybrids. While spring wind speed negatively affected Beaupre, winter, summer, and autumn wind speeds were positively associated with PAIC in Trichobel (Table 2.1). Moderate wind can enhance turbulent air exchange around leaves, reducing boundary-layer resistance and facilitating CO_2 diffusion and water vapour exchange (Charrier et al., 2015). Increased airflow may also improve transpiration-driven nutrient transport and promote evaporative cooling during warm periods (Schymanski & Or, 2015). Similar relationships between wind conditions and transpiration have been reported for temperate forest trees (Jackson et al., 2021).

However, wind can also negatively affect tree performance through mechanical stress, stem deformation, canopy damage, and disturbances to hydraulic functioning (Burgess et al., 2005; Gardiner et al., 2019). Therefore, the positive effects observed in Trichobel most likely reflect moderate wind conditions that enhanced physiological functioning rather than damaging wind events. The contrasting responses of Beaupre and Trichobel further emphasise genotype-specific climatic adaptation among poplar hybrids, and important consideration depending on the landscape and wind and moisture conditions likely to be experienced.

An important finding was that weather variables from multiple biological seasons influenced PAIC. In particular, the influence of winter and autumn weather variables on Trichobel suggests that growth responses are not determined solely by climatic conditions during the active growing season. Instead, climatic conditions preceding growth may affect dormancy release, carbohydrate storage, bud development, and subsequent cambial activity, resulting in carry-over effects on annual growth. Similar lagged climatic responses have been reported in temperate forest species, where growth is influenced by both current and preceding seasonal conditions (Gudynaitė-Franckevičienė et al., 2022).

Overall, the predominance of negative temperature-growth relationships in Beaupre and Gibecq, coupled with the positive associations between rainfall, wind speed, and PAIC observed in Trichobel, indicates clear differences in climatic sensitivity among hybrids. The positive responses of Trichobel to several climatic variables suggest greater physiological flexibility and resilience to environmental variability. In contrast, Beaupre and Gibecq appeared more vulnerable to conditions associated with heat and moisture

stress. These findings highlight the importance of genotype selection for maintaining productivity and C sequestration under changing climatic conditions and suggest that Trichobel may possess greater long-term potential for biomass production, and therefore C accumulation, under future climatic conditions than Beaupre and Gibecq.

2.5. Conclusions

This study provides one of the first long-term assessments of AGB C accumulation, MAIC, and PAIC in a mature poplar-based silvoarable systems. The findings demonstrate that C sequestration dynamics vary substantially among poplar hybrids and are influenced by both climatic conditions and the configuration of agroforestry systems. Together, these factors shaped both the magnitude and temporal patterns of AGB C accumulation over the 30-year study period.

Among the four hybrids evaluated, Trichobel exhibited the greatest long-term C sequestration potential, achieving the highest AGB C stock (65.8 Mg C ha^{-1}), almost double that of the lowest-performing hybrid, Gibecq (33.03 Mg C ha^{-1}). Although Beaupre exhibited greater C accumulation during the first decade, Trichobel outperformed all other hybrids over the full 30-year study period through sustained productivity during later stages of stand development. These contrasting trajectories demonstrate that early growth performance does not necessarily predict long-term C sequestration potential and highlight the importance of long-duration assessments when evaluating the climate change mitigation potential of agroforestry systems.

The study also demonstrated the value of incorporating PAIC alongside MAIC when evaluating AGB C sequestration performance. While MAIC provided a useful measure of long-term C accumulation, PAIC revealed considerable year-to-year variation in productivity and C uptake that would not have been captured by MAIC alone. Comparison of PAIC and MAIC trajectories revealed clear differences in growth dynamics among hybrids, with Trichobel maintaining productivity for longer and reaching maturity later than Beaupre, Gibecq, and Robusta. These findings emphasise that short-term monitoring may not adequately capture long-term trends in AGB C sequestration potential.

The results further revealed distinct responses of poplar hybrids to climatic variability. Summer climatic conditions exerted the strongest influence on PAIC, particularly for Trichobel and Beaupre, highlighting the importance of weather during the primary growing season in regulating annual AGB C accumulation. Climatic responses were strongly hybrid-specific. Beaupre and Gibecq showed reduced growth under higher summer soil temperatures, whereas Trichobel responded positively to summer rainfall and seasonal wind conditions. These results indicate substantial variation in climatic sensitivity among poplar hybrids and suggest that hybrid selection may play an important role in maintaining productivity under changing environmental conditions.

Management practices also influenced AGB C sequestration, particularly during the active silvoarable phase of the experiment (1993-2024). During this period, the crop-free treatment exhibited the highest MAIC and PAIC values, indicating that the absence of crop competition promoted greater tree growth and C uptake. In contrast, trees subjected to greater crop competition accumulated less biomass and accumulated less C. Following the cessation of cropping, treatment differences diminished suggesting that competitive effects were largely restricted to periods of active crop-tree interaction and that previously suppressed trees were able to partially compensate for earlier reductions in growth. Overall, these findings highlight the importance of minimising crop-tree competition when maximising AGB C sequestration is a primary management objective.

Overall, the findings show that mature poplar-based silvoarable systems can make a meaningful contribution to climate change mitigation by accumulating C in their AGB. However, the magnitude of this contribution is strongly influenced by hybrid selection, climatic conditions, and management practices. Trichobel consistently exhibited the strongest C accumulation performance and the greatest resilience to climatic variability, highlighting the importance of selecting appropriate genotypes when designing agroforestry systems for C sequestration objectives. More broadly, the study demonstrates the value of long-term monitoring for accurately quantifying agroforestry AGB C stocks and sequestration rates, and provides evidence that can help improve C accounting, inform agroforestry policy, and guide the development of climate-smart land-use systems.

Chapter 3. Soil carbon sequestration in a poplar silvoarable system and the potential contribution of agroforestry expansion to UK net-zero targets

Abstract

Agroforestry systems are increasingly promoted as a nature-based solution for climate change mitigation, yet their potential contribution to national 'net-zero' targets remains uncertain, particularly for temperate countries. This study quantified soil organic carbon (SOC) stocks and sequestration within a mature poplar silvoarable system and evaluated the potential carbon (C) sequestration of large-scale silvoarable expansion across UK croppable and arable land, with a focus on 2050. Soil organic carbon stocks (0–30 cm) averaged 71.70 Mg C ha^{-1} in tree rows and 75.71 Mg C ha^{-1} in alleys, compared with 58.92 Mg C ha^{-1} in the adjacent arable field. Consequently, the silvoarable system stored 27% more SOC than the conventional arable system. The average net SOC sequestration rate attributable to silvoarable agroforestry was 0.55 Mg C ha^{-1} yr^{-1}, with rates of 0.43 Mg C ha^{-1} yr^{-1} and 0.56 Mg C ha^{-1} yr^{-1} in tree rows and alleys, respectively. Three large-scale planting scenarios were modelled to assess the contribution of poplar silvoarable agroforestry to national C sequestration. Across all scenarios, aboveground biomass (AGB) represented the largest C pool, accounting for 66% of total C uptake, while belowground biomass (BGB) and SOC both contributed 17%. Carbon sequestration varied substantially among poplar hybrids, with Trichobel consistently outperforming the other hybrids and sequestering approximately 52% more C than the least productive hybrid, Gibecq, across all scenarios and land-use types. Estimated annual CO_2 removals ranged from 0.87 to 7.16 Tg CO_2 yr^{-1}, highlighting the strong influence of agroforestry adoption rates on mitigation potential. Overall, mature poplar silvoarable systems provided substantial long-term C sequestration, with 83% stored in biomass and 17% in SOC. However, hybrid selection, planting density, and deployment scale strongly influenced sequestration outcomes. As modern silvoarable systems typically employ wider tree spacing and lower tree densities than the system examined here, future C sequestration rates are likely to be lower than those reported in this study. Nevertheless, strategically deployed poplar silvoarable agroforestry could

make a valuable contribution to UK agricultural decarbonisation and net-zero objectives when integrated within a broader landscape-scale climate mitigation strategy.

3.1. Introduction

In recent years, agroforestry has gained increasing attention as a climate change mitigation strategy due to the integration of woody species into agricultural landscapes. This approach enhances C sequestration and contributes to the achievement of national net zero targets while supporting the goals of the Paris Agreement (IPCC, 2002; Zomer et al., 2016; Cardinael et al., 2018a; Mayer et al., 2022). Consequently, agroforestry is increasingly recognised within national climate policy frameworks and is often included in Nationally Determined Contributions under the Paris Agreement (Westaway et al., 2023).

Within the European Union (EU), agroforestry was formally incorporated into the Common Agricultural Policy (CAP) in 2005 under both Pillars I and II, marking an important policy milestone (Mosquera-Losada et al., 2018). Since then, it has received increasing attention as a multifunctional land-use system that supports climate mitigation, biodiversity conservation, and sustainable agricultural production (Augère-Granier, 2020). Agroforestry currently covers approximately 6.5% of utilised agricultural land in the EU, with most systems located in the Mediterranean region (den Herder et al., 2017). More recently, the EU Forestry Strategy set a target of planting an additional 3 billion trees across Europe by 2030, explicitly supporting the promotion of agroforestry as a multifunctional land-use system (European Commission, 2021). Furthermore, Gabourel-Landaverde et al. (2025) estimated that adopting agroforestry on 30.4% of European agricultural land would be required to achieve significant climate change mitigation and environmental benefits.

In the UK, and in particular England, policy frameworks in the 21st century have increasingly focused on tree planting as part of its climate mitigation strategy. As of 2020, approximately 1,310,000 ha, or 10% of the UK's total land area, was covered by woodland, making it one of the least forested countries in Europe (Image et al., 2023). This low woodland cover highlights the need to increase tree presence across the landscape, including integrating trees within productive agricultural systems. Agri-

environment schemes such as Countryside Stewardship and the Sustainable Farming Initiative (SFI) have therefore been used to encourage farmers to plant trees on agricultural land, by providing grants and ongoing management payments to landowners (Defra, 2022 and Forestry Commission & Natural England, 2024).

The UK has set a target to achieve net-zero greenhouse gas (GHG) emissions by 2050 under the Climate Change Act (UK Government, 2019). Agriculture currently accounts for approximately 12% of total UK GHG emissions, a proportion that is expected to increase as emissions from other sectors continue to decrease. Consequently, both the government's Net Zero Strategy and the Carbon Budget Delivery Plan emphasise that land-use change, particularly the expansion of tree and woodland cover, has significant mitigation potential. Consequently, there is an ambition to increase tree cover by 30,000–70,000 ha annually, including greater integration of trees within agricultural systems through agroforestry (Image et al., 2023). More specifically, the UK Climate Change Committee has proposed planting trees on 10% of agricultural land by 2050 as part of its pathway towards net-zero emissions (CCC, 2021). Despite this policy support, progress toward these agroforestry targets in the UK remains limited, and silvoarable agroforestry is still rare in practice. It currently occupies only 0.05% of arable land in the UK, compared to 0.41% across the rest of Europe (den Herder et al., 2017).

Alongside policy development, C offsetting and insetting have become increasingly important mechanisms for organisations seeking to achieve net-zero emissions targets (e.g., Black et al., 2022; Phelan et al., 2024). Carbon offsetting involves compensating for emissions through investments in external C sequestration or avoidance projects, whereas insetting focuses on reducing emissions within an organisation's own supply chain. Both approaches have contributed to increased interest in large-scale tree planting on agricultural land as a strategy for mitigating climate change (Awazi et al., 2025a). However, it remains unclear to what extent the UK's Climate Change Committee target of a 10% increase in farmland tree cover will contribute to achieving net-zero emissions by 2050. In addition, despite growing awareness of the benefits of silvoarable agroforestry and increasing policy support, uptake of this practice in the UK remains very limited (Felton et al., 2023). Robust monitoring and verification of agroforestry C

sequestration potential are needed to support climate incentive mechanisms, including C markets and government-funded schemes (Feliciano et al., 2018).

There is a significant knowledge gap regarding the amount of C stored and sequestered in different agroforestry systems, as well as the rates at which C is sequestered in both the biomass and soil over time following establishment. Previous studies have demonstrated that silvoarable agroforestry systems possess considerable C sequestration potential. For example, Graves et al. (2005) and Kay et al. (2019) estimated that silvoarable systems can sequester approximately 2-8 tonnes of CO_2 ha^{-1} yr^{-1}. However, empirical data, particularly for temperate regions, remain limited. Kay et al. (2019) reported that the AGB C accumulation in European silvoarable systems ranged from 0.16 to 2.80 Mg C ha^{-1} yr^{-1}. Few studies have simultaneously quantified C stocks in both soil and biomass in temperate silvoarable systems (Dollinger and Jose, 2018 and Bateni et al., 2021). Thus, improving our understanding of the C sequestration potential of temperate agroforestry systems is urgently needed. for improving the accuracy of climate change mitigation models and facilitating the wider promotion and adoption of agroforestry as a practice for atmospheric CO_2 capture and long-term storage.

As a fast-growing tree species, poplar is particularly well suited to C farming and serves as an effective nature-based solution for climate change mitigation through its high C sequestration potential (Zhang et al., 2020; Joshi et al., 2024). However, there is limited information on the total C sequestration potential and CO_2 uptake of the most representative poplar hybrids grown under UK conditions. Consequently, substantial uncertainty remains regarding the contribution that poplar-based silvoarable systems could make towards national net-zero targets.

The present research aims to quantify soil C sequestration in a temperate poplar-based silvoarable system and combine it with biomass C sequestration presented in Chapter 2 to assess its potential contribution to achieving the UK Climate Change Committee's net-zero target. Specifically, the study will assess the extent to which agroforestry adoption could contribute to C sequestration by 2050. This will be achieved by the following objectives:

1. Quantify SOC stocks within different components of a poplar-based silvoarable system (tree rows and alleys) and compare these with the SOC stocks in an adjacent conventional arable field.

2. Determine the mean annual SOC sequestration rate of a 30-year old poplar silvoarable system in northern England.

3. Integrate SOC sequestration rates with above- and belowground biomass C sequestration estimates to model the potential contribution of silvoarable expansion to UK net-zero targets by projecting total C and CO_2 sequestration under a range of adoption scenarios designed to achieve either 5% or 10% agroforestry coverage on croppable and arable land by 2050.

3.2. Methods

3.2.1. Study site

This research was carried out at the commercially-run University of Leeds farm located near Tadcaster, northern England. The long-term (1993-2023) mean annual temperature was 10.9°C, and mean annual rainfall was 767 mm (Bramham station 534, Met Office, 2023). The soil is dominated by well-drained calcaric brown earths, which correspond to Calcaric Endoleptic Cambisols of the Aberford Series. These soil types are widely associated with Permian and Jurassic limestone in England (Pun et al., 2024).

The 2.3 ha poplar silvoarable system was established in 1992 in one of the arable field on the farm, as outlined by Burgess et al. (2003). Briefly, it comprises four poplar hybrids: Trichobel (T), Gibecq (G), Beaupre (B), and Robusta (R) and was initially combined with three cropping treatments: continuously fallow (CF; i.e. no crops), continuously cropped (CC), and alternately cropped (AC; i.e. cropping on one side of the tree row only) (1993 -2004). All crops were managed according to standard commercial practices, receiving appropriate applications of fertiliser and pesticides. The combinations of the four hybrids and three cropping treatments were arranged in three replicated blocks (Fig 3.1).

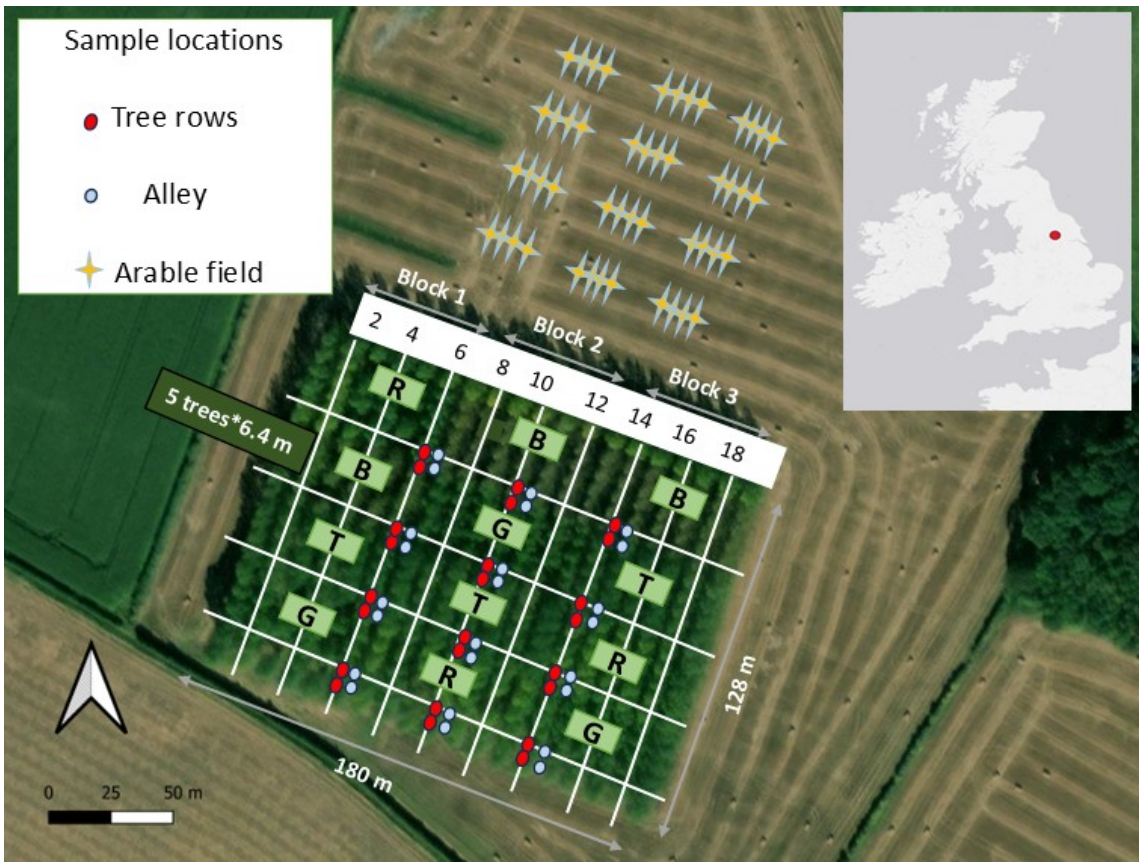


Figure 3.1 Soil sampling locations within each of the three locations. Base image source: Esri (2025). World Imagery [Basemap]. © Esri, Maxar, Earthstar Geographics, and the GIS User Community

Nineteen north–south-oriented tree rows were established, with 6.4 m spacing between trees within rows and 10 m spacing between adjacent tree rows. Each tree row consisted of groups of five consecutive trees of each hybrid, with a guard tree planted at each end. This design resulted in a planting density of 166 trees per ha. The entire experimental site covers 2.3 ha, with tree rows occupying 0.49 ha and alleys covering 1.84 ha (see Chapter 2 for more site details). Cropping within alleys ceased after the 2004 harvest due to increasing competition between trees and crops for resources. Since then, unmanaged natural vegetation has replaced the original cropping treatments in the alleys.

3.2.2. Sampling strategy

Soil sampling was conducted in January 2024 using a modified all-terrain vehicle equipped with a hydraulic core extractor to collect intact 36-mm-diameter soil cores.

Soil cores were sampled to bedrock. The mean (n=72) total core depth was 50 (±16) cm. The vehicle was equipped with GPS trackers that recorded the sampling location of each soil core. Soil cores were kept intact in air and water-sealed tubes and refrigerated until processing. Before soil analysis, soil cores were subdivided into three depths corresponding to 0-15 cm, 15-30 cm, and 30+ cm. However, we report total SOC stock only for the 0–30 cm layer because soil depth below 30 cm was inconsistent between locations due to shallow soils overlying limestone, particularly in the arable field. In addition, many studies recommend sampling topsoil to a depth of 0–30 cm, as it represents the most active zone for plant growth.

Eight soil cores were collected from tree rows 6, 10, and 14, which meant samples were collected from each of the three blocks and each of the poplar hybrids. Eight soil cores were taken from the alleys 5 m to the centre-right of the cores collected in each of the tree rows (6, 10, and 14, looking north). For each selected tree row, two soil cores were sampled from each of the four hybrids, resulting in eight cores per tree row (24 cores in total, with 6 cores per hybrid across all rows). By sampling the alleys to the right of the tree rows, eight cores from each of the different cropping management histories: CF, CC, and AC were included. In addition, eight soil cores were sampled out from the tree rows and alleys into the adjacent arable field (Fig. 3.1), resulting in the collection of 48 soil samples from the arable field.

We assume that the arable field is representative of the soil conditions prior to the establishment of the silvoarable system. It is a space-for-time substitution approach, where existing arable land is considered a record of the pre-silvoarable situation. As the agroforestry plot was established on land that had previously been managed as arable, the surrounding arable field is considered an appropriate reference for baseline soil conditions in the absence of pre-conversion measurements of SOC.

3.2.3. Soil analysis

3.2.3.1. Bulk density, SOC content and SOC stock

Once soil cores were divided into three samples, individual soil samples were dried at 75°C for a minimum of 1.5 hours and sieved to 2 mm to determine bulk density (g cm^{-}

^{3}), and SOC content (SOCcon, g kg^{-1}). Bulk density was calculated as the mass of the dry sample divided by its volume. Gravel (>2 mm) was removed from each sample, and its mass was recorded ($Mass_{coarse}$, g). By using a standard constant coarse fraction density (ρcoarse) of 2.65 g cm^{-3}, the fine soil fraction bulk density (also known as fine soil stock), which will be referred to as bulk density from here onwards, was calculated following Poeplau et al. (2017):

$$Bulk\ density\ = \frac{\text{dry sample mass} - \text{Masscoarse}}{\text{sample volume} - \left(\frac{Masscoarse}{\rho coarse}\right)} \quad \text{(Equation 3.1)}$$

SOCcon of the fine earth fraction was determined for a subsample of dried soil (< 2 mm) ground to a fine powder. Inorganic C was removed from soil samples by reaction with 10% (v/v) hydrochloric acid, and SOCcon was measured by dry combustion using an automated C: N analyser. Results were reported to two decimal points (e.g. 0.01 g kg^{-1}). Finally, SOC stock (Mg C ha^{-1}) was calculated as:

$$SOC\ stock = SOCcon * BD * depth\ interval \quad \text{(Equation 3.2)}$$

3.2.4. Data and statistical analysis

The normality of bulk density, soil organic carbon concentration (SOCcon), and SOC stock data was assessed using the Shapiro-Wilk test. As the data were not normally distributed, non-parametric statistical analyses were used. The Kruskal-Wallis test was applied to evaluate differences in bulk density, SOCcon, and SOC stock among soil depth intervals (0-15 and 15-30 cm) and sampling locations (alleys, tree rows, and arable field). In addition, differences among cropping treatments were assessed using a block-based experimental design, in which each block represented one of the three cropping treatments (CF, CC, or AC). Where significant differences were detected, Dunn's post hoc multiple comparison test was performed to identify pairwise differences between groups. Statistical significance was accepted at $P < 0.05$. All statistical analyses were conducted in RStudio (R version 4.3.3).

3.2.5. SOC sequestration rate

The annual soil sequestration (Mg C ha^{-1} yr^{-1}) rate for the tree and alley location was determined as:

$$SOCseqalleys = \frac{SOCalleys - SOCarable}{years\ since\ planting} \quad \text{(Equation 3.3)}$$

$$SOCseqtreerows = \frac{SOCtreerows - SOCarable}{years\ since\ planting} \quad \text{(Equation 3.4)}$$

To determine the overall annual average SOC sequestration rate for the whole silvoarable site (2.3 ha and 166 trees ha^{-1}), the total SOC stored in the silvoarable site ($SOC_{silvoarable}$) to 30 cm depth was determined by adding the amount of SOC stored in the area occupied by tree rows with the amount of SOC stored in the area occupied by alleys. The mean ($\bar{X}$) SOC storage of the tree rows and alleys was calculated as:

$$SOCalleys = \bar{X}\, SOC\ stockalleys * Areaalleys \quad \text{(Equation 3.5)}$$

Then the average SOC stock (Mg C ha^{-1}) of the silvoarable site was calculated by using:

$$\frac{SOCstocksilvoarable}{totalareasilvoarable} \quad \text{(Equation 3.6)}$$

The change in SOC stock that has occurred over 30 years since establishment of the silvoarable system was determined by calculating the difference between the SOC stock in silvoarble and arable field:

$$\Delta SOC\ stock = SOC\ stocksilvoarable - SOC\ stockarable\ field \quad \text{(Equation 3.7)}$$

Finally, the mean annual SOC sequestration rate (Mg C ha^{-1} yr^{-1}) was calculated as:

$$SOC\ sequestration\ rate = \frac{\Delta SOCstock}{number\ of\ years\ since\ establishment} \quad \text{(Equation 3.8)}$$

3.2.6. Upscaling of results to estimate the climate change mitigation potential of silvoarable system

We followed the UK Climate Change Committee's target of converting 10% of croppable/arable land to agroforestry by 2050 (CCC, 2020). In the UK, the total croppable area, defined as the land utilised for agriculture, including crop production and temporary grass, is approximately 6,000,000 ha, of which 4,100,000 ha is dedicated to annual arable crops (Defra, 2025). Applying the CCC target to all croppable land corresponds to the establishment of 600,000 ha of silvoarable agroforestry, whereas applying the target only to land currently under annual arable crops corresponds to 410,000 ha. Given the relatively low uptake of silvoarable in the UK, an additional adoption target of 5% of croppable and arable land was also evaluated.

Three planting scenarios were evaluated:

Scenario 1: all 600,000 or 410,000 ha (representing 10% of total croppable or arable land) are converted to silvoarable systems in a single year (2025).

Scenario 2: a total of 600,000 or 410,000 ha of silvoarable agroforestry is established over the period 2025-2050, at annual rates of 24,000 or 16,400 ha yr^{-1}, respectively.

Scenario 3: a 5% adoption target is achieved over the period 2025-2050, corresponding to the establishment of 300,000 ha of croppable land or 205,000 ha of arable land as silvoarable systems. This is equivalent to an annual planting rate of 12,000 and 8200 ha yr^{-1}, respectively.

For each scenario, total C and CO_2 sequestered by AGB, belowground biomass (BGB), and SOC were estimated to 2050 for each poplar hybrid, representing 25 years from the start of the scenarios. AGB sequestration rates were based on the periodic annual increment in carbon (PAIC) derived in Chapter 2. BGB sequestration was calculated using a shoot-to-root ratio of 0.26 (FAO, 2004), equivalent to 26% of the AGB PAIC (Mg C ha^{-1} yr^{-1}). Soil organic carbon sequestration was assumed to occur at a constant rate of 0.55 Mg C ha^{-1} yr^{-1}. Total C sequestration for each hybrid was obtained by summing the contributions from AGB, BGB, and SOC.

In Scenario 1, the entire target area (600,000 or 410,000 ha) is converted to silvoarable systems in a single year (2025). Consequently, the full area contributes to C sequestration from 2026 onwards. Annual AGB PAIC for each hybrid was multiplied by the total converted area for each year to 2050, and the cumulative AGB was calculated as the sum of annual sequestration values over the 25-year period. BGB and SOC sequestration were calculated using the same approach.

In Scenario 2, silvoarable targets are reached incrementally from 2025 to 2050 at annual establishment rates of 24,000 or 16,400 ha yr^{-1}. Carbon sequestration was therefore calculated using a cohort-based approach. In the first year, only one cohort contributes to sequestration, resulting in annual AGB sequestration of AGB PAIC × 24,000 or 16,400 ha. In the second year, two cohorts contribute; in the third year, three cohorts, and so on. BGB and SOC sequestration increments were calculated using the same cumulative cohort method.

Scenario 3 follows the same cohort-based approach as Scenario 2 but assumes incremental adoption on only 5% of croppable or arable land over 25 years. Annual and cumulative AGB, BGB, and SOC sequestration were calculated using annual establishment rates of 12,000 and 8,200 ha yr^{-1} for croppable and arable land, respectively.

For all scenarios, C uptake was converted to CO_2 using a factor of 3.67 (IPCC, 2006). Total CO_2 uptake for each hybrid was calculated by summing the CO_2 equivalents derived from AGB, BGB, and SOC sequestration.

3.3. Results

3.3.1. Bulk density

Bulk density was significantly affected by both soil depth and location within the silvoarable system ($p < 0.001$; Table 3.1). Across all locations, bulk density increased with depth, with values at 15–30 cm significantly higher than those at 0–15 cm. Tree rows exhibited the lowest bulk density at both depths (1.01 g cm^{-3} at 0–15 cm; 1.16 g cm^{-3} at 15–30 cm). At 0-15 cm soil depth, bulk density showed highly significant differences between tree rows and the arable field ($p < 0.001$) and significant

differences between alleys and the arable field ($p < 0.05$), while tree rows and alleys were not statistically significant. At 15-30 cm depth, significant differences were observed between tree rows and alleys ($p < 0.01$) and tree rows and the arable field ($p < 0.05$), whereas higher values were observed in the alleys and the arable field (Table 3.1) did not differ significantly ($p > 0.05$).

Table 3.1 Mean and (median) bulk density, SOC%, and SOC stock by location and depth; n = 24 for all alley and tree-row measurements; n = 48 for all arable field measurements.

Location/Depth (cm)	Bulk density (g/cm^3)	SOC%	SOC stock (Mg ha^{-1})
alleys 0-15	1.11 (1.10) a*	2.66 (2.65) a*	44.16 (44.13) a*
alleys 15-30	1.36 (1.38) b*	1.56 (1.57) ab*	31.90 (31.54) ab
alleys 0-30			75.71 (76.02) a
tree rows 0-15	1.01 (0.98) a*	2.86 (2.83) a*	42.54 (43.54) a*
tree rows 15-30	1.16 (1.17) a*	1.69 (1.65) a*	29.16 (29.07) a
tree rows 0-30			71.70 (72.61) a
arable field 0-15	1.18 (1.21) b*	1.72 (1.69) b*	30.64 (31.14) b*
arable field 15-30	1.31 (1.32) b*	1.45 (1.51) b*	28.28 (29.82) b
arable field 0-30			58.92(60.96) b

Different lowercase letters within a column indicate significant differences among locations at a given soil depth ($p < 0.05$). Values sharing at least one common letter are not significantly different. An asterisk (*) indicates a significant difference between soil depths within the same location.

3.3.2. SOC concentration

SOCcon was highest in the tree rows, averaging 2.86% and having a median of 2.83% in the 0–15 cm layer. At the 0-15 cm soil depth, SOCcon was significantly greater under tree rows and alleys compared with the arable field ($p < 0.001$), whereas tree rows and alleys did not differ. At 15-30 cm depth, SOCcon was comparable in the alleys and arable field (Table 3.1), while it was higher under tree rows than in the arable field ($p < 0.001$). Across all locations, SOCcon was greater in the surface layer than at 15-30 cm depth, with a significant effect of depth on SOCcon ($p<0.001$).

3.3.3. SOC stock

At depths of 0–15 cm, SOC stock varied significantly among locations. The arable field exhibited significantly lower values than both the alleys ($p < 0.001$) and tree rows ($p <$

0.001, Fig 3.2), while no significant difference was detected between the tree rows and alleys (p = 0.725). In contrast, at a depth of 15–30 cm, SOC stock did not differ significantly among locations (p = 0.87) (Fig. 3.2, Table 3.1). When the SOC stock for the entire soil profile (0-30 cm) is considered, the tree rows and alleys contained 21.7% and 28.5% more C, respectively, than the arable field. In addition, the arable field had significantly lower SOC stocks than in both the alleys and tree rows (p<0.001, while no significant difference was observed between the tree rows and alleys (p = 0.21, Fig. 3.2). Moreover, at 0-30 cm, SOC stock differed significantly between the AC and CC treatments (p = 0.02), with the treatment effect being consistent in Blocks 2 and 3.

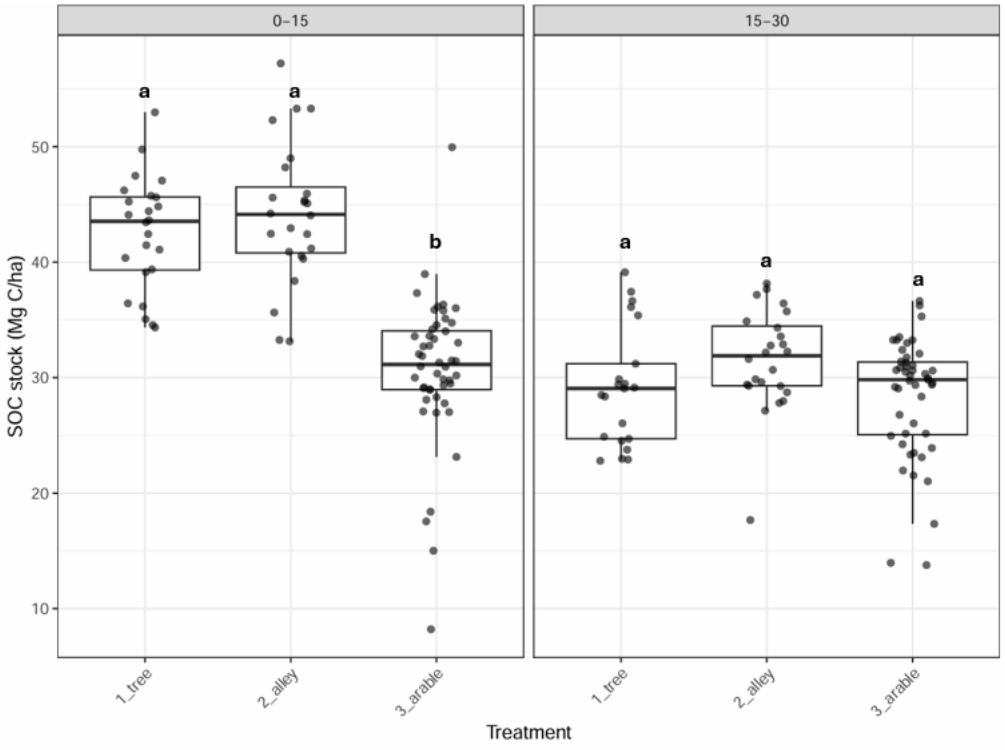


Figure 3.2 Distribution of SOC stock at 0–15 and 15-30 cm depths across three treatments: (tree rows (n=24), alleys (n=24), and arable field (n=48)), illustrated using a box-and-whisker plot. The box plot shows the median and first and third quartiles (the 25th and 75th percentiles). Different letters indicate statistically significant differences according to post hoc Dunn's tests within the same soil depth ($p < 0.05$).

3.3.4. SOC storage

Table 3.2 shows that within the silvoarable system, SOC was distributed between tree rows (35 Mg C) and alleys (140 Mg C). This represented a mean SOC stock of 75 Mg C ha^{-1} within the 0–30 cm soil layer, representing an increase of 16.58 Mg C ha^{-1} (27%) compared with the adjacent arable field (59 Mg C ha^{-1}). Estimated SOC sequestration rates were 0.46 and 0.58 Mg C ha^{-1} yr^{-1} for the tree rows and alleys, respectively. Relative to the arable field, the additional SOC stock associated with the whole silvoarable system corresponded to a net SOC sequestration rate of 0.55 Mg C ha^{-1} yr^{-1} (Table 3.2).

Table 3.2 SOC storage and sequestration rates associated with the silvoarable agroforestry system at 0–30 cm soil depth

Locations at 0-30 cm	
SOC stored in the tree rows (Mg C)	35
SOC stored in alleys (Mg C)	140
SOC stock in the silvoarable system (Mg C ha^{-1})	75
SOC stock in arable field (Mg C ha^{-1})	59
Additional stock (Mg C ha^{-1}) in silvoarable system	16.84
SOC sequestration in tree rows (Mg C ha^{-1} yr^{-1})	0.43
SOC sequestration in alleys (Mg C ha^{-1} yr^{-1})	0.56
SOC sequestration in silvoarable system (Mg C ha^{-1} yr^{-1})	0.55

3.3.5. Climate change mitigation potential of establishing silvoarable systems

The cumulative C uptake in AGB, BGB and SOC for each poplar hybrid under the three scenarios is presented in Fig. 3.3 for croppable land and Fig. 3.4 for arable land. Although the temporal trends were similar across hybrids and scenarios, the magnitude of C uptake differed, as shown by the differing y-axis scales in Figs 3.3 and 3.4. This indicates that while the pattern of response is consistent, the magnitude of the measurements differs between scenarios. Across all scenarios, total C uptake consistently ranked Trichobel> Beaupre > Robusta > Gibecq.

Under Scenario 1 for croppable land, Trichobel recorded the greatest cumulative C uptake over the 25-year period, totalling 48.77 Tg C. Of this, AGB accounted for 65.9% (32.16 Tg C) BGB for 17.1% (8.36 Tg C) and SOC for 16.9% (8.25 Tg C) (Table 3.2; Fig. 3.3). Beaupre and Robusta followed similar trends but achieved lower cumulative C uptakes of 34.34 and 32.59, respectively. Gibecq recorded the lowest total C uptake of 29.19 Tg C. Consequently, the total C sequestration potential of Gibecq was approximately 40% lower than that of Trichobel over the 25-year period.

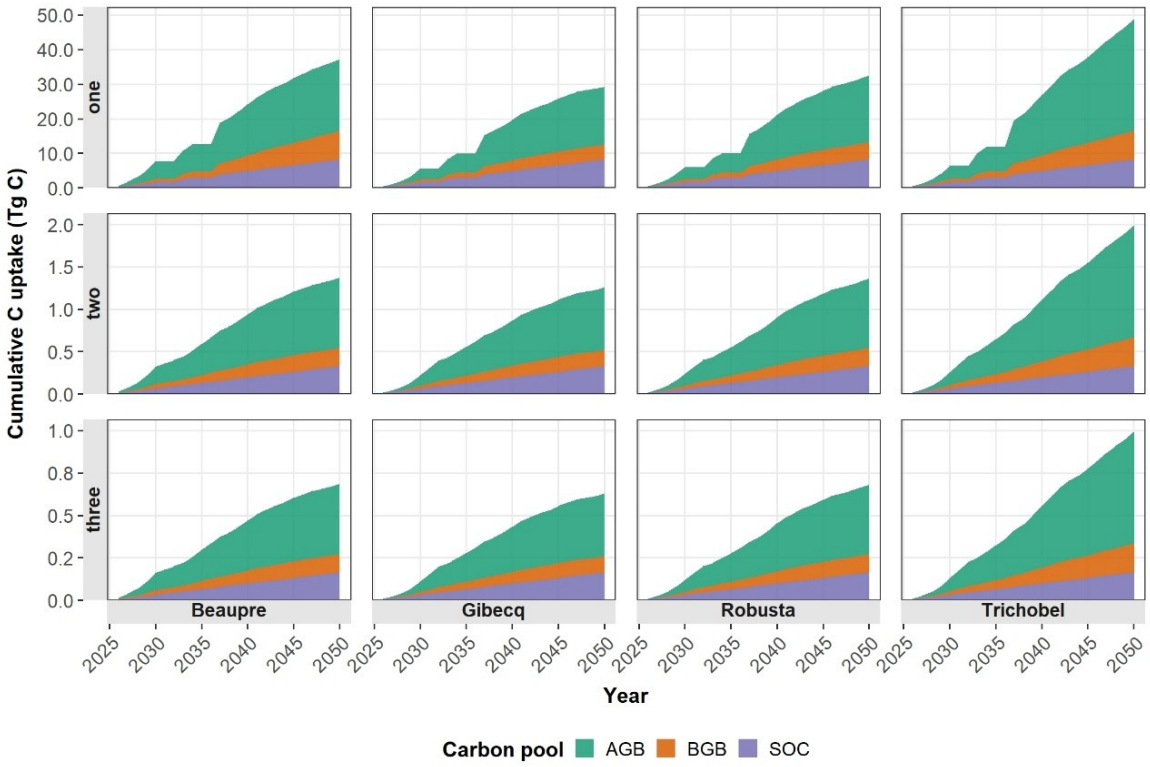


Figure 3.3 Croppable land: Cumulative C uptake by AGB, BGB and SOC for four poplar hybrids across three silvoarable establishment scenarios on UK croppable land: (1) establishment of all 600,000 ha in a single year, (2) an annual establishment rate of 24,000 $ha^{-1}yr^{-1}$ over 25 years and (3) an annual establishment rate of 12,000 $ha^{-1}yr^{-1}$ over 25 years.

Under Scenario 1 for arable land, Trichobel still had the highest cumulative C uptake, reaching 33.33 Tg C after 25 years (Table 3.3). However, this was 15.44 Tg C (or 31.7%) lower than the value estimated for croppable land. The remaining hybrids retained the same ranking, with cumulative C values of 23.47 Tg C for Beaupre, 22.27 Tg C for Robusta, and 19.95 Tg C for Gibecq. As a result, Trichobel sequestered

between 9.86 and 13.38 Tg C more C than the other hybrids under arable land conversion to silvoarable.

Scenario 3 assumes the establishment of silvoarable agroforestry on only half the area converted under Scenario 2; total C uptake by all hybrids was correspondingly 50% lower (Figs 3.3 and 3.4; Table 3.3). For example, in Scenario 2, Trichobel sequestered a total of 2.67 and 1.61 Tg C over 25 years when established on croppable and arable land, respectively. In Scenario 3, cumulative C uptake declined to 1.18 Tg C on croppable land and 0.8 Tg C on arable land. These reductions reflect the smaller area available for agroforestry establishment under Scenario 3 than Scenario 2.

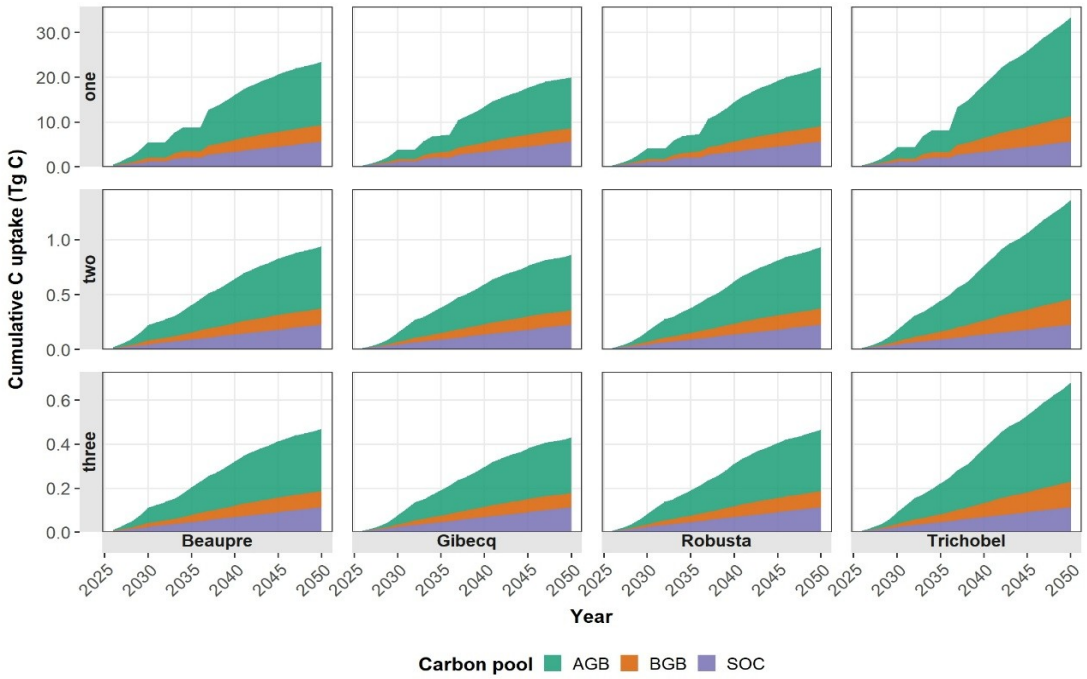


Figure 3.4 Arable land: Cumulative C uptake by AGB, BGB and SOC for four poplar hybrids across three silvoarable establishment scenarios on UK arable land (1) establishment of all 410,000 ha in one year, (2) an establishment of 16,400 ha yr^{-1} over 25 years, and (3) an establishment rate of 8,200 ha yr^{-1} over 25 years.

Furthermore, the rates of C uptake under Scenario 2 were approximately half those achieved under Scenario 1. This highlights the importance of the rate and extent of silvoarable agroforestry adoption in determining its overall contribution to C sequestration and net-zero targets. Although all scenarios demonstrated positive C sequestration benefits, the cumulative C uptake achieved over the 25-year period was

strongly dependent on the hybrids and area converted to agroforestry and the pace of establishment.

Table 3.3 illustrates the variation in total C and CO_2 uptake among scenarios and poplar hybrids. For all hybrids, total CO_2 uptake declined substantially from Scenario 1 to Scenario 3 as the area and rate of silvoarable agroforestry establishment decreased. Under Scenario 1, an optimal but unrealistic scenario in which all new silvoarable is established within a single year across either 600,000 ha of croppable land or 410,000 ha of arable land, Trichobel was projected to sequester 178.99 and 122.32 Tg of CO_2, respectively, by 2050. Under Scenario 2, where silvoarable establishment occurs steadily over 25 years at rates of 24,000 and 16,400 ha yr^{-1} on croppable and arable land, respectively, CO_2 sequestration by Trichobel was reduced by 51.44 and 51.49% less relative to Scenario 1 by 2050. Scenario 3 represented the most realistic (but still ambitious) scenario, given the very low area of silvoarable across the UK, where silvoarable is established on 5% of croppable/arable land. Under this scenario, Trichobel, the most productive hybrid, was projected to sequester 43.45 Tg of CO_2 on croppable land and 29.73 Tg of CO_2 on arable land by 2050. In contrast, Gibecq, the least productive hybrid, was projected to sequester 32.26 and 21.72 Tg of CO_2 on croppable and arable land, respectively, by 2050. For Trichobel, these values represent a 75.7% reduction in CO_2 sequestration potential relative to Scenario 1 for both land use categories.

Across all scenarios and land-use types, Trichobel consistently exhibited the greatest CO_2 sequestration potential, achieving approximately 52% greater sequestration than Gibecq (Table 3.3). These findings demonstrate that both the extent of agroforestry adoption and hybrid selection strongly influence the contribution of silvoarable systems towards achieving net-zero targets.

Table 3.3 Estimated C and CO_2 uptake and storage in AGB, BGB and SOC of four poplar hybrids over 25 years under three silvoarable establishment scenarios on UK croppable and arable land

		Croppable land				Arable land			
Scenario 1		*Trichobel*	*Beaupre*	*Robusta*	*Gibecq*	*Trichobel*	*Beaupre*	*Robusta*	*Gibecq*
	Total AGB C uptake (Tg C)	32.16	20.71	19.32	16.62	21.98	14.15	13.20	11.36
	Total BGB C uptake (Tg C)	8.36	5.38	5.02	4.32	5.71	3.68	3.43	2.95
	Total SOC uptake (Tg C)	8.25	8.25	8.25	8.25	5.64	5.64	5.64	5.64
	Total C uptake (Tg C)	**48.77**	**34.34**	**32.59**	**29.19**	**33.33**	**23.47**	**22.27**	**19.95**
	Total CO_2 uptake (Tg CO_2)	*178.99*	*126.03*	*119.61*	*107.13*	*122.32*	*86.13*	*81.73*	*73.22*
Scenario 2	Total AGB C uptake (Tg C)	15.39	11.83	11.21	10.53	10.51	8.06	7.66	7.20
	Total BGB C uptake (Tg C)	4 .00	3.07	2.91	2.74	2.73	2.10	1.99	1.87
	Total SOC uptake (Tg C)	4.29	4.29	4.29	4.29	2.93	2.93	2.93	2.93
	Total C uptake (Tg C)	**23.68**	**19.19**	**18.41**	**17.56**	**16.17**	**13.09**	**12.58**	**12.00**
	Total CO_2 uptake (Tg CO_2)	*86.91*	*70.43*	*67.56*	*64.45*	*59.34*	*48.04*	*46.17*	*44.04*
Scenario 3	Total AGB C uptake (Tg C)	7.69	5.91	5.61	5.27	5.26	4.04	3.83	3.60
	Total BGB C uptake (Tg C)	2 .00	1.54	1.46	1.37	1.37	1.05	1 .00	0.94
	Total SOC uptake (Tg C)	2.15	2.15	2.15	2.15	1.47	1.47	1.47	1.47
	Total C uptake (Tg C)	**11.84**	**9.6 0**	**9.22**	**8.79**	**8.10**	**6.56**	**6.30**	**6.01**
	Total CO_2 uptake (Tg CO_2)	*43.45*	*35.23*	*33.84*	*32.26*	*29.73*	*23.77*	*22.8*	*21.72*

3.4. Discussion

3.4.1. Bulk density and SOC

Bulk density was significantly lower in the tree rows than in both the alleys and the adjacent arable field at the 0-15 cm depth, and bulk density also differed significantly between the alleys and arable field (Table 3.1). At 15-30 cm depth, bulk density remained significantly lower in the tree rows, whereas values in the alleys and arable field were not significantly different. This pattern may reflect both historical machinery-induced compaction in the alleys and arable field during the early years of the silvoarable phase (Upson and Burgess, 2013; Correa et al., 2019) and the gradual recovery of soil structure within the tree rows where machineries cannot access. The absence of differences between the alleys and adjacent arable field after approximately 20 years suggests either persistent legacy effects of past management or similar rates of soil structural recovery. In addition, canopy-derived litter inputs may extend into the alleys, potentially reducing the magnitude of expected differences between these locations. Moreover, the lower bulk density observed within tree rows may result from greater organic matter inputs from litterfall, root turnover, and understorey vegetation, which can improve soil structure and reduce soil density. This interpretation is supported by the well-established negative relationship between soil organic matter and bulk density (Chaudhari et al., 2013; Robinson et al., 2025). Further SOC concentration results support these findings, as the greatest difference was observed between tree rows and arable field at the 15-30 cm soil depth, suggesting that greater accumulation of organic matter under tree rows may have contributed to the observed variation in soil physical properties.

SOC stocks within the silvoarable system (72.5–76.2 Mg C ha^{-1}, 0–30 cm) were similar to those reported from other temperate poplar agroforestry systems (Table 3.4), including the 78.5 Mg C ha^{-1} reported for a 13-year-old poplar system in Canada (Peichl et al., 2006). Furthermore, the SOC stocks measured in the present study (72.5-76.2 Mg C ha^{-1}, 0-30 cm) are comparable to those reported for mature poplar-based silvoarable agroforestry systems elsewhere, including the 78.5 Mg C ha^{-1} reported by Peichl et al. (2006) in Canada.

The relatively high SOC stock observed in this study may reflect the long-term development of the system (30 years old) and the cessation of cropping in 2004, which could have favoured continued organic matter inputs from tree litter and naturally established ground vegetation. However, these mechanisms were not directly assessed and therefore remain speculative. Thus, the land-use change context in this study needs to be noted. Direct comparisons between studies should also be interpreted with caution because SOC stocks are strongly influenced by initial land use, sampling depth, soil types, stand age, tree species, climate and management history (Cardinael et al., 2018b; Upson and Burgess, 2013). The studies summarised in Table 3.4 sampled soils at depths ranging from 15 to 100 cm, limiting direct quantitative comparisons. Compared with the conventional arable field, the silvoarable system exhibited significantly higher SOC stocks in the surface layer (0-15 cm; Table 3.1 and Fig. 3.2).

This suggests that enhanced litter inputs, root turnover, and microbial activity promoted higher C accumulation. These effects are most likely related to the development of the poplar canopy, which increased litter deposition not only in tree rows but also into adjacent alleys. Following termination of intercropping in 2004, the establishment of understorey vegetation may have provided additional sources of organic matter. Consequently, no significant difference in SOC stocks was observed between tree rows and alleys, suggesting that tree-derived C inputs were distributed across both locations rather than being confined to the tree rows. This is maybe because consistent with canopy litterfall extending into the alleys and may help explain the limited spatial variation in SOC stocks despite the lower bulk density observed in the tree rows (Cardinael et al., 2015; Cardinael et al., 2018ab; Ivezić et al., 2022). Because at the 0-30 cm depth, SOC stock differed significantly between the AC and CC treatments (p = 0.02), whereas SOC concentration and bulk density showed no significant treatment or block effects. This suggests that SOC stock still reflects the legacy of historical cropping practices, while more than two decades of natural vegetation growth have likely reduced treatment-related differences in SOC concentration and bulk density through continued litter inputs and root turnover.

In contrast, SOC stocks did not differ among locations at 15–30 cm depth, suggesting that agroforestry-induced increases in SOC were mainly restricted to surface soils. This finding is consistent with previous studies, showing that deeper soil layers respond more

slowly to changes in vegetation and C inputs (Lal, 2004; Germon et al., 2016; Mayer et al., 2022).

Table 3.4 SOC stock and sequestration rates reported for temperate silvoarable systems

Region	Species	Age	Tree density (ha^{-1})	Depths (cm)	SOC stock (Mg C ha^{-1})	SOC seq rate (Mg C $ha^{-1}yr^{1}$)	References
Guelph	Poplar TBI	21	111	0-15	34-38	0.19	Bambrick et al. (2010)
Canada	*Populus deltoides* × *Populus nigra* clone DN-177.	13	111	0-20	78.5	1.04	Peichl et al. (2006)
UK (alley)	Poplar hybrids	30	166	0-30	76.22	0.43	Present study
UK (tree rows)	Poplar hybrids	30	166	0-30	72.53	0.56	Present study
UK (tree rows)	Poplar hybrids	20	-	0-60	161	-	Upson and Burgess, (2013)
6 sites across France	Hybrid Walnuts & cherry	6-44	34-100	0-30		0.24 (0.09–0.46)	Cardinael et al. (2018b)
France	Hybrid walnuts (*Juglans regia* × *nigra* L.)	18	110	0-100		0.35	Cardinael et al. (2018a)

The 27% increase in SOC storage observed in the silvoarable system, when compared to the adjacent arable field, aligns with the typical global range of SOC stock increases (26-40%) reported following the conversion of conventional agricultural land to agroforestry systems, although responses vary according to soil depth and system type (De Stefano & Jacobson, 2018). The SOC sequestration rate recorded in this study (0.55 Mg C ha^{-1} yr^{-1}) exceeds the average rates reported for temperate alley-cropping systems in a recent global meta-analysis, which identified mean sequestration rates of 0.26 ± 1.15 Mg C ha^{-1} yr^{-1} in topsoils (0-20 cm) and 0.23 ± 0.25 Mg C ha^{-1} yr^{-1} in subsoils (20-40 cm) (Mayer et al., 2022). Although higher than the global averages, the sequestration rate is within the range reported for other temperate silvoarable systems (0.19-1.04 Mg C ha^{-1} yr^{-1}; Table 3.4).

Unlike most previous studies that reported a system-level average SOC sequestration rate, sequestration was quantified separately within tree rows and alleys in this study. This revealed pronounced spatial heterogeneity in the silvoarable system. While SOC

stocks were greater in the alleys, sequestration rates were highest in the tree rows (Tables 3.1 and 3.2), suggesting that active C accumulation is concentrated near trees where litterfall and root-derived inputs are greatest (Cardinael et al., 2018b). This within-system variability, together with differences in stand age, management and planting density, likely contributes to the wide range of sequestration rates reported across agroforestry systems (Table 3.4). Such complexity may also help explain why SOC sequestration is often omitted in broader assessments of agroforestry contributions to net-zero targets, despite its potential importance as a long-term C sink and increasingly promoted by governments as a nature-based climate solution and a means of achieving Nationally Determined Contributions (NDCs) (Defra, 2022).

The role of agroforestry system design should therefore be considered when evaluating future C sequestration potential on arable soils. Recent silvoarable systems increasingly adopt wider alley widths (>24m) to maximise crop production and improve operational efficiency (Staton et al., 2018; Judson et al., 2025; Reuse & Langhof, 2025). While wider alleys may improve agricultural productivity, they reduce tree density and may consequently lower system-level SOC stocks and long-term sequestration potential. Overall, these findings demonstrate that mature poplar based agroforesty systems can deliver measurable gains in SOC storage and sequestration relative to conventional arable land, while highlighting the importance of spatially explicit assessment and system design when maximising climate mitigation benefits.

3.4.2. Carbon sequestration potential of poplar silvoarable expansion under alternative adoption Scenarios

This study provides one of the first data-driven assessments of the national-scale C sequestration potential of poplar silvoarable systems, incorporating C stored in AGB, BGB, and SOC (Fig. 3.3 and 3.4). The results demonstrated that both the extent and rate of silvoarable establishment strongly influence total C sequestration potential. The UK CCC identifies agroforestry as a key component of the pathway towards net-zero, with a target of establishing agroforestry on 10% of cropland by 2050 alongside a 40% increase in hedgerow extent (AHDB, 2025; CCC, 2020). The upscaling results demonstrate that poplar-based silvoarable agroforestry could make a substantial contribution to mitigating agricultural CO_2 emissions, with estimated annual CO_2

removals ranged from 0.87 to 7.16 Mt CO_2eyr^{-1} across the different adoption scenarios and hybrids evaluated (Fig. 3.3 and 3.4). Variation in sequestration potential was mainly driven by differences in the area allocated to agroforestry in each scenario, highlighting that the scale of adoption is more important determinant of national mitigation outcomes than differences in sequestration rates among hybrids alone.

Relative to projected UK agricultural emissions of approximately 41 Mt $CO_2e\ yr^{-1}$ in 2050, the highest-performing hybrid (Trichobel under Scenario 1) was estimated to offset approximately 17.3% of annual agricultural emissions, whereas the lowest-performing hybrid (Gibecq under Scenario 3) still offset around 3.1%. Under Scenario 1, hybrid poplar systems offset between 10.3 and 17.3% of projected agricultural emissions, while under Scenario 2, which reflects the CCC land-use change pathway, offsets ranged from 6.2 to 8.4%. Under Scenario 3, representing a more realistic, although still ambitious, level of agroforestry adoption, offsets ranged from 3.1 to 4.2% of projected agricultural emissions. These findings highlight that even under the lowest level of adoption, agroforestry expansion could contribute meaningfully to agricultural decarbonisation, although the level of mitigation achieved remained highly dependent on deployment scale (Department for Energy Security and Net Zero, 2022).

The projected C trajectories should be interpreted in the context of tree growth dynamics. Biomass accumulation represents the largest proportion of total C sequestration (83%), and therefore overall C removal rates were strongly linked to patterns of tree growth. However, biomass accumulation is unlikely to increase indefinitely, because tree growth rates generally decline with increasing stand age, resulting in progressively slower rates of C accumulation over time (Truax et al., 2014). Consequently, the values presented here should be interpreted as plausible C sequestration pathways under current conditions, rather than predictions that sequestration rates will continue indefinitely into the future. Even so, the results demonstrate that mature silvoarable agroforestry can provide substantial long-term C storage and make a meaningful contribution to climate change mitigation.

The contrast between scenarios highlights an important trade-off between theoretical mitigation potential and practical implementation. Although Scenario 1 generated the greatest C sequestration, achieving widespread establishment at this scale may provide

challenging due to competing demands for food production, biodiversity conservation, and broader landscape multifunctionality. This challenge is reflected in the UK's historical difficulties in meeting national tree-planting targets (Westaway et al., 2023). Lower adoption rates, such as those represented by Scenario 3, may be easier to integrate within existing farming systems and therefore more achievable in practice. However, this came at the cost of substantially reduced mitigation potential. Across all hybrids, total sequestration under Scenario 3 was approximately 70-76% lower than under Scenario 1. For example, annual sequestration by Trichobel decreased from 7.09 Tg CO_2 yr^{-1} under Scenario 1 to 1.72 Tg CO_2 yr^{-1} under Scenario 3, a reduction of 75.7%. Consequently, Scenario 3 offset only 3.1-4.2% of projected UK agricultural emissions in 2050, compared with 10.3-17.3% under Scenario 1 (Table 3.3 and 3.4). These findings suggest that although agroforestry can provide a significant C sink, it is unlikely to deliver net-zero targets in isolation and should be implemented alongside other agricultural mitigation measures (Westaway et al., 2023).

Comparison with Burgess et al. (2022) further emphasises the importance of hybrid selection, system design, and implementation scale. Annual soil and tree C sequestration rates reported for the 19- and 30-year-old silvoarable systems in Burgess et al. (2022) were converted to Tg CO_2 yr^{-1} and scaled according to the land area available under each scenario (Table 3.5). Under Scenario 1, Trichobel exhibited the highest annual C sequestration rate (4.84 Tg CO_2 yr^{-1}), representing a 38.7% increase relative to the 19-year-old Burgess system (3.49 Tg CO_2 yr^{-1}) and a 27.4% increase relative to the 30-year-old system (3.80 Tg CO_2 yr^{-1}). This difference was primarily attributable to greater AGB C accumulation, whereas SOC sequestration rates were relatively similar among systems. Under Scenario 2, all poplar hybrids outperformed the Burgess reference systems, with annual sequestration ranging from 1.74 to 2.35 Tg CO_2 yr^{-1} compared with 0.77 to 1.11 Tg CO_2 yr^{-1} for the Burgess systems. Although similar trends were observed under Scenario 3, the large differences among systems were influenced not only by biological productivity but also by the assumptions used when scaling sequestration rates to national land areas. Consequently, these comparisons demonstrate that national mitigation potential is determined not only by per-hectare sequestration rates but also by the extent of agroforestry adoption. Together, the

findings identify hybrid selection and deployment scale as the principal factors controlling the overall C sequestration potential of silvoarable agroforestry.

Table 3.5 Predicted annual mean C and CO_2 sequestration potential of silvoarable expansion across UK arable land

	Age	SOC stock (Mg C ha^{-1})	Tree (Mg C $ha^{-1}yr^{-1}$)	Soil seq. (Tg CO_2 yr^{-1})	Tree (Tg CO_2 yr^{-1})	Total (Tg CO_2 yr^{-1})
Scenario 1						
Burgess et al., 2022	19	0.46	1.65	1.01	2.48	3.49
Burgess et al., 2022	30	0.29	2.10	0.64	3.16	3.80
Trichobel	25	0.55	2.74	0.78	4.06	4.84
Beaupre	25	0.55	0.72	0.78	2.62	3.40
Robusta	25	0.55	0.94	0.78	2.44	3.22
Gibecq	25	0.55	0.19	0.78	2.10	2.88
Scenario 2						
Burgess et al., 2022	19	0.46	1.65	1.01	0.1	1.11
Burgess et al., 2022	30	0.29	2.10	0.64	0.13	0.77
Trichobel	25	0.55	2.74	0.41	1.94	2.35
Beaupre	25	0.55	0.72	0.41	1.49	1.90
Robusta	25	0.55	0.94	0.41	1.42	1.83
Gibecq	25	0.55	0.19	0.41	1.33	1.74
Scenario 3						
Burgess et al., 2022	19	0.46	1.65	1.01	1.24	0.06
Burgess et al., 2022	30	0.29	2.10	0.64	1.58	0.07
Trichobel	25	0.55	2.74	0.20	0.97	1.18
Beaupre	25	0.55	0.72	0.20	0.75	0.95
Robusta	25	0.55	0.94	0.20	0.71	0.91
Gibecq	25	0.55	0.19	0.20	0.67	0.87

SOC stock in a poplar-based silvoarable system and the predicted total C and CO_2 uptake of all hybrids for the period 2025–2050 for arable land only. Tree (Mg C $ha^{-1}yr^{-1}$) represents the C sequestered through aboveground PAIC from Chapter 2. Soil sequestration (Tg CO_2 yr^{-1}) includes upscaled SOC sequestration over 25 years. Tree CO_2 uptake (Tg CO_2 yr^{-1}) is calculated from the combined AGB and BGB C uptake. Total CO_2 sequestration (Tg CO_2 yr^{-1}), includes C accumulated in AGB, BGB, and SOC over 25 years.

Comparison with hedgerow systems further highlights the substantial biomass accumulation potential of poplar-based agroforestry. Biffi et al. (2023) reported the accumulation of 0.67 Tg C in hedgerow AGB over a 27-year period, whereas estimated AGB C accumulation in the poplar-based systems ranged from 5.26 to 21.98 Tg C

across the scenarios evaluated. This represents approximately 8 to 33 times higher C uptake than hedgerows alone, likely due to the greater tree density (166 trees ha^{-1}) and biomass production of poplar systems. Even if tree density were halved, estimated C uptake would still be about 4 to 16 times higher than in hedgerows. While hedgerows provide numerous environmental benefits, including habitat provision, landscape connectivity, and erosion control, these results suggest that silvoarable agroforestry has considerably greater potential for biomass C sequestration at the landscape scale.

From a policy perspective, these findings suggest that predicting the impact of expanding agroforestry to net-zero targets should not rely on generic C sequestration estimates or a one-size-fits-all approach. The substantial variation in C uptake among poplar hybrids and implementation Scenarios demonstrates the importance of matching suitable tree genotypes to specific sites and management objectives. Policies should therefore support not only the establishment of agroforestry systems, but also their long-term management to maximise C sequestration benefits over time. In addition, robust monitoring frameworks are needed to capture variation in C outcomes among systems and ensure that reported climate benefits are evidence-based. This approach aligns closely with current UK agroforestry policy, which seeks to increase agricultural C sequestration while simultaneously delivering wider ecosystem services, including soil erosion control, water regulation, biodiversity enhancement, and the maintenance of food production (Areal et al., 2026). Ultimately, the results highlight that the climate mitigation potential of agroforestry depends not only on adoption rates, but also on system design, species selection and long-term management.

3.5. Conclusion

This study highlights three key findings:

1. Agroforestry can contribute meaningfully to net-zero targets but the magnitude of mitigation depends on how and where it is implemented. Upscaling estimates indicate that poplar-based silvoarable systems could remove between 0.87 and 7.16 Tg CO_2 yr^{-1}, equivalent to offsetting between 2.13% and 17.5% of projected UK agricultural CO_2emissions by 2050. However, the potential climate mitigation benefit varied substantially among adoption scenarios and hybrids, demonstrating that sequestration outcomes are highly dependent on system design, species selection, and the extent of agroforestry adoption. Consequently, agroforestry policies should move beyond generic sequestration estimates and instead promote evidence-base, site-specific approaches that prioritise high-performing hybrids, locally adapted tree varieties and long-term system management. Such targeted deployment would maximise climate mitigation benefits while simultaneously delivering wider environmental co-benefits, including improved soil health, biodiversity enhancement, and greater resilience of agricultural systems.

2. *Silvoarable agroforestry enhances long-term soil carbon storage.* Mature poplar-based silvoarable agroforestry systems accumulated significantly greater SOC stocks than an adjacent conventional arable field, indicating their capacity to increase long-term soil C storage. Although SOC accumulation was concentrated within the surface soil layer (0-15 cm), the average net sequestration rate of 0.55 Mg C ha^{-1} yr^{-1}, with rates of 0.43 and 0.56 Mg C ha^{-1} yr^{-1} in tree rows and alleys, respectively, indicates a sustained capacity for C storage over time. Furthermore, separating sequestration estimates between tree rows and alleys revealed substantial within-system heterogeneity, highlighting the importance of spatially explicit assessments when evaluating agroforestry soil C dynamics. Carbon sequestration within the system was not limited to soils; AGB accounted for the largest proportion of total C uptake (66%), while BGB (17%) and SOC (17%) provided additional long-term storage. Together, these findings demonstrate that poplar silvoarable systems can simultaneously increase C storage, improve soil quality and maintain productive agricultural land use.

3. Hybrid selection is a critical determinant of carbon sequestration potential. Carbon sequestration differed substantially among poplar hybrids, with Trichobel consistently achieved the highest C uptake. Across all expansion scenarios, Trichobel sequestering approximately 52% more C than the least productive hybrid, Gibecq, highlights the importance of selecting productive and site-adapted tree varieties when designing silvoarable systems. Although Scenario 1 delivered the greatest cumulative C sequestration over 25 years, its assumption of immediate large-scale agroforestry adoption makes it unlikely to be achieved in practice. In contrast, Scenarios 2 and 3 represent more realistic pathways for UK agroforestry expansion and demonstrate that both the rate of expansion and the choice of hybrid strongly influence long-term climate mitigation outcomes. These findings emphasise that successful agroforestry policy should focus not only on increasing adoption, but also on promoting the most effective tree varieties capable of maximising C sequestration for local environmental conditions.

Overall, this study demonstrates that mature poplar-based silvoarable agroforestry systems can provide substantial long-term C sequestration through the combined accumulation of C in soil, AGB, and BGB. However, the magnitude of climate change mitigation achieved depends on interactions among hybrid performance, system design, and the scale of agroforestry adoption. The results therefore highlight the need for strategically targeted agroforestry expansion supported by robust C accounting, long-term monitoring, and appropriate hybrid selection to maximise the contribution of agroforestry to UK net-zero ambitions.

Chapter 4. Spatial variability in soil nutrients, fine root traits and belowground carbon dynamics in a mature temperate poplar silvoarable system

Abstract

This study investigated spatial variability in soil nutrients, root traits, and soil physical properties within a mature poplar-based silvoarable system in northern England. Soil and root measurements were compared across three management zones: crop alleys (Alley), tree rows (Row), and an adjacent conventional arable field (Arable). Available nitrogen (AN) and Olsen phosphorus (P) declined significantly with soil depth and exhibited distinct spatial patterns, with AN generally higher in agroforestry zones and P highest in Arable. Both nutrients were positively associated with live and dead root length density, highlighting the linkage between nutrient availability and root distribution. In contrast, bulk density was negatively correlated with key root traits, including live and dead root length, biomass density and root C stocks, and was generally lower in Row and Alley, suggesting improved soil structure in biologically active zones. Root traits varied significantly with depth and location, with the highest (80-90%) live root length density, biomass density, and root C stocks occurring within agroforestry zones. Soil organic carbon (SOC) concentrations increased significantly with depth, being 29.9% greater at 45 cm than at 15 cm in Alley and 14.4% greater in Row. Although SOC stocks did not differ significantly among locations, they were strongly associated with root biomass and root C stocks. Dead roots dominated topsoil C inputs in Row, accounting for 89.2% of root biomass density and 97.1% of root C stocks, Similar patterns were observed for live root biomass density and root C stocks, with 86-99% occurring in the topsoil whereas live roots extended into the subsoil in Alley and Row (9.6% and 11.1% of root length density, respectively) but were absent in the subsoil of Arable. These patterns suggest complementary C pathways, with live roots supporting nutrient cycling and rhizosphere processes and dead roots contributing to necromass accumulation and longer-term soil C storage in agroforestry. Overall, mature silvoarable agroforestry enhanced belowground spatial heterogeneity, promoted deeper rooting, and increased the potential for soil C sequestration.

4.1. Introduction

In terrestrial ecosystems, including woodlands, fine and coarse roots, together with their interactions with soil microorganisms, play a fundamental role in biomass production and the regulation of nutrient and carbon (C) cycling (Clark et al., 2001; Raich et al., 2014; Smyth et al., 2013). It is estimated that approximately 22-63% of organic C fixed through total gross primary productivity in forests is allocated to below ground (Litton et al., 2007; Prescott, 2010). Recognising these functions, both the IPCC (Ngo et al., 2019) and UNFCCC (Rosenstock et al., 2019) have identified agroforestry as a multifunctional land-use approach that enhances agroecological processes, contributes to climate change mitigation through C sequestration, and supports a wide range of interconnected ecosystem services. Consequently, agroforestry systems have been identified as a key component of Europe's C farming scheme (COWI, Ecologic Institute & IEEP, 2021). However, despite increasing recognition of their potential, considerable uncertainty remains regarding the soil C sequestration potential of complex, biodiversity-rich systems such as agroforestry, particularly in comparison with conventional agricultural and forestry systems.

Silvoarable agroforestry systems consist of three principal vegetation zones: (1) trees planted in rows, (2) an understory strip, typically a few metres wide, located within the tree rows, and (3) arable crops cultivated in the inter-row alleys, which are typically 10s of metres wide. While these systems can enhance resource use efficiency and ecosystem functioning, competition for light, water, and nutrients can reduce crop yields due to interspecific competition, particularly where tree root systems extend laterally into adjacent crop alleys (Livesley et al., 2004; Zhang et al., 2019; Guillot et al., 2021). The root systems of these vegetation types differ in their chemical characteristics, including C, N, and C/N ratio, as well as in key functional traits such as root length density, root biomass density, root diameter, and root turnover (Prieto et al., 2015). These differences influence interactions with soil physical and chemical properties (Freschet et al., 2021), thereby regulating below-ground processes such as nutrient acquisition, organic matter decomposition and nutrient mineralisation - immobilisation dynamics (Smith et al., 2013).

Silvoarable systems are characterised by spatially heterogeneous root distributions, which can enhance soil C storage across the system, although these effects often vary among vegetation zones and soil depth (Bonanomi et al., 2021a; Villarino et al., 2021). In these systems, both tree and crop roots may adapt their functional traits such as root diameter and root length density to optimise nutrient uptake and soil exploration under varying levels of competition and resource availability (Freschet et al., 2021; Houde et al., 2020). In addition to this spatial heterogeneity, root-derived C inputs display pronounced temporal dynamics. Annual crops contribute substantial belowground C inputs during crop maturation and harvest, whereas perennial trees provide more continuous inputs through root growth, exudation, and turnover processes (Abbas et al., 2020). Consequently, the diversity and temporal variability of root systems in silvoarable agroforestry play a critical role in regulating soil C inputs and influencing long-term soil C sequestration.

Root development is strongly influenced by the availability of both C and nitrogen (N). C provides the structural resources required for root construction and expansion, particularly under nutrient-limited conditions, whereas N availability regulates root growth and nutrient acquisition strategies. In silvoarable systems, nutrient availability is spatially heterogeneous, with fertilised crop alleys typically containing greater nutrient inputs than tree rows, creating varying conditions for root growth (Sharp et al., 1990). Consequently, both trees and crops may adjust their root morphology and foraging strategies in response to these spatial gradients.

Interactions among living roots, dead roots, and soil microbial communities underpin SOC storage through mechanisms such as rhizodeposition, root turnover, and rhizosphere-mediated microbial activity (Dijkstra et al., 2021; Villarino et al., 2021). Root-derived C inputs (exudates and decomposing root material) play an important role in SOC stabilisation because they can be incorporated into stable soil C pools more than aboveground litter plant inputs. Living root inputs have been reported to be 2–13 times more effective than litter inputs in forming SOC (Sokol et al., 2019), while root-derived C typically has a longer residence time in soil than shoot-derived C (Rasse et al., 2005).

Mature agroforestry systems offer a valuable opportunity to investigate the long-term dynamics of soil resource acquisition and its interaction with root traits, particularly in

systems containing fast growing tree species. Despite growing recognition of the importance of root-driven processes for ecosystem functioning and soil C sequestration, very few studies have examined the relationships between root functional traits and soil functionality including C storage and nutrient cycling in mature agroforestry alley-cropping systems (Prescott, 2010; Raich et al., 2014). Among these systems, mature poplar-based agroforestry systems are of particular interest because of their high productivity, extensive rooting networks, and potential resilience to environmental variability (Enescu et al., 2026). Although numerous studies have demonstrated the capacity of agroforestry systems to enhance soil C sequestration (Kaur et al., 2000; Sharrow & Ismail, 2004; Cardinael et al., 2018a; Cardinael et al., 2018b), no studies, to date, have comprehensively examined horizontal soil gradients in conjunction with root functional traits in mature poplar agroforestry systems in temperate regions. Furthermore, it remains unclear the extent to which root-trait nutrient acquisition strategies influence C sequestration (Prieto et al., 2015; Pausch & Kuzyakov, 2018). Hence, this study aims to fill this knowledge gap by investigating root–soil interactions and their implications for nutrient cycling and C sequestration in a mature silvoarable system.

This study aims to investigate the spatial variation in plant-available soil nutrients, including available nitrogen (AN), phosphorus (P), total nitrogen (N%), SOC%, SOC stock, and bulk density, as well as root traits such as live and dead root length density, root biomass density, root C stock, and root C and N content. These parameters will be assessed across three vegetation zones: Alley, Row, and Arable and at different soil depths within the soil profile. We hypothesise that (i) soil nutrient availability would differ among locations and soil depths, (ii) root traits would vary in response to these nutrient gradients, and (iii) interactions between soil nutrient availability and root dynamics would influence belowground C inputs, turnover, and storage, resulting in spatially distinct soil environments within the agroforestry system.

4.2. Methods

4.2.1. Study site

The research was conducted at the University of Leeds farm located near Tadcaster (53° 51′ 44.45″ N, −1° 19′ 58.14″ W), northern England, a mixed arable and pasture system (Guest et al., 2022; Judson et al., 2024). The farm comprises both conventional arable fields and some mature agroforestry plots established in the 1990s. The region has a temperate oceanic climate under the Köppen system, characteristic of much of Western Europe (Beck et al., 2018). It has a mean annual temperature of 10.9°C and a mean annual rainfall of 767 mm (1993-2023 data for Bramham station, UK Met Office, 2023). Soils at the farm are classified as calcaric endoleptic cambisol from the Aberford series, well-drained calcareous brown earth developed on Permian and Jurassic limestone in central and northern England (British Geological Survey, 2018).

The poplar silvoarable experimental system was established in 1992 and had four poplar hybrids, Trichobel, Gibecq, Beaupre, and Robusta, which represented the major groups of poplars being grown in Britain at the time (Burgess et al., 2005). Three arable cropping systems were implemented: continuously fallow (CF), continuously cropped (CC), and alternatively cropped (AC). Each combination of poplar hybrid and cropping system was arranged in three replicated blocks in a randomised block design to account for spatial variability in the field and edge effects throughout the experiment.

The site consisted of 19 rows of trees oriented approximately NNE-SSW. Trees were planted at 6.4 m intervals within rows, with 10 m spacing between rows. Within each row, hybrids were planted in groups of five consecutive trees, with a guard tree positioned at each end of the row. The planting design resulted in an overall tree planting density of 166 trees per ha. Further information on the establishment and management of the trees at the site is provided by Burgess et al. (2005).

Between 1992 and 2003, the 10-metre-wide alleys were ploughed annually in autumn in 8-metre strips. This resulted in tree root damage to a depth of approximately 20-30 cm in the cultivated alleys, while leaving a 2 m-wide uncultivated strip along the tree rows (Burgess et al., 2005). In the CF treatment, two adjacent alleys were regularly

cultivated. In the CC treatment, a rotation of cereals and non-cereal crops was cultivated on both sides of the measurement tree row. The AC treatment involved continuous cropping on one side of the measurement tree row and the fallow treatment on the other side (Burgess et al., 2003).

All crops were managed following standard commercial practices, including the application of fertilisers and pesticides as appropriate. However, crop cultivation in the alleys ceased after the 2004 harvest due to declining yields, largely resulting from competition for light, nutrients, and water. Since then, the alleys have been unmanaged, allowing natural vegetation to establish and develop.



Figure 4.1 Map of the experimental site, highlighting the poplar silvoarable system, and adjacent arable field and the red dots indicate the sampling locations where soil pits were excavated. (Base image source: Esri (2025). World Imagery [Basemap]. © Esri, Maxar, Earthstar Geographics, and the GIS User Community).

One soil pit per location measuring approximately 2.5 m × 1 m, was excavated to a depth of 50 cm in each of Row, Alley, and Arable using a small mechanical excavator. The pits in Alley and Row were positioned between evenly spaced rows, 10-12 of the 19 total rows (Fig. 4.1). In the Row location, the pit was located in the centre of the Row, at the mid-point between two adjacent rows. From this location, the pit in Alley was established 2 m away, also positioned in the centre of the alley. The Arable pit was

located in a field adjacent to the silvoarable system (Fig. 4.1). The vertical faces of each pit were cleaned using a shovel and measured with a tape to facilitate soil sampling. This preparation also allowed precise observation of soil colour and identification of soil horizons (British Society of Soil Science, 2025).

As shown in Fig. 4.2a and b, soil layers varied irregularly between Row and Alley. In Alley, limestone-rich soil was more prominent than in Row. In both locations, shallower soil horizons were associated with a more complex root network. Observations from the soil pits, measured using a tape, indicated that root diameters generally did not exceed 3 cm in Alley and 5 cm in Row. In contrast, Arable (Fig. 4.2c) exhibited shallower soil compared with the silvoarable locations. Limestone was encountered at a depth of ~10 cm, and the soil appeared more compact than in the silvoarable systems, as indicated by bulk density.

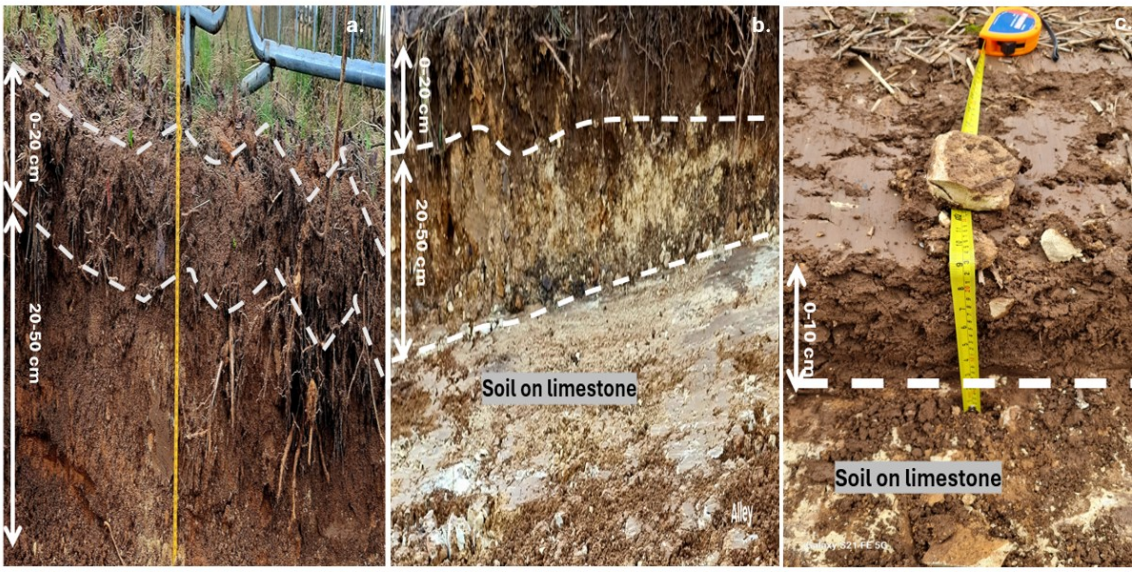


Figure 4.2 Soil profile for a. Row, b. Alley, c. Arable

4.2.2. Sampling strategy

Soil sampling was conducted in January 2024 from each pit. Samples were collected using a grid system within each pit, aligned with predefined soil sampling locations on the pit face. Soil cores were extracted horizontally from depths of 5-45 cm at 10 cm intervals using a cylindrical core sampler (5 cm diameter, 5.1 cm length; volume = 100 cm^3). This resulted in five nominal sampling depths per pit face (2.5–7.5, 12.5–17.5,

22.5–27.5, 32.5–37.5, and 42.5–47.5 cm) hereafter referred to as 5, 15, 25, 35, and 45 cm depths based on their central positions. To minimise soil compaction, which can significantly affect bulk density measurements, cores were taken diagonally across face of the soil pit to avoid repeated disturbance at the same location. Additional 100 cm^3 soil cores were collected from the same depth interval for root analysis. For the analysis of soil nutrients, including AN, P, SOC, and N%, loose soil was taken from the same locations and depth intervals.

A total of 60 soil samples for each determinant (5 depths × 4 pit faces × 3 locations) was initially targeted. However, sampling depth was constrained by the presence of limestone bedrock, which occurred between approximately 40 and 70 cm in the Alley and Row locations, and at shallower depths (less than 30 cm) in the Arable location. As a result, several samples could not be sampled: one from Alley (face 4, 35 and 45 cm depth), eleven from Arable (all faces, 25, 35 and 45 cm depths), and one from Row (face 4, 35 and 45 cm depth). Hence, a total of 47 soil samples were collected. Consequently, subsequent analysis considered a depth range of 5-45 cm for Alley and Row location and 5-25 cm for the Arable location. All soil samples were stored at 4 °C and analysed in the laboratory within one week of collection.

4.2.3. Soil analysis

Bulk density was calculated as the oven-dried mass of soil divided by the volume of the soil core, following the method described by Poeplau et al. (2017). Available nitrogen was measured by extracting fresh, moist soil with 1 M KCl at a 1:5 soil-to-solution ratio. The resulting extracts were analysed for nitrate (NO_3^-–N) and ammonium (NH_4^+–N) using a SKALAR SAN++ auto-analyser, following standard procedures outlined by (Carter and Gregorich,2008; Motsara and Roy, 2008). For the determination of available P (Olsen P), SOC and N%. Soil samples were oven-dried at 40 °C for 24 hours and subsequently passed through a 2 mm sieve. Available phosphorus was extracted using Olsen's extraction method (Olsen, 1954) and the phosphate (PO_4-P) content determined using a Skalar San ++ (Skalar Analytical B.V., Netherlands) continuous flow auto-analyser. Total SOC and N contents were quantified using a dry combustion elemental analyser (EuroEA 3000). To account for the presence of inorganic C associated with calcareous soils, SOC samples were pre-treated by sieving

to <100 µm, acidified with 30 µL of 15% HCl to remove inorganic C, and dried at 80 °C for 2 hours prior to analysis. Additional HCl was added where necessary due to a high carbonate content from underlying limestone.

SOC stock (Mg ha^{-1}) was calculated using the following equation:

SOC stock = SOC (%) * BD (bulk density) * depth (Equation 4.1)

where BD is expressed in g cm^{-3}, and depth in cm (10 cm intervals).

4.2.4. Root washing

Within three days of sample collection, roots were separated from soil using a washing and hand-sieving procedure (Suppl. Fig. B.1). Approximately 100–200 g of field-moist soil was placed in a tray and submerged in water. The soil–water mixture was gently agitated to disperse the soil and release root material. Floating roots and suspended organic matter were decanted onto a series of sieves with mesh sizes of 500 µm and 250 µm, followed by filtration through a 53 µm sieve. The washing and sieving process was repeated until no visible roots remained in the soil sample. Each sample was subsequently rinsed four times under running tap water to remove adhering mineral particles and organic debris. The selected sieve combination was based on preliminary trials, which demonstrated its effectiveness in removing soil particles while retaining intact roots. After washing, roots retained on the sieves, along with those floating in the water, were manually collected with tweezers (Frasier et al., 2016; Ristova and Barbez, 2018). Roots were subsequently separated into live and dead fractions based on morphological characteristics, such as colour and elasticity, following established criteria (Berhongaray et al., 2013; Brassard et al., 2013; Fiala et al., 2017; Fortier et al., 2019; Yang et al., 2010). All separated root samples were placed in tins, sealed in plastic bags and stored at 4 °C prior to root image scanning and subsequent C and N analysis.

4.2.5. Root image acquisition and interpretation

For root trait analysis, all root samples were scanned within 48 hours of washing using an EPSON scanner at a resolution of 400 dpi. Prior to scanning, samples were carefully spread out in a transparent tray filled with clear water and placed on the scanner glass to ensure proper separation and high-quality image acquisition (Slota et al., 2016). For large root samples, multiple images were taken, with roots evenly spread across the scanning area to avoid overlapping and maintain an appropriate root density (Ristova and Barbez, 2018).

Root diameters were measured using WinRHIZO 2020 software (Regent Instruments Inc., Canada) with the Regent method (WinRHIZO, 2020). However, due to the presence of heterogeneous root material, including broken and fine absorptive roots, WinRHIZO was unable to measure root length accurately. Therefore, ImageJ software was used to quantify root length (Pierret et al., 2013). Root length density was calculated as root length (cm) per unit soil volume (cm^3).

4.2.6. Root biomass and nutrients

Immediately after scanning, live and dead roots were air-dried for 2 days, then oven-dried at 80 °C for 24 hours. The dry weight of each sample was recorded to calculate root biomass. The oven-dried roots were then ground using hand homogenisers to a particle size of approximately 100 µm for C and N analysis via combustion using elemental analyser (EuroEA 3000).

For longer roots, samples were weighed directly into 5 × 5.5 mm tin capsules, using tweezers and scissors for handling and cutting. Ground root samples were similarly weighed into tin capsules before analysis. Root biomass density was calculated as oven-dry mass (mg) per unit soil volume (cm^3). Root C stock (g ha^{-1}) was determined by multiplying the root C concentration (g kg^{-1}) by the oven-dry mass (g) and scaling by the area represented by the soil core (ha) (Drexler and Don, 2024).

4.2.7. Statistical analysis

The Shapiro-Wilk test was performed to assess the normality of the soil and root datasets. The SOC stock and bulk density were found to be normally distributed. For variables that did not meet the assumption of normality, data transformations were applied to both soil (AN, P, SOC%, N%, SOC/N ratio) and root datasets. Log transformation successfully normalised several variables, including soil AN and SOC%, as well as root traits such as live root length density, live root biomass density, live root C stock, dead root length density, dead root nitrogen content, and dead root C/N ratio. However, other soil variables, such as P, N%, and C/N ratio, along with additional root traits, did not achieve normality despite applying different transformations.

As distributions of soil nutrient and root morphological trait data were approximately symmetrical around the mean, a two-way ANOVA was conducted to assess the effects of soil depth and location, followed by post hoc Tukey's tests or pairwise comparisons. Relationships between soil nutrient variables and root traits were assessed using Spearman's rank correlation analysis. Statistical significance was determined based on p-values obtained via the rcorr() function from the Hmisc package in R. Principal Component Analysis (PCA) was subsequently applied to identify patterns of variation and associations between soil nutrients and root morphological traits across soil depths. Prior to PCA, soil data were standardised using the scale() function in RStudio. RStudio version 4.3.3 was used to undertake all statistical analyses. All statistical analyses were conducted using RStudio (version 4.3.3). Hereafter, topsoil is considered 5-25 cm and subsoil 35-45 cm.

4.3. Results

4.3.1. Soil nutrients

Available nitrogen was highest in the topsoil layers (5-25 cm, 90.25%) and declined with increasing depth across all locations (Fig. 4.3a). Available nitrogen was similar at depths of 5 cm and 15 cm, showing a slight increase of 4.4%, but declined sharply by 80.9% between 15 cm and 25 cm. These results indicate that most of the AN is retained

within the upper 15 cm of the soil profile, with substantially lower concentrations at greater depth. A two-way ANOVA revealed a significant effect of soil depth on AN concentration ($F = 19.76$, $p < 0.001$), indicating substantial variation across soil layers (Suppl. Table B.1). In contrast, location had no significant effect ($F = 0.32$, $p = 0.728$), nor was there a significant depth × location interaction ($F = 0.81$, $p = 0.568$). Similarly, P concentrations were highest in the topsoil (90.02%) and decreased with depth. Two-way ANOVA indicated significant effects of both depth ($F = 28.28$, $p < 0.001$) and location ($F = 10.01$, $p < 0.001$), while the interaction effect was not significant ($F = 1.39$, $p = 0.240$) (Suppl. Table B.1). Mean P concentrations decreased by 55.9% from 5 to 15 cm and by 22.0% from 15 to 25 cm depth, with topsoil concentrations significantly greater than those at deeper depths (Tukey's test, $p < 0.001$; Suppl. Table B.1, Fig. 4.3b). At 5 cm depth, median P concentration followed the trend Arable>Row>Alleys (median = 41 mg kg^{-1}, 27 mg kg^{-1} and 24 mg kg^{-1}, respectively) (Suppl. Table B.1, Fig. 4.3b), while no significant differences were observed among locations in the deeper soil layers (Suppl. Table B.1).

Bulk density was significantly affected by location ($F = 5.33$, $p = 0.010$), but not by depth ($F = 1.80$, $p = 0.151$) or the interaction between factors ($F = 0.48$, $p = 0.818$) (Suppl. Table B.1, Fig. 4.3d). Tukey's test showed that bulk density was significantly higher in Arable compared with both Alley and Row.

Soil organic carbon concentration varied significantly with soil depth ($F = 2.70$, $p = 0.047$), whereas neither location ($F = 0.51$, $p = 0.607$) nor the depth × location interaction ($F = 0.55$, $p = 0.768$) had a significant effect. Although the difference was only marginally significant ($p = 0.090$), post hoc Tukey comparisons showed higher SOC concentrations at 45 cm than at 15 cm (Fig. 4.3e), with increases of 29.9% in the Alley and 14.4% in the Row. In contrast, N% and the C/N ratio showed no significant effects of depth, locations, or their interaction (Fig. 4.3f, g).

Soil organic carbon stock was not significantly affected by location ($F = 1.36$, $p = 0.271$) or the depth*location interaction ($F = 0.67$, $p = 0.678$) but showed a marginal effect of depth ($F = 1.03$, $p = 0.070$) (Suppl. Table B.1; Fig. 4.3c). Although the post hoc Tukey test revealed only a marginally significant depth effect (15 vs. 45 cm), SOC

stocks at 45 cm were 67.8% and 48.4% greater than those at 15 cm in the Alley and Row locations, respectively.

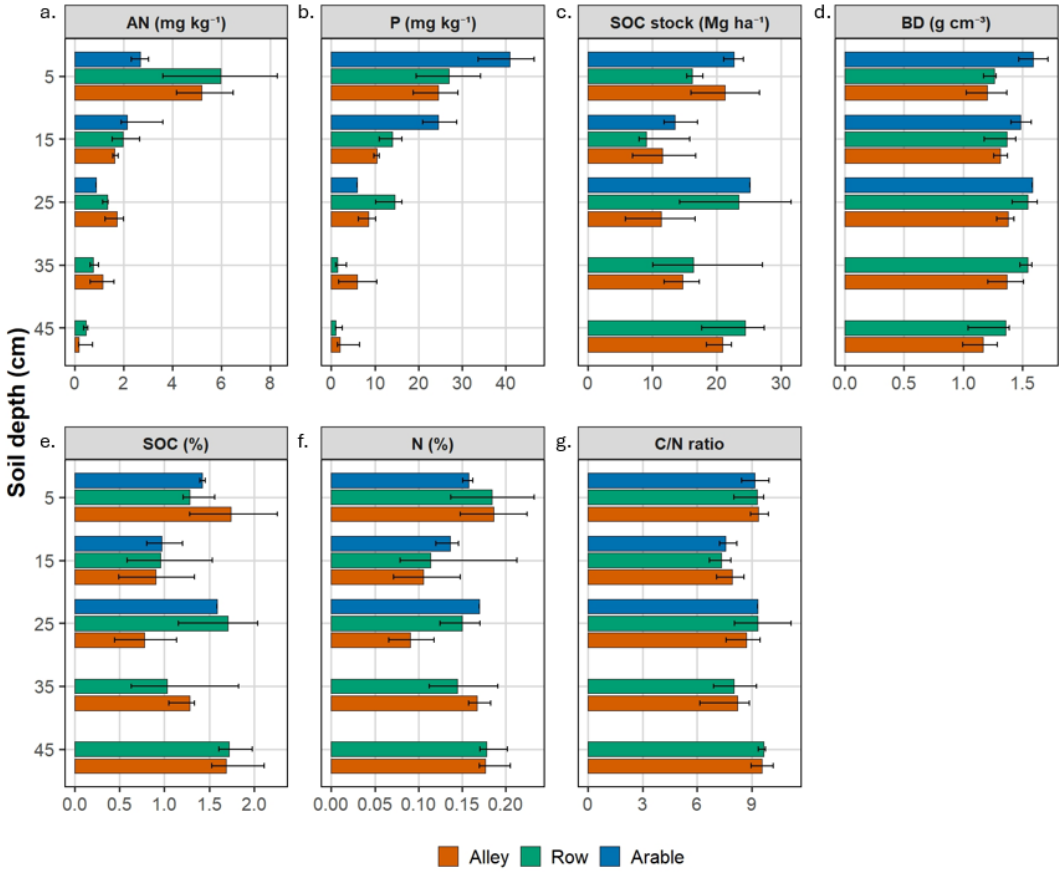


Figure 4.3 Variation in soil properties with depth and location: (a) Available nitrogen (AN), (b) Olsen phosphorus (P), (c) SOC stock, (d) bulk density, (e) SOC%, (f) N%, (g) C/N ratio, at five soil depth intervals under three locations. Bar show the median (n=4), error bars indicate the interquartile range (25th and 75th percentiles)

4.3.2. Root traits

Both soil depth and location had significant effects on live root length density, live root biomass density, and live root C stock (Fig. 4.4 b, c, and f), whereas their interaction was not significant. This indicates that location effects were consistent across depths. Tukey's HSD test revealed significant pairwise differences among depth intervals (5, 15, 25, and 35 cm; $p < 0.001$) and between locations, specifically Arable–Alley and Row–Arable ($p < 0.001$) for both live root length density and live root biomass density (Suppl. Table B.2). Across locations, mean live root length density and mean live root

biomass density were highest in the Alley, followed by the Row and lowest in the Arable, especially at 5 cm depth.

For live root C stock, significant differences were primarily observed between the 5 cm layer and the deeper soil layers (35 cm and 45 cm), with location contrasts similar to those observed for live root length density and live root biomass density (Suppl. Table B.2). As shown in Fig. 4.4f, the highest live root C stock occurred in Alley (2.62 Mg ha^{-1}), followed by Row (1.29 Mg ha^{-1}), while Arable had the lowest value (0.12 Mg ha^{-1}) at 5 cm depth. Relative to the 5 cm depth, only 9.92% and 5.43% of the live root C stock remained at 45 cm in Alley and Row, corresponding to declines of 90.08% and 94.57%, respectively.

For live root N, soil depth ($F = 3.28$, $p = 0.024$), locations ($F = 7.56$, $p = 0.002$), and their interaction ($F = 3.78$, $p = 0.006$) all had statistically significant effects (Fig. 4.4d). Tukey's post-hoc test revealed significant location differences at specific depths, particularly between Arable and Alley ($p < 0.01$) and between Row and Arable ($p < 0.001$) within the 5 and 15 cm soil layer. Additional significance contrasts were observed among depth-location combinations (e.g., 15 cm: Arable vs. 5cm: Alley, 45cm: Rows vs. 15cm: Arable; $p < 0.001$) (Fig. 4.4d, Suppl. Table B.2). Live root N concentrations were highest in Arable within the upper soil layers (1.84% at 5 cm and 3.33% at 15 cm) compared with Alley and Rows. However, vertical patterns varied among locations, with live root N showing an irregular distribution across the soil profile and a slight increase with depth in the Row location.

In contrast, live root C concentration was not significantly influenced by soil depth ($F = 2.43$, $p = 0.070$) or location ($F = 1.89$, $p = 0.168$) (Fig. 4.4e). However, live root C/N ratio was significantly affected by location ($F = 9.60$, $p < 0.001$), whereas neither depth ($F = 1.94$, $p = 0.130$) nor the interaction between factors ($F = 1.52$, $p = 0.206$) had a significant effect (Fig. 4.4g). Tukey's post hoc test confirmed significant differences between locations, particularly between Arable and Alley, and between Row and Arable ($p < 0.001$). The live C/N ratio exhibited location-specific patterns, with the highest values observed in the Row at 15 cm depth and in the Alley at 5 cm depth, followed by a general decline with increasing soil depth.

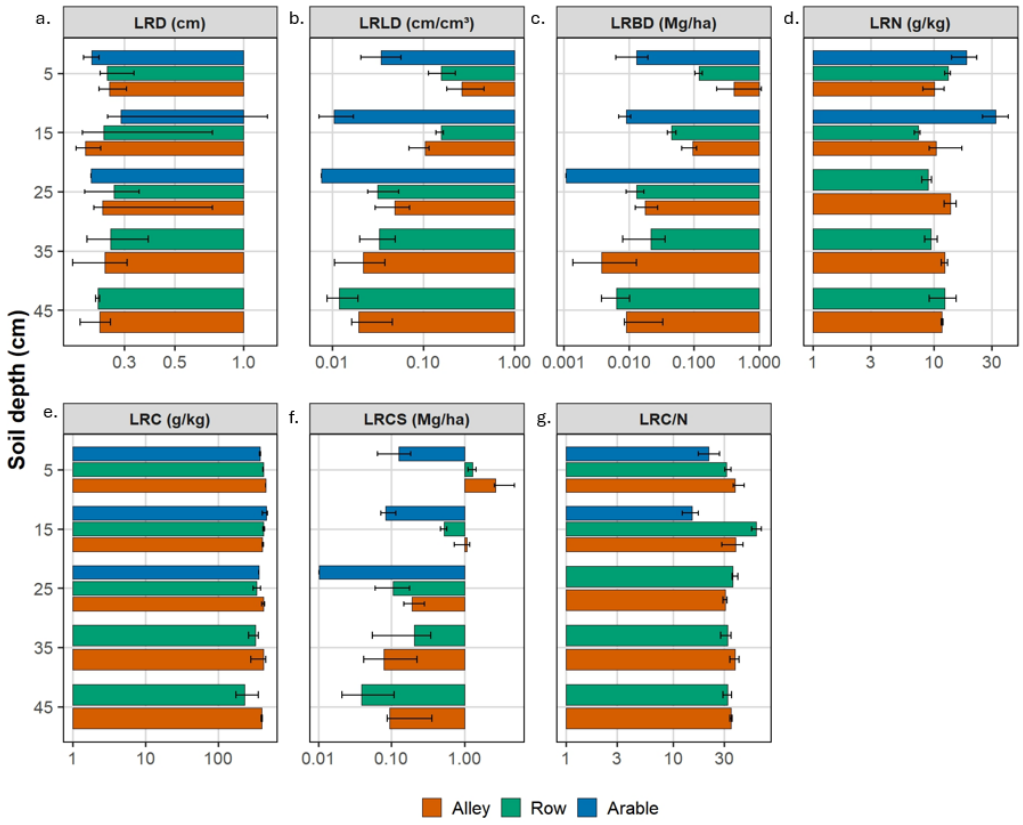


Figure 4.4 Variation in live root traits with soil depth and location: (a) live root length diameter (cm), (b) live root length density (LRLD), (c) live root biomass density (LRBD), (d) live root N (LRN), (e) live root C (LRC), (f) live root C stock (LRCS), (g) live root C/N ratio (LRCN) at five soil depth intervals and at three locations. Bars represent median values and error bars indicate the interquartile range (25th to 75th percentiles)

Finally, ANOVA revealed no significant effects of soil depth, location, or their interaction on either live or dead root diameter (Fig. 4.4 and 4.5a). While Tukey's post hoc test indicated significant pairwise differences between specific location combinations, most notably between Arable and Alley and between Row and Arable ($p < 0.001$), these differences should be interpreted with caution given the absence of significant main effects or depth × location interactions in the overall ANOVA.

Soil depth ($F = 6.48$, $p < 0.001$) and location ($F = 32.90$, $p < 0.001$) had significant effects on dead root length density, whereas the interaction between these factors was not significant ($F = 0.66$, $p = 0.683$) (Fig. 4.5b; Suppl. Table B.3). The highest dead root length density was observed in Row (0.36 cm cm-3), followed by Alley (0.14 cm/³), whilst the lowest occurred in Arable within the 5 and 15 cm soil layer (Fig. 4.5b; Suppl.

Table B.3). Tukey's post hoc test showed significant differences between deeper soil layers (25–35 cm and 35- 45 cm), and among locations, particularly between Alley and Arable and between Row and Arable ($p < 0.001$, Fig. 4.5b, Suppl. Table B.3). The interaction between depth and locations was not significant ($F = 0.66$, $p = 0.683$).

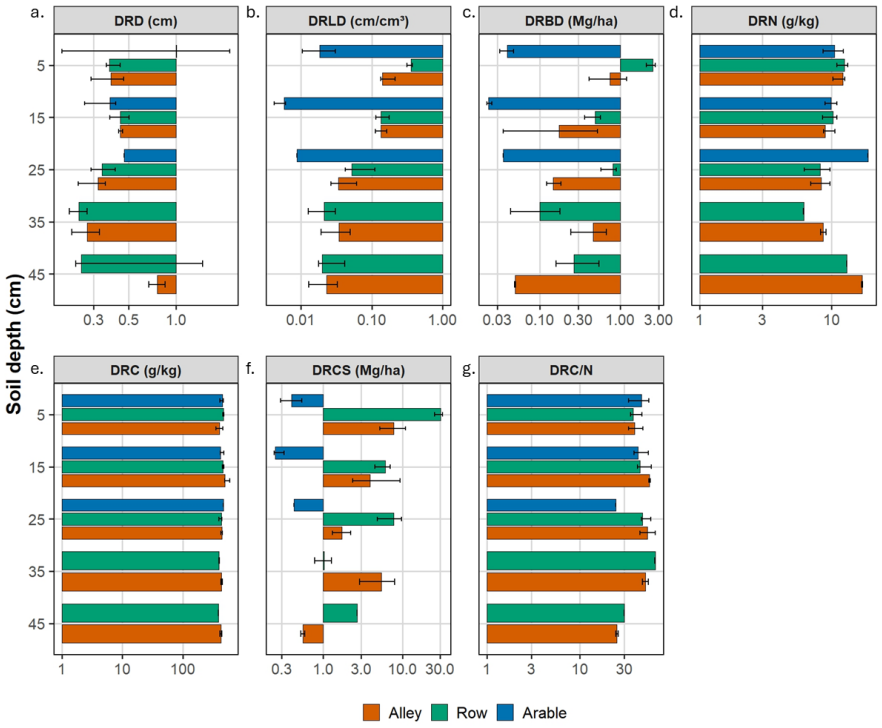


Figure 4.5 Variation in dead root traits with soil depth and location: (a) dead root diameter (DRD), (b) dead root legnth density (DRLD), (c) dead root biomass density, (d) dead root N (DRN), (e) dead root C (DRC), (f) dead root C stock (DRCS), (e) dead root C/N ratio (DRCN) measured at five soil depth intervals and at three locations. Bar show the median (n=4), and error barss indicate the interquartile range.

Similarly, dead root biomass density was significantly influenced by both soil depth ($F = 6.11$, $p = 0.001$) and locations ($F = 15.94$, $p < 0.001$), with a significant depth x location interaction ($F = 3.89$, $p = 0.005$) (Fig. 4.5c, Suppl. Table B.3), indicating that location effects varied across the soil profile (Fig. 4.5c; Suppl. Table B.3). Relative to the surface layer (5 cm), only 17.03% and 5.68% of dead root biomass remained at 45 cm in Row and Alley, representing declines of 82.97% and 94.32%, respectively. This indicates a steeper reduction of dead root biomass with depth in Alley than in Row ($p < 0.001$, Fig. 4.5c). Significant differences were observed among all locations; comparisons were significant, with the largest differences between Rows at 5 cm and

Arable at 25 cm depth. Several depth-location contrasts were also significant (e.g., Row at 5 cm depth and Alley at 15 cm depth, and Row at 15 cm depth and Alley at 5 cm depth (Suppl. Table B.3).

Dead root C concentration was not significantly affected by soil depth ($F = 2.02$, $p = 0.122$), locations ($F = 0.81$, $p = 0.456$), or their interaction ($F = 1.10$, $p = 0.388$) (Fig. 4.5e; Suppl. Table B.3). In contrast, dead root C stock was significantly influenced by soil depth ($F = 5.53$, $p = 0.002$), location ($F = 16.67$, $p < 0.001$), and the depth x location interaction ($F = 4.63$, $p = 0.003$) (Fig. 4.5f). Relative to a depth of 5 cm, dead root C stock in Alley declined by 23.1%, 78.2%, and 93.5% at 15, 25, and 45 cm, respectively, while in Row it declined by 80.0%, 74.9%, and 90.0%. Tukey's post hoc test showed that dead root C stock was significantly higher at 5 cm than at 15, 25, and 45 cm ($p < 0.001$). particularly evident in the Alley and Row locations (Suppl. Table B.3). Significant differences were observed among all locations, following the trend: Arable < Alley < Row, with the strongest positive effect in the Row (Fig. 4.5f). Furthermore, several significant depth × location interactions were observed, particularly within the Row soil profile (e.g., 5 cm: Rows vs. 25 cm: Alleys, 35 cm: Alley, and 5 cm: Arable; $p < 0.001$) (Fig. 4.5f), highlighting pronounced location effects at specific depths.

Soil depth also had a significant influence on dead root N concentration ($F = 3.75$, $p = 0.016$), whereas location ($F = 1.00$, $p = 0.383$) and the depth × location interaction ($F = 1.54$, $p = 0.205$) were not significant (Fig. 4.5d; Suppl. Table B.3). Tukey's post hoc test revealed significant differences between 45 and 15 cm, and between 25 and 45 cm ($p < 0.001$) soil depths. Dead root N generally increased with depth, with the highest values observed at 45 cm in Alley and Row, and at 25 cm for Arable. At 45 cm, it was 72.7% higher in Alley and 40.1% higher in Row than at 15 cm depth (Fig. 4.5d).

Similarly, the dead root C: N ratio decreased with soil depth across all locations, reflecting a significant main effect of depth ($F = 3.32$, $p = 0.026$). In contrast, neither location ($F = 0.98$, $p = 0.391$) nor the depth x location interaction ($F = 0.76$, $p = 0.608$) significantly affected dead root C:N ratio (Fig. 4.5g, Suppl. Table B.3). These results indicate that variation in dead root stoichiometry was primarily driven by vertical changes within the soil profile with differences among locations.

4.3.3. Relationships between soil properties and root traits

Live and dead root traits exhibited significant relationships with several soil properties (Fig. 4.6). Overall, the correlation matrix identified 11 significant associations between soil characteristics and root morphological and biochemical traits, highlighting the important role of soil nutrient availability and physical conditions in shaping belowground plant development (Fig. 4.6c). The scatterplots further illustrate that AN and BD were the primary factors influencing root abundance-related traits (Fig. 4.6a, b).

Among the live root traits, live root length density, live root biomass density, and live root C stock showed strong responses to both AN and BD. Live root length density was positively correlated with AN ($\rho = 0.552$, $p < 0.001$; Fig. 4.6b) and P ($\rho = 0.352$, $p < 0.001$) but negatively correlated with BD ($\rho = –0.296$, $p = 0.048$; Fig. 4.6a). Similarly, live root biomass density exhibited a positive correlation with AN ($\rho = 0.498$, $p < 0.001$; Fig. 4.6b) and a negative correlation with BD ($\rho = –0.373$, $p = 0.010$; Fig. 4.6a). Likewise, live root C stock increased with increasing AN ($\rho = 0.466$, $p < 0.001$; Fig. 4.6b) and decreased with increasing BD ($\rho = –0.331$, $p = 0.030$; Fig. 4.6a).

In contrast, dead root traits displayed weaker and more variable relationships with soil properties (Fig. 4.6a–c). Although some negative trends with BD and positive trends with AN were evident, these relationships were generally less pronounced than those observed for live roots. In addition to the relationships among soil properties and nutrients, a strong correlation was observed between dead root N and dead root C/N ratio ($\rho = –0.92$, $p <0.001$). This suggests that the distribution of dead roots is influenced not only by current soil conditions but also by root turnover, mortality, and decomposition processes (Fig. 4.6).

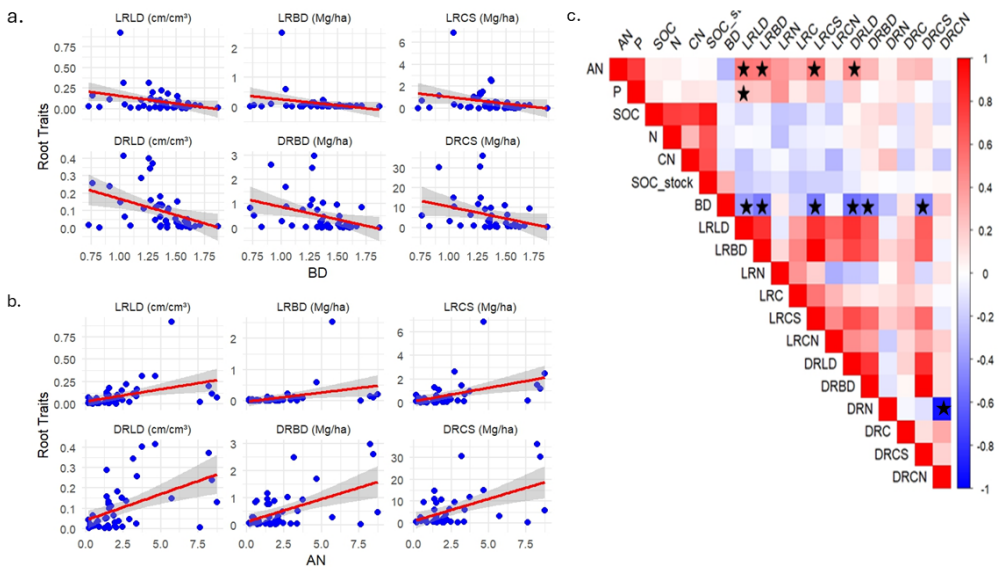


Figure 4.6 Spearman correlation analysis of soil properties and root traits: (a) BD versus live and dead root traits. (b) AN versus live and dead root traits, showing the most influential and abundant traits based on the correlation matrix. (c) Spearman's rank correlation matrix between soil properties and root traits. Colours indicate correlation coefficients from −1 (blue) to +1 (red), and black stars denote significant correlations ($p < 0.05$)

Principal component analysis (PCA) was used to explore the multivariate relationships among soil nutrients and root traits, and to summarise the overall variability in the dataset. For soil properties (Fig. 4.7a), the first two principal components explained 66.5% of the total variance, with PC1 accounting for 38.0% and PC2 for 28.5%. SOC% and SOC stock were strongly aligned and contributed primarily to PC1, indicating that C-related variables were major drivers of variation in soil characteristics. Variables such as C/N ratio, AN, and P contributed to both principal components, reflecting their influence across multiple environmental gradients. In contrast, soil depth and BD were negatively associated with SOC%, SOC stock, N%, and soil C/N ratio, suggesting an inverse relationship between soil compaction and increasing depth on one hand, and C and nutrient-rich conditions on the other.

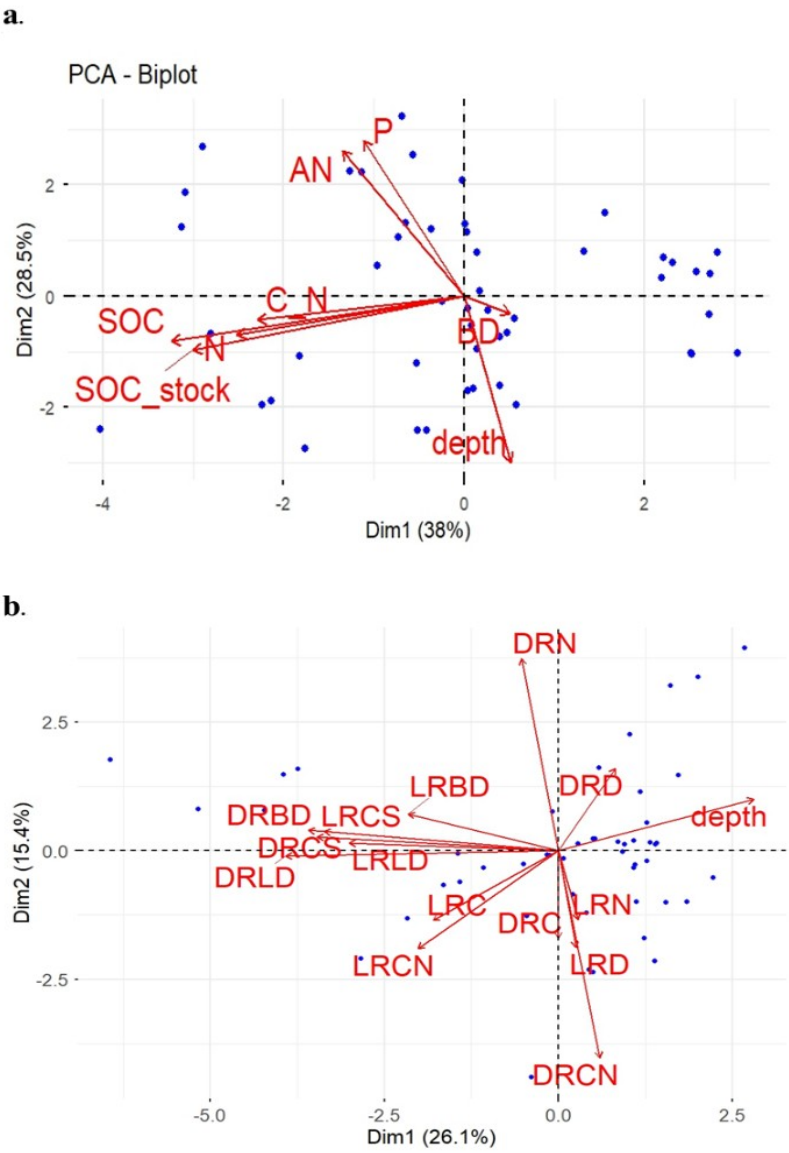


Figure 4.7 PCA biplot showing the distribution of (a) soil variables: SOC, SOC stock, N, AN, P, bulk density, and soil depth along the first two principal components (Dim1: 38%, Dim2: 23.5%) and (b) live and dead root variables: live root length density, live root biomass density, live root C stock, live root N, live root C, live root C/N, live root diameter, dead root length density, dead root biomass density, dead root C stock, dead root N, dead root C, dead root C/N, dead root diameter with soil depths (Dim1: 26.3%, Dim2: 15.4%). Red vectors represent the direction and magnitude of each variable's contribution, with blue points indicating individual observations; variables pointing in the same direction are positively correlated.

For root traits (Fig. 4.7b), PC1 accounted for 26.3% of the total variance, and PC2 explained 15.4%, together accounting for 41.7% of the overall variability. Several live root traits, including live root C, live root C/N, live root biomass density, live root

length density, and live root C stock, were strongly aligned along the principal axes, indicating a strong positive association among these variables. Dead root traits such as dead root biomass density, dead root length density, and dead root C concentration also clustered with this group, together with live root N concentration, suggesting coordinated patterns of variation in root morphology and chemistry. In contrast, live and dead root diameter, dead root C/N ratio, and soil depth were negatively associated with the main cluster of root traits, indicating contrasting functional and structural patterns with the root system (Fig. 4.7b). Overall, the PCA highlights clear gradients in both soil and root trait distributions. The scree plots (Suppl. Fig. B.2ab) supported the retention and interpretation of the first two principal components, showing that explained variance declined substantially after PC2, with an evident elbow around PC2-PC3.

4.4. Discussion

This study demonstrates that silvoarable agroforestry systems generate pronounced spatial heterogeneity in soil properties and root functional traits, with important implications for nutrient dynamics, belowground resource dynamics and C storage. Across the Row, Alley, and Arable locations, clear patterns emerged in nutrient and C distribution, root biomass allocation, and root–soil interactions, highlighting the capacity of silvoarable systems to modify biogeochemical processes relative to conventional arable management.

4.4.1. Soil nutrient stratification and management effects

Soil properties exhibited clear vertical stratification across all locations. Soil organic carbon, AN, and P were most abundant in the surface soil (5 cm) and generally declined with depth, whereas BD increased with depth, reflecting higher compaction in deeper soil layers (Fig. 4.3a-g). Although these depth-related trends were consistent across locations, their magnitude varied according to land use and management practices. PC1, which explained 33.2% of the total variance, was primarily driven by SOC (-2.74), SOC stock (-2.54), N (-2.13), and C/N ratio (-1.94) (Fig. 4.7). This indicates that differences in soil C and nutrient dynamics played a key role in shaping the vertical and spatial distribution of soil resources (Judson et al., 2025). Available nitrogen declined significantly with increasing soil depth across all locations (Fig. 4.3a). While

concentrations were generally similar in the Row and Alley and were lower in the Arable (Suppl. Table B.1), overall depth effects were substantially larger than location effects. The decrease in AN with depth may support results from previous investigations suggesting that N availability is preferential at surface soils, where nutrient cycling, organic matter input, litter decomposition, microbial activity and microbial composition are most significant (Kim and Isaac, 2022; Tonucci et al., 2023; Judson et al., 2025). The presence of litter and root biomass, along with organic matter accumulation in the upper soil layers of silvoarable systems, can promote biological activity and N cycling, thereby ensuring higher N availability near the soil surface (Kim and Issac, 2022). Thus, the vertical distribution of AN reflects the extent of organic matter inputs and nutrient turnover within the soil profile, with the highest concentrations occurring near the surface and declining with increasing depth.

In contrast to AN, N% did not vary significantly with either depth or location (Fig. 4.3a and f), indicating a relatively stable N pool throughout the soil profile and across locations. This divergence between AN and N% highlights the distinction between labile and more stable N fractions, suggesting that management practices primarily influence N availability rather than the total amount of N stored in the soil (Kaye et al., 2003). These findings indicate that N cycling processes are more dynamic in surface soil layers, where biological activity is greatest, while deeper layers maintain a more conserved N pool.

Phosphorus availability exhibited a clear spatial gradient across the study locations, with concentrations decreasing in the order Arable > Alley > Row. Significant differences were observed between the Arable location and both silvoarble locations (Alley and Row) (Figure 4.3b; Suppl. Table B.1). This variation probably reflects the legacy effects of historical fertilisation at the Arable location, where P inputs can persist in soils for extended periods because of the relatively low mobility of P compared with other nutrients. Previous studies have shown that fertiliser-derived P applied can accumulate within soil and remain available for many years, contributing to elevated P concentrations in arable soils (Johnston et al., 2014; Solangi et al., 2023). However, the relatively low P availability observed in the silvoarable system might indicate a reduced supply of fertiliser-derived P, especially in tree-based (Row) systems for increased plant uptake. But this finding is contrary to some of the results of Judson et al. (2025), who

reported higher P concentrations in both Alley and Row positions than in their corresponding control location. These contrasting findings between studies suggest that P dynamics in agroforestry systems are highly site-specific and influenced by factors such as management history, fertiliser legacy, tree species composition, vegetation cover, and nutrient redistribution processes. In the present study, the long-term reduction in cropping and fertiliser inputs in Alley, together with sustained nutrient uptake by trees and associated vegetation, may have ultimately contributed to lower P availability compared to Arable.

Alley recorded the highest SOC% at 5 cm depth, followed by Row and Arable, with relatively small variation across soil depths (Fig. 4.3c, Suppl. Table B.1). These patterns likely reflect differences in root activity, organic matter inputs, and microbial processes, which are influenced by vegetation structure and management history (Coleman et al., 2000; Berhongaray et al., 2013). While no significant differences in SOC% and SOC stock were observed among locations in the present study (Suppl. Table B.1), the spatial distribution of SOC stock suggests that the silvoarable system may play an important role in soil C cycling, particularly when compared with the adjacent arable system (Fig. 4.3c). This interpretation is consistent with the findings reported in Chapter 3, where SOC concentration and stock differed significantly between the silvoarable and arable systems, although no significant differences were detected between the tree row and alley positions within the silvoarable system. In established silvoarable systems, similar spatial patterns have been reported, with SOC stock often concentrated in tree-influenced areas due to greater litter inputs, enhanced root turnover, and reduced soil disturbance (Cardinael et al., 2015; Lorenz and Lal, 2014; Udawatta and Jose, 2011). Together, these findings suggest that trees enhance localised above- and belowground C inputs, contributing to spatial variation in soil C retention and nutrient cycling (Jose et al., 2004; Nair et al., 2009a). Furthermore, cropping in Alley ceased in 2004, allowing natural vegetation to establish. This transition in vegetation composition is likely to have altered both the quantity and distribution of below-ground C inputs, while changes in rooting distribution may have affected nutrient cycling and soil organic matter turnover. Consequently, contemporary differences in SOC and nutrient dynamics within the Alley and Row may reflect not only tree effects but also the legacy influences of vegetation succession and changes in land management.

4.4.2. Root trait variation and belowground resource acquisition through root soil interactions

Root distribution with soil depth is largely controlled by nutrient availability, water accessibility, and physical constraints imposed by the soil environment (Bardgett et al., 2014). In the present study, root length density and biomass density declined with increasing depth across all locations, corresponding with reductions in AN, P, and SOC (Fig. 4.3-4.6). This pattern suggests that roots were concentrated in nutrient-rich surface horizons where resource acquisition is most efficient. The co-occurrence of higher root densities and elevated nutrient concentrations in surface soils indicates that nutrient availability may have been an important driver of rooting distribution. Similar vertical rooting patterns have been widely reported in agroforestry and agricultural systems, where fine roots are concentrated in upper soil layers to maximise nutrient uptake and exploit organic matter-rich layers (Pilon et al., 2013; Chen et al., 2013ab; Drexler et al., 2023).

By contrast, P appeared to influence the spatial distribution of roots more strongly than overall root abundance. Although live root length density was positively correlated with available P, the relationship was weaker than that observed for AN (Fig. 4.6c), suggesting that P was not the primary driver of root biomass accumulation across locations. Instead, plants may have responded to P limitation through changes in root architecture and nutrient foraging strategies (Lu et al., 2024). The greater root proliferation observed between the Row and Alley indicates that roots explored larger soil volumes to compensate for localised nutrient limitations, a response consistent with the findings of Forde and Lorenzo (2001), who demonstrated that root architecture is highly responsive to nutrient heterogeneity. Furthermore, P concentrations declined markedly with soil depth, reinforcing its stratification within surface soil horizons. Given the low mobility of P in soils, efficient uptake depends on the development of dense surface-root networks capable of exploiting nutrient-rich topsoil layers (Schachtman et al., 1998). Together, these findings suggest that nutrient stratification contributed to the high root densities observed in surface soils.

Significant variation in root N concentrations among locations and depths further highlights the influence of soil nutrient dynamics on root functioning (Fig. 4.4 and 4.5).

Live root N was greatest at 15 cm in Arable, whereas dead root N peaked at 25 cm depth (Suppl. Table B.2, B.3). These patterns likely reflect differences in historical fertiliser inputs, decomposition processes, and microbial activity, all of which can influence root nutrient content and turnover rates (van Noordwijk et al., 1995; Huang et al., 2021). Fertiliser-derived N can modify root mortality, microbial community structure, nutrient cycling processes, and competition for belowground resources, resulting in heterogeneous patterns of root nutrient distribution through the soil profile (Nadelhoffer, 2000; Treseder, 2008).

In contrast, root diameter remained relatively stable or increased slightly with depth (Fig. 4.4 and 4.5), suggesting a shift in root function from nutrient acquisition in surface soils to resource transport and structural support in deeper horizons (McCormack et al., 2012). Deeper soil layers are typically characterised by lower nutrient availability, greater mechanical resistance, and reduced biological activity, favouring the development of thicker, longer-lived roots that facilitate water uptake and resource transport (Geng & Jin, 2022; Gao et al., 2023). The persistence of larger-diameter roots at depth may therefore represent an adaptive strategy that enables plants to access subsoil resources while reducing the costs associated with high rates of root turnover (Pregitzer et al., 2002; McCormack et al., 2012; Lynch, 2019).

Spatial variation in root distribution was evident across the silvoarable system. Live roots were strongly concentrated within the topsoil (5-25 cm) across all locations, accounting for 90.4%, 88.9%, and 100% of live root length density in the Alley, Row, and Arable, respectively (Fig. 4.4, Suppl. Table B.2). Similar patterns were observed for live root biomass density and root C stocks, with 86-99% occurring in the topsoil (5-25 cm, Table B.2). These values exceed the 50-80% range commonly reported for fine roots within the upper 30 cm of soil and are greater than the 51% fine-root biomass C reported by Drexler et al. (2024) for hedgerows in Germany, indicating a particularly strong concentration of roots in surface soil layers. Surface-root distribution likely reflects greater nutrient availability and biological activity in the topsoil (Keiblinger et al., 2023).

Dead roots were also predominantly concentrated in the topsoil; however, in the Row, dead root biomass density (89.16%) and dead root C stocks (97.07%) exceeded those of

live roots (86.34% and 88.29%, respectively). This pattern suggests greater root turnover and accumulation of root-derived residues, which may enhance nutrient recycling and contribute to the formation and stabilisation of soil organic C through microbial processing and mineral associations (Riley et al., 2009; Rasse et al., 2005). Despite the predominance of roots in surface soils, the Alley and Row maintained live roots in the subsoil, accounting for 9.6% and 11.1% of root length density, respectively, whereas no live or dead roots were detected below the topsoil in the Arable. This finding suggests that tree integration promoted deeper rooting and greater utilisation of subsoil resources, a characteristic commonly associated with silvoarable agroforestry (Upson and Burgess, 2013; Thorup-Kristensen et al., 2020).

No significant relationship was observed between root distribution and soil C stocks. This finding supports previous findings that root abundance alone is a poor predictor of vertical SOC distribution, which is also regulated by microbial transformations, mineral associations and C stabilisation mechanisms (Rasse et al., 2005; Lorenz and Lal, 2014). Nevertheless, the presence of both live and dead roots at depth may indicate a greater potential for long-term C sequestration potential in the silvoarable system than in the conventionally managed Arable location, as deeper C inputs are generally subject to slower turnover and may contribute disproportionately to persistent soil C pools.

Correlation analysis further demonstrated that live root length density and biomass density were strongly and positively associated with AN (Fig. 4.6 c), indicating that N availability was a key driver of belowground growth. These root traits were also negatively associated with BD, suggesting that both nutrient supply and soil physical structure influenced rooting patterns. Similar relationships have been widely reported, whereby higher AN has been shown to stimulate root proliferation, increase biomass allocation, and enhance nutrient acquisition efficiency (Poorter et al., 2012; Giehl and von Wirén, 2014). The strong positive correlations observed between live root traits and AN (Fig. 4.6 a and c), together with their negative relationships with BD (Fig. 4.6b and c), therefore suggest that both nutrient availability and soil physical structure played important roles in determining root distribution patterns. Although, low N availability in poplar-based silvoarable systems has been shown to induce various adaptive responses, including increased fine root growth and enhanced N uptake efficiency (Coleman &

Aubrey, 2018), responses are often context dependent and influenced by species composition, and resource availability (Poorter et al., 2012).

Overall, these findings demonstrate that mature silvoarable agroforestry system create a more heterogeneous root distribution both vertically and horizontally compared with conventional Arable location. Greater root presence in subsoil layers and distinct root dynamics within tree-influenced zones indicate that trees substantially modify belowground resource acquisition and allocation patterns. These changes highlight the important role of agroforestry in influencing soil nutrient cycling and C dynamics in temperate agricultural landscapes.

4.3.3. Root–soil interactions and implications for C storage

The relationships between root traits and soil properties indicate that roots both respond to and influence soil conditions. Significant negative correlations between BD and live and dead root length density, biomass density, and root C stocks (Figs. 4.6 and 4.7) suggest that greater root abundance contributes to improved soil structure through enhanced porosity, aggregation, and the incorporation of root-derived organic matter. In the silvoarable system, these effects are likely reinforced by reduced soil disturbance and continuous organic matter inputs. Similar relationships between root abundance, improved soil structure, and reduced compaction have been reported in other agroforestry systems (Tracy et al., 2013; Correa et al., 2019; He et al., 2022).

Root traits were strongly interrelated, particularly live and dead root length density, biomass density, and associated C stocks, indicating coordinated belowground resource allocation and C cycling (Fig. 4.7). Live roots were most abundant in the Alley, suggesting greater inputs of C through root growth, turnover, and rhizodeposition, whereas the greater abundance of dead roots in the Row indicates a larger contribution of root-derived organic matter through decomposition. These contrasting distributions suggest that Alley and Row contribute to belowground C cycling through complementary pathways.

Although SOC stocks declined with depth, both live and dead roots were present throughout the soil profile, demonstrating continued inputs of root-derived C

belowground. Relationships between root traits and SOC stocks further suggest that root abundance and turnover contribute to soil C dynamics within the system. However, Chapter 3 found no significant effect of location on SOC stocks at 15-30 cm, despite the clear spatial differences in root distribution observed between Row and Alley in present study. This indicates that while root-derived C inputs are associated with SOC storage, spatial variation in root abundance has not yet resulted in distinct SOC stocks among locations. The absence of a location effect may reflect the relatively slow accumulation and stabilisation of soil C compared with the more dynamic processes of root growth, turnover, and decomposition.

The accumulation of dead roots within the upper 5 cm of the Row may provide an important source of root necromass for SOC formation, whereas the greater abundance of live roots in the Alley suggests more active rhizosphere processes and continuous C inputs through root turnover and exudation (Liang et al., 2017; Sokol et al., 2019). These findings indicate that different zones within the agroforestry system contribute to soil C cycling through distinct but complementary mechanisms, with dead roots acting as a source of organic matter inputs and live roots sustaining ongoing C fluxes belowground.

Live and dead roots perform complementary functions in regulating soil processes. Live roots facilitate nutrient and water uptake while stimulating microbial activity through root exudation, whereas dead roots contribute to nutrient recycling, organic matter formation, and C storage through decomposition and incorporation into soil organic matter (Ehrenfeld et al., 1997). Overall, the combined findings from Chapters 3 and 4 suggest that mature silvoarable agroforestry systems influence belowground C dynamics. While significant differences in SOC stocks among locations were not detected, relationships between root traits and SOC highlight the important role of both live and dead roots in shaping soil structure, nutrient cycling, and C dynamics within the system.

4.4. Implications for agroforestry systems

The findings of this study highlight the important role of mature silvoarable agroforestry systems in regulating belowground C dynamics, nutrient cycling, and soil physical properties. The greater abundance of both live and dead roots in agroforestry zones indicates enhanced C inputs to the soil through root growth, turnover, and decomposition. In particular, the accumulation of dead root biomass within Row, suggests increased inputs of root-derived organic matter, which may contribute to necromass formation and the stabilisation of SOC. At the same time, greater live root abundance in the Alley supports active nutrient uptake, rhizosphere processes, and continuous C inputs through root exudation. These biological processes are further facilitated by lower BD in agroforestry zones, which enhances soil porosity, facilitates root growth, and promotes the movement of water, air, and nutrients through the soil.

The presence of live and dead roots below the topsoil in both the Row and Alley, combined with the absence of roots in deeper soil layers under the Arable location, suggests that tree integration extended rooting depth and increased utilisation of subsoil resources. This deeper rooting may also increase the delivery of C to deeper soil horizons, where turnover rates are typically slower and the potential for long-term C stabilisation is greater. Consequently, silvoarable systems may enhance not only the quantity of belowground C inputs but also their potential persistence within the soil profile.

Collectively, these findings suggest that mature silvoarable agroforestry systems develop multifunctional rooting networks that combined nutrient-foraging in surface soils with nutrient acquisition and transport functions at depth. The spatial heterogeneity generated by tree–crop interactions appear to promote complementary resource use, improve nutrient cycling, and enhance soil structural conditions. s These processes may contribute to greater ecosystem resilience and increased C sequestration potential relative to conventional arable management. This highlights the value of silvoarable systems as a promising sustainable land-use strategy for maintaining soil health, improving resource-use efficiency, and contributing to climate change mitigation through increased belowground C storage.

4.5. Limitations and future research directions

A key limitation of this study is the limited spatial replication, with only one soil pit sampled per location. Furthermore, it was not possible to distinguish between tree and grass roots; therefore, the results represent the combined root system within each sampling location. Similar difficulties in separating tree and understory roots have been reported in temperate silvoarable agroforestry systems (Upson and Burgess, 2013). Consequently, caution should be exercised when interpreting species-specific root contributions and extrapolating the findings beyond the sampled locations.

A further limitation was the presence of shallow limestone bedrock, which restricted the maximum sampling depth, particularly in the Arable location. This constrained the comparability of soil profiles among locations and may have limited the assessment of deeper root distributions and subsoil C storage. Given the importance of deep roots in belowground C sequestration, a greater sampling depth would provide a more comprehensive understanding of C dynamics throughout the soil profile.

Coarse roots were not investigated in the present study because very few were recovered from soil samples collected at the 5 and 45 cm depth in the Alley and Row. This constitutes a limitation of the study, as coarse roots can contribute to belowground C inputs through root turnover, decomposition, and rhizodeposition, thereby potentially influencing the formation and stabilisation of mineral-associated organic carbon (MAOC). Given that MAOC represents a relatively stable soil C pool, the contribution of coarse roots to MAOC dynamics warrants further investigation in future studies.

Future research should incorporate greater spatial replication across locations to better quantify within- and between-locations variability in soil properties and root functional traits. Additional sampling locations and the inclusion of multiple agroforestry systems would improve the robustness and general applicability of the findings. Integrating measurements of microbial communities, soil enzyme activity, and isotopic tracing techniques would also help clarify the mechanisms through which root-derived C is transferred, transformed, and stabilised within soil organic C pools.

Further research should focus on the role of root exudates from both live and decomposing roots, and on their interactions with root architecture, soil properties, and microbial processes. Such investigations would improve understanding of rhizosphere dynamics and their contribution to nutrient cycling and C sequestration. Evaluating these relationships across different tree species, crop combinations, and climatic conditions would provide valuable insights into species-specific effects on belowground ecosystem functioning. This knowledge could support the design of more effective agroforestry systems by identifying crop–tree combinations that maximise rhizosphere activity, enhance soil C storage, improve nutrient-use efficiency, and contribute to low-emission, climate-resilient agricultural production systems.

4.6. Conclusion

The findings of this study demonstrate that soil nutrients, root traits, and soil physical properties are closely linked within a mature poplar agroforestry system. Although originally established as a silvoarable system with active cropping between 1993 and 2004, crop production subsequently ceased and the alleys became dominated by grasses and herbaceous vegetation. Within this system, AN and P were most strongly associated with root traits in the topsoil, where approximately 80-90% of root biomass was concentrated. Compared with the Arable location, the agroforestry zones had lower BD, greater root abundance, and higher root-derived C stocks. Clear differences were also observed between the Row and Alley, indicating greater spatial heterogeneity in belowground conditions and root distribution.

Both live and dead roots contributed to belowground C dynamics as live roots were more abundant in the Alley, whereas dead roots and associated C stocks were greater in the Row. These contrasting patterns suggest spatial variation in root growth, turnover, and decomposition across the agroforestry system. Although SOC stocks did not differ significantly among the Arable, Row, and Alley locations, significant relationships between root traits and SOC indicate that root distribution is closely linked to soil C dynamics. Overall, this study shows that the mature agroforestry system promoted greater belowground heterogeneity than conventional arable management, particularly within the topsoil. These findings highlight the importance of root systems in shaping soil structure, nutrient availability, and C cycling within temperate agroforestry systems.

Chapter 5. Leaf litter decomposition and nutrient dynamics beneath hedgerows in a temperate agricultural landscape

Abstract

Leaf litter decomposition is a vital process that regulates nutrient cycling, carbon (C) turnover and ecosystem function. Despite the widespread presence of hedgerows in temperate agri-ecosystems, their contribution to litter decomposition and nutrient dynamics remains poorly studied. This study investigates leaf litterfall, decomposition and nutrient release of two common UK hedgerow species, hawthorn (*Crataegus monogyna*) and hazel (*Corylus avellana*), using a 366-day litterbag experiment conducted in northern England. Hazel produced significantly greater annual litterfall than hawthorn, indicating higher litter production at the species level. Decomposition followed a biphasic pattern, characterised by rapid mass loss (40-60%) during the initial phase (0-223 days) and slower decomposition during the second phase (224-366 days), reflecting the progressive depletion of readily decomposable material. Litterbag position significantly influenced decomposition dynamics ($p<0.001$), with buried litter generally decomposing more rapidly than surface litter. Decomposition was strongly influenced by litter placement (litterbag position), with buried litter decomposing more rapidly than surface litter. Species effects on mass loss and decay constants were limited, suggesting that environmental conditions associated with litterbag placement were the primary drivers of decomposition. Initial litter quality differed between species, with hazel exhibiting higher nitrogen (N) concentrations and lower C: N ratios than hawthorn. However, nutrient dynamics were influenced by both litter chemistry and litterbag placement. Nitrogen concentrations and C: N ratios fluctuated throughout decomposition, and litter mass loss was positively correlated with N concentration and negatively correlated with C:N ratio. Carbon and N release were strongly associated with litter mass loss, demonstrating a close link between decomposition and nutrient turnover. Hence, the results show that litter decomposition in temperate hedgerows is regulated by the interaction among litter quality, litter placement, and the associated decomposition environment. The substantial mass loss and nutrient release observed over the study period highlight the important role of hedgerows in supporting nutrient turnover and C cycling within agricultural landscapes.

5.1. Introduction

Leaf litter decomposition is a fundamental ecological process that regulates energy flow and nutrient cycling in terrestrial ecosystems (Liao et al., 2022; Zhao et al., 2022). Globally, terrestrial vegetation produces nearly 100 billion tonnes of organic matter annually, of which approximately 90% enters the detrital pool through senescence and litterfall. Through decomposition, this organic matter is processed by complex detritus food webs, facilitating the cycling of carbon (C) and essential micro- and macronutrients (Didion et al., 2016; Petraglia et al., 2019). Approximately 70% of leaf litter contributes to nutrient and energy flows via decomposition, playing a key role in terrestrial biogeochemical cycles (Bradford et al., 2016; Cárdenas et al., 2017; Zan et al., 2021). During this process, C is released into the atmosphere while a proportion is stabilised in soils, and nutrients such as nitrogen (N) and phosphorus (P) are mineralised and made available for plant uptake (Freschet et al., 2012). Consequently, litter decomposition plays a critical role in maintaining soil fertility, nutrient availability, and C dynamics.

In addition, litterfall and decomposition processes offer valuable insights into soil-tree interactions, which underpin the functioning and management of tree-based systems (Gray & Schlesinger, 1981; Andivia et al., 2010). Leaf litter serves as a major source of organic matter and nutrients to both topsoil and subsoil layers, linking primary production to nutrient cycling and ultimately influencing vegetation productivity (Giweta, 2020). The rate of leaf litter decomposition is primarily influenced by the interaction of three key factors: litter quality (Djukic et al., 2018; Sun et al., 2002), environmental factors (Qu et al., 2021; Vitousek et al., 1994), and soil biota (Riutta et al., 2012; García-Palacios et al., 2013; Cotrufo et al., 2015). Litter quality, commonly characterised by its nutrient content and C:N ratio, strongly influences decomposition rates and nutrient release dynamics (Vitousek et al., 1994; Dawoe et al., 2018; Vigulu et al., 2019). These properties determine the balance between N mineralisation and immobilisation processes, thereby regulating nutrient availability in soils (Seneviratne, 2000). Climate factors, such as temperature, moisture, and daylight hours, further modulate decomposition by influencing microbial activity and litter-processing organisms (García-Palacios et al., 2013; Bradford et al., 2016).

Hedgerows are a common feature of agricultural landscapes in temperate regions, particularly across western and northwestern Europe (Image et al., 2023). They provide multiple ecosystem services, including soil erosion control, reduction of runoff, enhancement of biodiversity and regulation of local microclimatic conditions (Biffi et al., 2023). In the UK, hedgerows are increasingly recognised for their ecological importance, with policy initiatives aiming to restore and expand the national hedgerow network by 30,000 miles (48,280 km) by 2037 and 45,000 miles (72,420 km) by 2050 (Defra, 2023a; CPRE, 2024). Species composition within hedgerows can significantly influence soil processes and broader ecosystem functioning (Holden et al., 2019). For example, hawthorn (*Crataegus monogyna*), a dominant native species in the UK, contributes to C sequestration and habitat provision (Axe et al., 2017), while hazel (*Corylus avellana*) is valued for its ecological adaptability, biodiversity benefits and economic importance (Woodland Trust, 2025). As two of the most common native hedgerow shrubs in the UK, these species provide an ideal model for investigating species-specific influences on litter decomposition and nutrient cycling.

Despite growing recognition of hedgerows as important components of sustainable agricultural landscapes, relatively little is known about their nutrient cycling through leaf litter decomposition and nutrient release. Existing research, to date, has largely focused on broad functional comparisons, such as deciduous versus evergreen, or on forest and agroforestry systems (Cornwall et al., 2008; Zeng et al., 2010; Prescott, 2010; Hossain et al., 2011; Hasanuzzaman & Hossain, 2015; Akinyele & Donald-Amaeshi, 2021). While these studies provide valuable general insights into decomposition processes, they may not adequately represent the environmental conditions and species assemblages found in temperate hedgerow ecosystems. Empirical data on litterfall production, decomposition rates, and nutrient release within hedgerows remain limited, especially under UK conditions.

Differences in litter quantity, chemical composition, and decomposition environment (e.g., surface versus subsurface litter placement) may significantly affect nutrient turnover and soil fertility, yet these interactions remain underexplored for hedgerow systems. Seasonal variation and microenvironmental heterogeneity within hedgerows may further influence decomposition processes, contributing to variability in C and N dynamics over time. This knowledge gap constrains our ability to quantify the

contribution of hedgerows to nutrient cycling, soil fertility, and C sequestration, and consequently limits the development of evidence-based management strategies aimed at maximising ecosystem service provision.

Given the increasing emphasis on hedgerow restoration and sustainable land management, there is a clear need to quantify decomposition processes and nutrient dynamics in these systems. Therefore, this study aims to determine the rate of leaf litter decomposition for two common native shrub species, hawthorn and hazel, over a one-year period in a temperate environment. Specifically, the study sought to: (i) evaluate how litter placement (litterbag position) influences decomposition; litter was placed either on the soil surface or buried within the soil. This distinction is important because different agroecosystems vary in the extent to which litter is incorporated into the soil through management practices (e.g., tillage) and biological activity (e.g., earthworms). Consequently, surface and buried litter represent two common decomposition pathways. Surface litter decomposes under greater exposure to aboveground environmental conditions and surface-dwelling decomposers, whereas buried litter has closer contact with the soil matrix, higher moisture availability, and greater interaction with soil biota, all of which may influence decomposition rates. (ii) determine changes in C and N content throughout the decomposition process. By addressing these objectives, this research contributes to a better understanding of species-specific nutrient cycling dynamics and supports the development of evidence-based approaches to hedgerow management and restoration.

5.2. Methods

5.2.1. Study site

The study was conducted at the University of Leeds research farm (53.8667° N, -1.2833° W) in northern England, which covers an area of 317 ha. Approximately 75% of the farm is managed for arable production, while 7.7 hectares are dedicated to agroforestry systems (University of Leeds, 2019). The farm lies at an elevation of 40 to 70 metres above sea level and experiences a temperate climate. During the study year (2024), the mean annual temperature was 10.7 °C and total rainfall was 786.8 mm (Fig. 5.1), indicating climate conditions very similar to the long-term average (see Chapter 2).

The soils at the site are shallow (30–70 cm depth) and classified as Calcaric Endoleptic Cambisols of the Aberford series, overlying a dolostone aquifer of the Cadeby Formation (LandIS, 2025). These soils are well-drained and calcareous, typical of brown earths associated with gently sloping Permian and Jurassic limestone landscapes in central and northern England (Holden et al., 2019; Austen et al., 2022).

To characterise local environmental conditions, soil temperature and moisture were monitored using two Decagon 5TM sensors installed at a depth of 5 cm. Sensors were positioned midway along two of the four hedgerows included in the study. However, due to a sensor malfunction, data were available for only four months of the study. Mean soil temperature was 16.8°C in June, 18.8°C in July, 17.3°C in October, and 11.4°C in November. Corresponding mean soil moisture values were 11.2%, 8.7%, 16.4%, and 9.4%, respectively. These measurements indicate warmer, drier soil conditions during summer, followed by cooler, wetter conditions during autumn.

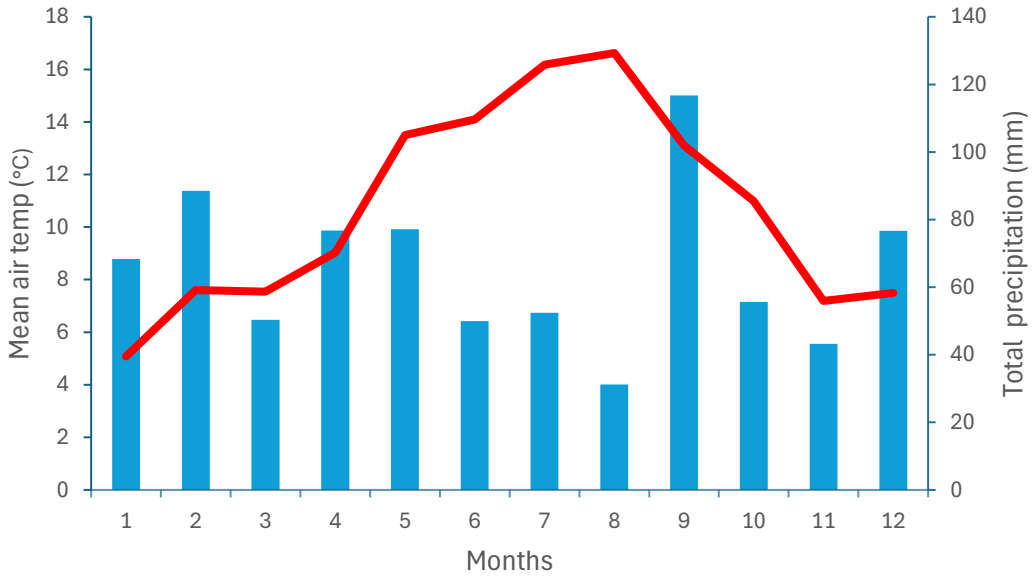


Figure 5.1 Mean monthly air temperature (red line) and total monthly precipitation (blue bars) at the study site in 2024 (Spen Farm, COSMOS-UK monitoring network, UK Centre for Ecology and Hydrology, 2026). Month 1 corresponds to January and month 12 to December.

5.2.2. Hedgerow characteristics

Four hedgerows were selected for the study, ranging from 1.0–2.5 m in width and 1.8–4.8 m in height. The hedgerows bordered a single arable field managed under a four-

year crop rotation system consisting of winter wheat, spring or winter barley, and oilseed rape. The hedgerows were continuous along all four boundaries of the field, forming a well-established linear network within the agricultural landscape. Species composition within the hedgerows was diverse, comprising 12 woody species. *Crataegus monogyna* (hawthorn) was the dominant species, accounting for an average of 60% of the composition (range: 0-100%). Other commonly occurring species included *Sambucus nigra* (elder) and *Ilex aquifolium* (holly), each contributing approximately 10% of the composition (range: 0-20%), while *Corylus avellana* (hazel) typically accounted for <10%. Additional species present in varying proportions (0-20%) included *Cornus sanguinea* (common dogwood) and *Rosa canina* (dog rose), along with less frequent species such as *Prunus spinosa* (blackthorn), *Acer campestre* (field maple), *Fraxinus excelsior* (ash), *Euonymus europaeus* (European spindle), and *Rhamnus cathartica* (buckthorn) (Holden et al., 2019). The dominance of hawthorn, combined with the presence of hazel and a range of other native woody species, is characteristic of established agricultural hedgerows in the UK. Consequently, the selected hedgerows provide a representative system for investigating litter decomposition and nutrient cycling processes within temperate agricultural landscapes.

5.2.3. Collection of leaf litter

Leaf litter was collected by hand from four hedgerows in late September to early October 2023, prior to natural leaf fall. The location of each hedgerow was recorded using a handheld GPS device, and a single coordinate was assigned to represent the approximate centre of each sampling area (Hedge 1 = 53.86994 N, 1.33232 W; Hedge 2 = 53.86990 N, 1.33223 W; Hedge 3 = 53.87077 N, 1.33256 W; Hedge 4 = 53.87069 N, 1.33262 W). Leaves of hawthorn (HW) and hazel (HZ) were randomly collected across all four hedgerows to capture natural variability in litter characteristics and decomposition potential between species.

Following collection, leaves were taken to the laboratory and manually sorted to remove non-target material and other organic debris. Samples were then rinsed with water to remove adhering soil particles, air-dried for 48 hours and subsequently oven-dried at 65°C until a constant weight was reached. This procedure ensured a uniform moisture content prior to chemical analysis and the decomposition experiment.

To determine initial litter chemistry, subsamples (10-15 g) of oven-dried material were randomly selected and finely ground using a mixer miller (Retsch MM400 ball mill; RETSCH GmbH, Germany). Total C and N concentrations were determined by weighing approximately 100 µg of the homogenised fine powder into aluminium tins and analysing them using an elemental combustion analyser (Elementar Analysensysteme GmbH, Germany). Four replicate samples were analysed for each species, and mean C and N contents were calculated. The C:N ratio was subsequently derived and used as an indicator of initial litter quality prior to the decomposition experiment.

5.2.4. Preparation of litterbags

Litterbags measuring 20 cm × 20 cm were constructed using 2 mm nylon mesh and sealed along all edges. Nylon was chosen for its durability and resistance to UV degradation, making it well suited for field-based decomposition studies (Krishna & Mohan, 2017). The mesh size (2 mm) was selected to limit access of larger soil macrofauna (>2 mm), thereby allowing decomposition to be driven primarily by microbial communities and smaller soil fauna (Sulkava and Huhta, 1998; Njoroge et al., 2022).

A total of 112 litterbags were prepared, comprising 56 bags for HZ and 56 for HW. The amount of litter added to each bag reflected differences in the quantity and morphology of leaf material collected for the two species. Hazel litterbags were filled with between 1.5 and 2.0 g of dried leaf material, whereas each HW litterbag contained 1.0 g. Litter mass per bag was determined using the total available material, differences in leaf size among species, and litterbag dimensions, following established litterbag methodologies (Rahman & Tsukamoto, 2013).

Litterbags were deployed during the first week of January 2024 beneath each of the four study hedgerows to assess decomposition under contrasting microenvironmental conditions. At each hedgerow, seven HW and seven HZ litterbags were placed on the soil surface at 50 cm intervals. An additional seven HW and seven HZ litterbags were buried at a depth of 5 cm directly beneath the corresponding surface positions. This design resulted in seven paired surface and subsurface litterbags for each species at each

hedgerow, allowing direct comparison of decomposition rates between litter placement treatments and species.

Subsurface litterbags were installed following the method of Harman et al. (1999). I inserted a spade at an angle of approximately 45° and gently lifted the soil to create a cavity. The litterbag was then placed within the opening beneath the lifted soil, after which the spade was carefully removed, and the soil pressed firmly back into place to ensure good contact between the litterbag and the surrounding soil.

Surface litterbags were placed directly above their buried counterpart to maintain consistency in location and environmental exposure. This paired placement design enabled a controlled comparison of litter placement effects on decomposition dynamics, particularly with respect to soil contact, moisture, and microclimatic conditions. Each litterbag was fitted with a durable plastic tag indicating species identity, hedgerow location (Hedge 1–4), and a unique bag number. This ensured reliable identification and tracking of individual samples throughout the decomposition experiment.

5.2.5. Litterbag retrieval and laboratory analysis

Litterbags were retrieved on seven occasions during the decomposition experiment: April, June, August, October, November, and December 2024, and January 2025. On each sampling occasion, two paired surface and subsurface litterbags of each species were collected from each of the four hedgerows.

In some cases, leaf litter could not be fully recovered from the retrieved litterbags. Extensive decomposition and mass loss had transformed the litter into fine organic residues indistinguishable from soil organic matter, making complete recovery impossible. Consequently, the amount of remaining material in some samples was insufficient for subsequent mass-loss and nutrient analyses. However, during August, rapid hedgerow growth obscured some litterbags, preventing their recovery during the scheduled sampling event. A follow-up collection was therefore conducted five days later to retrieve the remaining litterbags, resulting in a short sampling interval between Days 218 and 223.

Immediately following collection, litterbags were placed in sealed plastic bags and transported to the laboratory. There, each litterbag was carefully opened using scissors, and any adhering soil or other extraneous material was gently removed using a soft brush. Any macro- and mesofauna (e.g., snails, earthworms, spiders) present within the litterbags were manually removed prior to processing.

The remaining leaf litter was then rinsed 2–3 times with water to remove residual soil particles and organic debris, and then oven-dried at 65°C to a constant mass. The oven-dried mass was then recorded and used to calculate litter mass remaining throughout the decomposition period. After drying, litter samples were ground to a fine powder (~100 µm) using a mixer mill (Retsch MM400). Subsamples were subsequently analysed for C and N content using an elemental CN analyser (as described previously), thereby enabling assessment of temporal changes in litter chemistry during decomposition.

5.2.6. Estimation of leaf fall

To quantify mean leaf litterfall rates of HW and HZ, litter was collected from each of the four hedgerows during the autumn defoliation period between October and December 2024. At each hedgerow, five replicate hessian litter traps (1 m × 1 m) were placed centrally on the ground beneath the hedge canopy to capture falling litter. Collections were undertaken once in October and twice in each of November and December, until leaf fall had ceased.

Following collection, litter samples were sorted by species to isolate HW and HZ leaf litter. The leaves were thoroughly washed with tap water to remove adhering soil particles and debris, air-dried, and then oven-dried at 65 °C to a constant mass. Dry weights were recorded to calculate litterfall production for each species.

Litterfall for each species was standardised to a hedge length of 1.0 m and subsequently extrapolated to a per-hectare basis. This enabled comparison of annual litter inputs between HW and HZ within the study system and provided an estimate of their relative contributions to organic matter and nutrient inputs.

5.2.7. Data analysis

Leaf litter decomposition was quantified using standard mass-loss and exponential decay models. The percentage of leaf mass loss over time was calculated using Equation 5.1 (Bohara et al., 2019):

$$\left[\frac{X0-Xt}{X0}\right] \times 100 \qquad \text{(Equation 5.1)}$$

where X0 is the initial oven dried leaf litter mass, and Xt is the final oven-dried leaf litter mass. The leaf decomposition rate (mg day^{-1}) was determined using:

$$\frac{Leaf\ litter\ mass\ lost(mg)}{time\ (days)} \qquad \text{(Equation 5.2)}$$

Where $leaf\ litter\ mass\ lost\ (mg) = \frac{leaf\ litter\ mass\ loss\%}{100} * X0$ (Equation 5.3)

This provided a standardised measure of decomposition rate over the experimental period. Additionally, the decay rate constant (k) was estimated using a negative exponential decay model (Bohara et al., 2019):

$$\frac{k = -\ln\left(\frac{Xt}{X0}\right)}{t} \qquad \text{(Equation 5.4)}$$

This model assumes that decomposition follows a first-order decay process. The time required for 50% and 95% mass loss of leaf litter was estimated using the decay rate constant (k). These were denoted as t50 and t95, respectively, and were calculated using equations 5.4 and 5.5:

$$t50 = \frac{0.693}{k} \qquad \text{(Equation 5.5)}$$

$$t95 = \frac{5}{k} \quad \text{(Equation 5.6)}$$

Higher k values indicate faster decomposition rates and, therefore, shorter times required to reach 50% and 95% mass loss. These indices correspond to the time required for 50% and 95% decomposition of the initial litter mass, respectively. Importantly, *k*, t_{50}, and t_{95} represent model-derived measures of decomposition rather

than simple empirical estimates of mass loss (%) or decomposition rate (mg day^{-1}). The decomposition constant (k) is estimated from the exponential relationship between remaining litter mass and time, which reflects the rate of decay. Importantly, t_{50} and t_{95} represent model-derived measures of decomposition rather than simple empirical estimates of mass loss (%) or decomposition rate (mg day^{-1}). The decomposition constant is estimated from the exponential relationship between remaining litter mass and time, which reflects the rate of decay. The values of t_{50} (time required for 50% mass loss) and t_{95} (time required for 95% mass loss) are calculated directly from k and provide standardised measures of decomposition that are independent of initial litter mass. However, because t_{50} and t_{95} are derived from k, they remain dependent on the dataset's quality and duration used to estimate the decay constant (Adair et al., 2010; Wieder & Lang, 1982).

The percentage of nutrient released was also calculated to assess changes in nutrient content during decomposition (Linggi et al., 2024)**.** The percentage of nutrient remaining in the litter was determined using:

$$Nutrient\ remaining\ (\%) = \frac{Nf}{Ni} * \frac{Xt}{X0} * 100 \qquad \text{(Equation 5.7)}$$

Where *Nf* is the final nutrient concentration (C or N), and *Ni* is the initial nutrient concentration.

$$Nutrient\ released\ (\%) = 100 - nutrient\ remaining\% \qquad \text{(Equation 5.8)}$$

5.2.8. Statistical analysis

Data normality was assessed using the Shapiro-Wilk test. All measured variables, including leaf litter mass loss (%), k (day^{-1}), residual C and N concentrations, and C:N ratio, remained non-normally distributed even after data transformation. Consequently, non-parametric statistical tests were applied for subsequent analyses. The Kruskal-Wallis test was used to assess the effect of species, sampling time, and litter placement (surface vs. buried) on leaf litter mass loss, k and residual nutrient concentrations. Spearman's rank correlation analysis was conducted using the Spearman function in RStudio to examine relationships between leaf litter mass loss and litter nutrient dynamics (residual and released C, N and C:N ratio) during the decomposition process.

Differences in initial litter quality (C, N and C:N ratio) between HW and HZ were evaluated using Welch's two-sample *t*-test, using the t.test() function in RStudio, as initial litter C and N concentrations and C:N ratios were normally distributed. Welch's t-test was selected because it is robust to unequal variances and suitable for small sample sizes. Results were reported as t-statistics, degrees of freedom, p-values, 95% confidence intervals, and treatment mean, with statistical significance evaluated at $\alpha = 0.05$. All statistical analyses were performed in RStudio (version 4.3.3; R Core Team, 2024).

5.3. Results

5.3.1. Phases of leaf litter decomposition rate

Litter decomposition was monitored through successive retrieval intervals (0–110, 111–169, 170–217, 218–223, and 224-273, 274-316, 317-338, 339-366 days). However, examination of cumulative litter mass loss revealed two broad phases of decomposition. An initial phase (0–223 days) was characterised by rapid litter mass loss, whereas a later phase (223–366 days) exhibited slower and more gradual decay. These phases describe the overall decomposition dynamics and do not replace the interval-specific decomposition rate calculations.

Across both species, leaf litter mass loss was greatest during the first phase of decomposition (0–223 days) and subsequently decreased during the later phase (Fig. 5.2). Litterbag position and time since deployment had significant effects on mass loss ($p < 0.001$). During the initial decomposition phase, litter mass loss was consistently greater in buried than in surface-placed litterbags (Fig. 5.2). Mean mass loss in buried HZ litterbags reached 64.68 ± 63.33%, compared with 36.58 ± 37.52% in surface litterbags. Similarly, buried HW litterbags exhibited a mean mass loss of 81.52 ± 75.31%, substantially higher than the 41.75 ± 36.73% observed in surface litterbags (Fig. 5.2).

Decomposition proceeded rapidly for both species during days 0-223. Buried HZ litterbags had lost more than 90% of their initial mass by approximately day 217. By the beginning of the later decomposition phase (days 224-273), mass loss in surface HZ

litterbags had reached approximately 85% (Fig. 5.2). Although HZ generally exhibited higher mass-loss values than HW during much of the decomposition period, species identity had no significant overall effect on litter mass loss ($p = 0.51$).

By the end of the experiment (day 366), decomposition was nearly complete across all treatments, with litter mass losses exceeding 98% in every case. Mean mass loss was lowest in surface HZ litterbags (98.24%), followed by buried HW litterbags (91.16%), and buried HZ litterbags (99.45%), while surface HW litterbags exhibited the greatest mass loss (99.73%) (Fig. 5.2). Despite these small differences, treatment values converged towards the end of the study, indicating that most decomposition occurred during the initial phase and that decomposition rates declined as the remaining litter became increasingly recalcitrant.

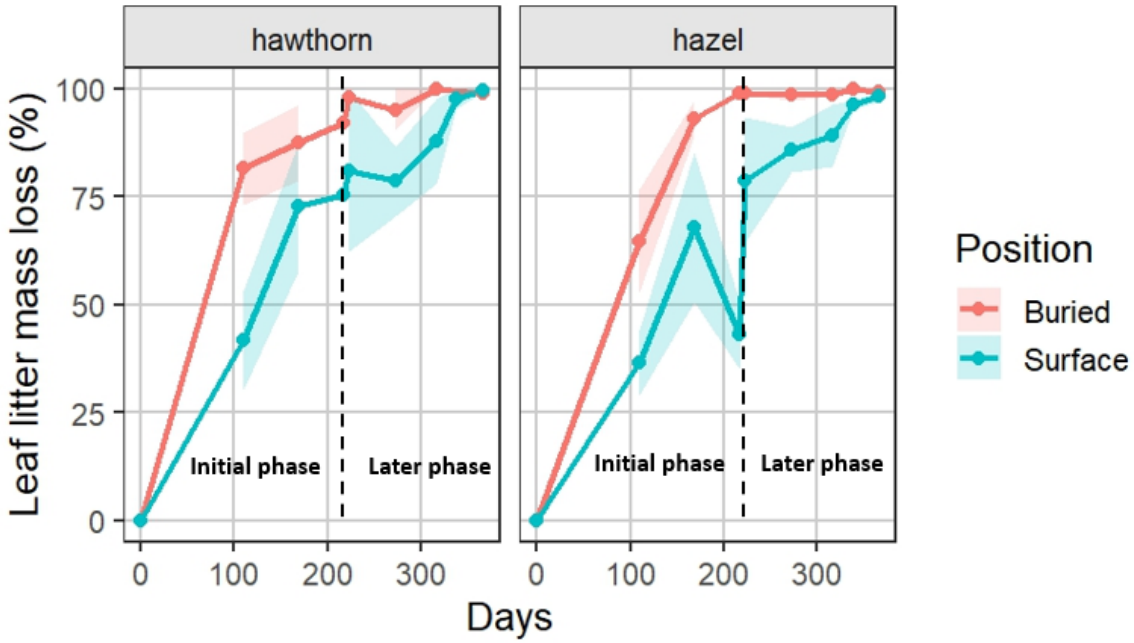


Figure 5.2 Leaf litter mass loss (%) for HW and HZ leaves, at the soil surface or buried at 5 cm depth. The shaded areas around the lines represent the 95% confidence interval. The vertical dashed (or dotted) lines indicate the transition point between the two decomposition phases identified in the analysis: an initial phase characterised by rapid mass loss and a later phase characterised by a slower decomposition rate. The position of each dashed line marks the boundary between these phases for the respective species and position of leaf litter bags

Based on leaf litter mass loss, decomposition rates (mg day^{-1}) were calculated for each sampling interval (Table 5.1). Decomposition rate differed significantly between species throughout the study period ($p < 0.001$), whereas litterbag position had no significant effect on decomposition rate ($p = 0.550$). Although decomposition rates did not differ statistically between surface and buried litterbags, temporal variation was observed during the decomposition process ($p = 0.041$).

During the early stages of decomposition (0–223 days), both HW and HZ generally exhibited higher decomposition rates in buried litterbags than in surface litterbags. However, decomposition rates declined over time for all treatments. By days 218- 223, surface HW litter exhibited a decomposition rate of 3.57 mg day^{-1}, exceeding that of buried HW litter. This pattern likely reflects the fact that much of the buried litter had already decomposed during earlier stages of the experiment, resulting in lower rates later in the decomposition process. A similar trend was evident at day 274-316, when surface HW litter exhibited a decomposition rate of 2.04 mg day^{-1}, compared with 1.52 mg day^{-1} for buried HW litter (Table 5.1). By the final sampling date (366 days), decomposition rates in surface and buried HW litter had converged, indicating that differences associated with litter placement became less pronounced as decomposition progressed.

Table 5.1 Mean (n=4) and median of leaf litter decomposition rate of both species over the period of decomposition. Empty spaces indicate that no residual leaf litter was found in the bag, or that there were insufficient data to calculate the standard deviation and median.

Species	Position of bags	Leaf decomposition rate (mg day^{-1})							
		Phase 1				Phase 2			
		Day 0-110	Day 111-169	Day 170-217	Day 218-223	Day 224-273	Day 274-316	Day 317-338	Day 339-366
HW	Surface	3.7	3.17	1.69	3.57	2.12	2.04	2.12	1.33
HW	Buried	5.46	3.81	2.08	2.18	2.56	1.52		1.34
HZ	Surface	5.88	7.17	3.52	6.28	5.58	4.95	5.02	2.43
HZ	Buried	7.87	7.39	4.01	7.85	4.81	4.19	3.74	3.44

Overall, HZ exhibited higher decomposition rates than HW in both surface and buried litterbags, consistent with the significant effect of species identity on decomposition rate

($p < 0.001$; Fig. 5.2). In contrast, litterbag position did not significantly influence decomposition rate, despite some variation in decomposition patterns through time.

The *k* presented in Table 5.2 represents the Phase 1 and Phase 2 values reported for 223 and 366 days of decomposition, respectively. During Phase 1, higher k values were observed for buried HW and buried HZ litter compared to surface litter. These higher decomposition rate constants were associated with lower estimated times required to achieve 50% (t_{50}) and 95% (t_{95}) mass loss. This indicates that buried HW and HZ litter decomposed more rapidly during the initial stage of decomposition, resulting in greater litter turnover than surface litter.

Table 5.2 Mean decay rate constant (day^{-1}) required for 50 and 95% of leaf litter decomposition of HW and HZ at the end of 223 days (phase 1) and 366 days (phase 2)

Species	Position of bags	Temporal pattern	k (day^{-1})	t_{50} (days)	t_{95} (days)
HW	Surface	Phase 1	0.016	91.90	663.08
HW	Buried		0.018	39.09	282.04
HZ	Surface		0.003	255.1	1840.53
HZ	Buried		0.021	32.74	236.19
HW	Surface	Phase 2	0.019	38.96	281.06
HW	Buried		0.014	50.63	365.33
HZ	Surface		0.012	60.64	437.51
HZ	Buried		0.016	45.81	330.51

In contrast, during Phase 2, lower *k* values were observed for buried HW and surface HZ litter. These lower decay constants corresponded to longer estimated times required t_{50} and t_{95}, indicating slower decomposition rates and reduced litter turnover. Conversely, surface HW and buried HZ litter exhibited higher *k* values and shorter t_{50} and t_{95} values, reflecting more rapid decomposition and faster litter turnover (Table 5.2). Overall, litterbag placement had a significant effect on decay dynamics ($p < 0.01$), whereas no significant differences in *k* were observed between species ($p = 0.19$). Furthermore, the decomposition phase (Phase 1 versus Phase 2), identified from the decomposition pattern, did not significantly influence *k*-derived parameters ($p = 0.17$).

Mean annual leaf litterfall was higher for HZ (235.21 kg ha^{-1}) compared to HW (216.4 kg ha^{-1}), indicating a greater potential contribution of HZ to organic matter inputs within the hedgerow system.

5.3.2. Nutrient dynamics of litter

Significant inter-species differences were observed in the initial N content and litter C:N ratio, whereas initial C content was similar between species (Table 5.3). Hawthorn contained significantly less N than HZ (mean: 15.04 g kg^{-1} vs. 20.55 g kg^{-1}; $t = -5.115$, $df = 4.60$, $p < 0.001$). In contrast, initial C contents were nearly identical between the two species (370.37 g kg^{-1}vs. 370.57 g kg^{-1}t = -0.062, df = 4.57, p = 0.953), indicating no significant difference. Consequently, the initial C: N ratio was markedly higher in HW (24.63) than in HZ (18.08), reflecting the lower N concentration relative to C in HW litter ($t = 5.701$, $df = 5.97$, $p < 0.001$; 95% CI: 3.75 to 9.40) (Table 5.3).

Carbon concentration fluctuated throughout the decomposition period and differed among litterbag positions, indicating non-linear changes relative to the initial C concentration (Table 5.3, Fig. 5.4). Nitrogen concentrations generally increased during the initial stages of decomposition in both species compared to their initial values, before fluctuating throughout the remainder of the study. Similar temporal patterns were observed in the C:N ratio, although responses differed among species and litterbag placements. Notably, the C:N ratio of surface-placed HZ litter increased substantially over the 366-day decomposition period and remained consistently higher than that of buried HZ litter (Table 5.3; Fig. 5.4). Overall, changes in C:N ratio exhibited pronounced species- and treatment-specific patterns, with no consistent trend relative to the initial litter values.

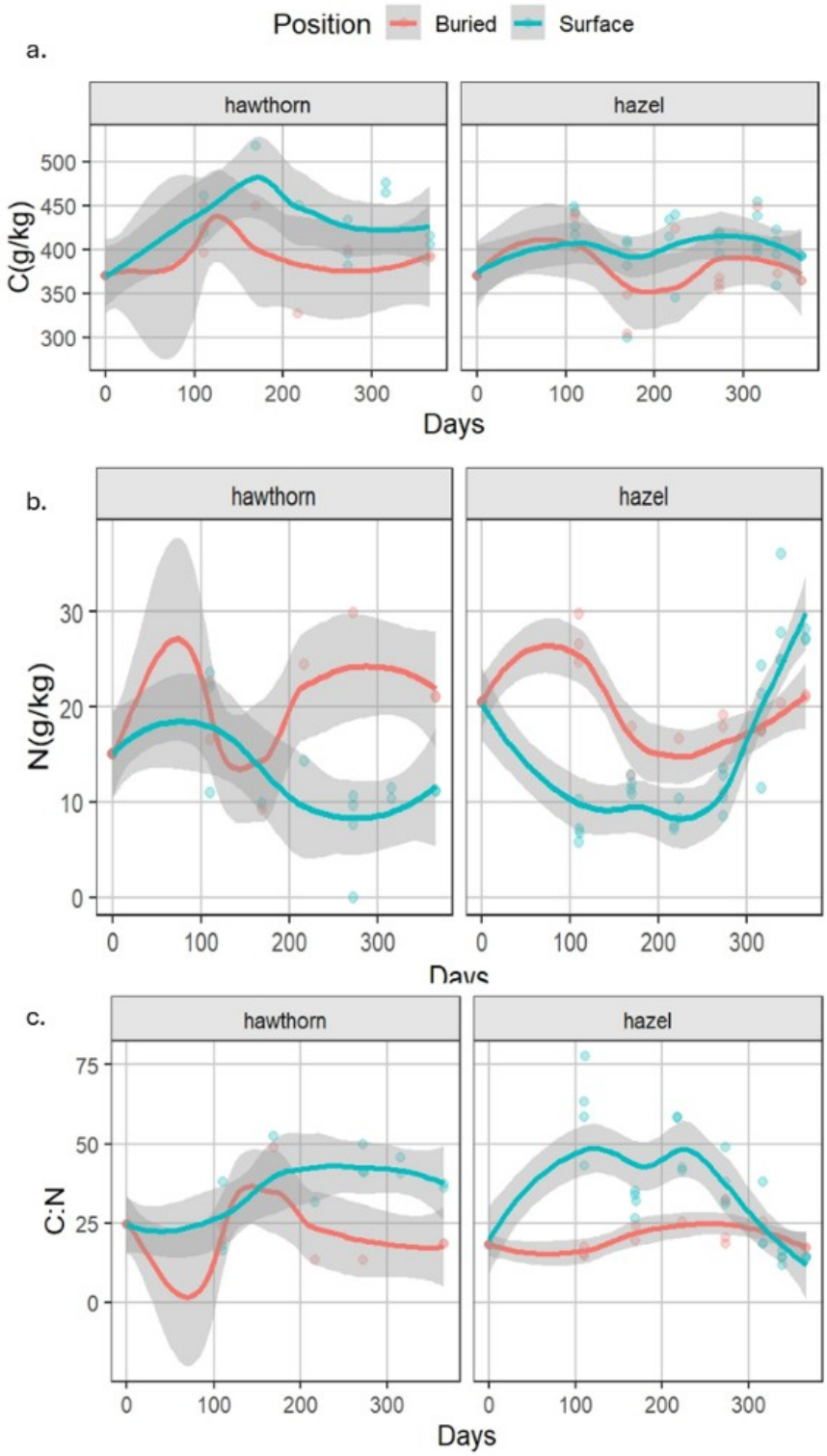


Figure 5.3 Residual nutrient concentration – (a) C (g/kg), (b) N (g/kg) and (c) C:N ratio for HW and HZ, placed on the surface or buried over 366 days. The shaded areas around the lines represent the 95% confidence interval.

Table 5.3 Mean (n= 0-4, depending on position of leaf litterbags; as leaf litter was not present in all four bags due to mass loss) of initial and residual C (g kg^{-1}), N (g kg^{-1}), and C: N ratio during decomposition over 0-366 days. Blank cells indicate that insufficient litter remained for nutrient analysis because of high mass loss.

Litter	Bag position	Nutrient	Day 0	Day 110	Day 169	Day 217	Day 223	Day 273	Day 316	Day 338	Day 366
HW	Surface	C	370.37	435.05	518.13	450.76		403.18	470.48		392.2
HW	Buried	C	370.37	423.31	449.84	327.33		399.34			410.96
HZ	Surface	C	370.57	433.33	374.33	424.65	392.31	411.14	422.95	396.49	392.50
HZ	Buried	C	370.57	425.97	326.63		423.11	361.34	427.40	372.64	364.97
HW	Surface	N	15.04	18.92	9.91	14.29		9.27	10.93		21.08
HW	Buried	N	15.04	19.58	9.23	24.45		29.87			
HZ	Surface	N	20.55	7.48	11.75	7.28	9.31	11.36	18.75	28.48	21.16
HZ	Buried	N	20.55	27.00	15.36		16.66	16.05	17.51	20.41	27.50
HW	Surface	C: N	24.63	25.61	52.28	31.54		44.02	43.18		18.61
HW	Buried	C: N	24.63	21.93	48.74	13.39		13.37			36.75
HZ	Surface	C: N	18.03	60.59	31.87	58.33	42.11	37.38	24.51	14.28	17.25
HZ	Buried	C: N	18.03	15.86	21.65		25.40	23.91	24.41	18.26	14.27

By Day 366, clear differences in residual litter chemistry were evident between species and litterbag positions. In HW, buried litter retained a higher C concentration than surface-placed litter, which contained 392.20 g kg^{-1}. In contrast, surface HZ litter retained a higher C concentration (392.50 g kg^{-1}) than buried HZ litter (364.97 g kg^{-1}) (Table 5.3; Fig. 5.4).

Nitrogen concentrations also varied among treatments at the final sampling days. Surface HW litter contained a higher N concentration (21.08 g kg^{-1}) than buried HW litter (11.19 g kg^{-1}). In contrast, buried HZ litter had a higher N concentration (27.50 g kg^{-1}) than surface HZ litter (21.16 g kg^{-1}) (Table 5.3; Fig. 5.4). Consequently, the highest C:N ratio was in buried HW litter (36.75) and the lowest in buried HZ litter

(14.27). Surface HW and HZ litter showed intermediate values of 18.61 and 17.25, respectively (Table 5.3; Fig. 5.4).

Overall, litterbag position and decomposition time significantly influenced C, N, and the C: N ratio ($p < 0.001$). Changes in the C:N ratio reflected concurrent fluctuations in both C and N concentrations throughout decomposition. Despite differences in residual litter chemistry among treatments, cumulative nutrient losses were substantial, with both HW and HZ releasing approximately 99% of their initial C and N by day 366 regardless of litterbag placement (Table 5.4).

Table 5.4 Percentage mean nutrient release from leaf litter at the end of 223 and 366 days of decomposition. Blank cells indicate that insufficient litter remained for nutrient analysis because of high mass loss.

Species	Position of bags	Temporal pattern	C%	N%
HW	Surface	Phase 1		
HW	Buried			
HZ	Surface		43.0	75.7
HZ	Buried		98.5	98.9
HW	Surface	Phase 2	99.7	99.6
HW	Buried		99.0	99.4
HZ	Surface		98.2	98.2
HZ	Buried		99.8	99.7

5.3.3. Relationship between nutrients and leaf litter mass loss

Leaf litter mass loss was significantly correlated with residual N and the C: N ratio in both species, but not with residual C, highlighting the contrasting roles of these elements in regulating litter breakdown. Spearman's rank correlation analysis showed a significant negative correlation between litter mass loss and C: N ratio ($\rho = -0.329$, $p = 0.01$), indicating that greater decomposition was associated with lower C: N ratios (Fig. 5.4). In contrast, litter mass loss was also significantly positively correlated with N concentration ($\rho = 0.34$, $p = 0.01$), suggesting that litter with higher N concentrations decomposed more extensively. No significant relationship was found between litter mass loss and C concentration ($\rho = -0.17$, $p = 0.21$) (Fig. 5.4).

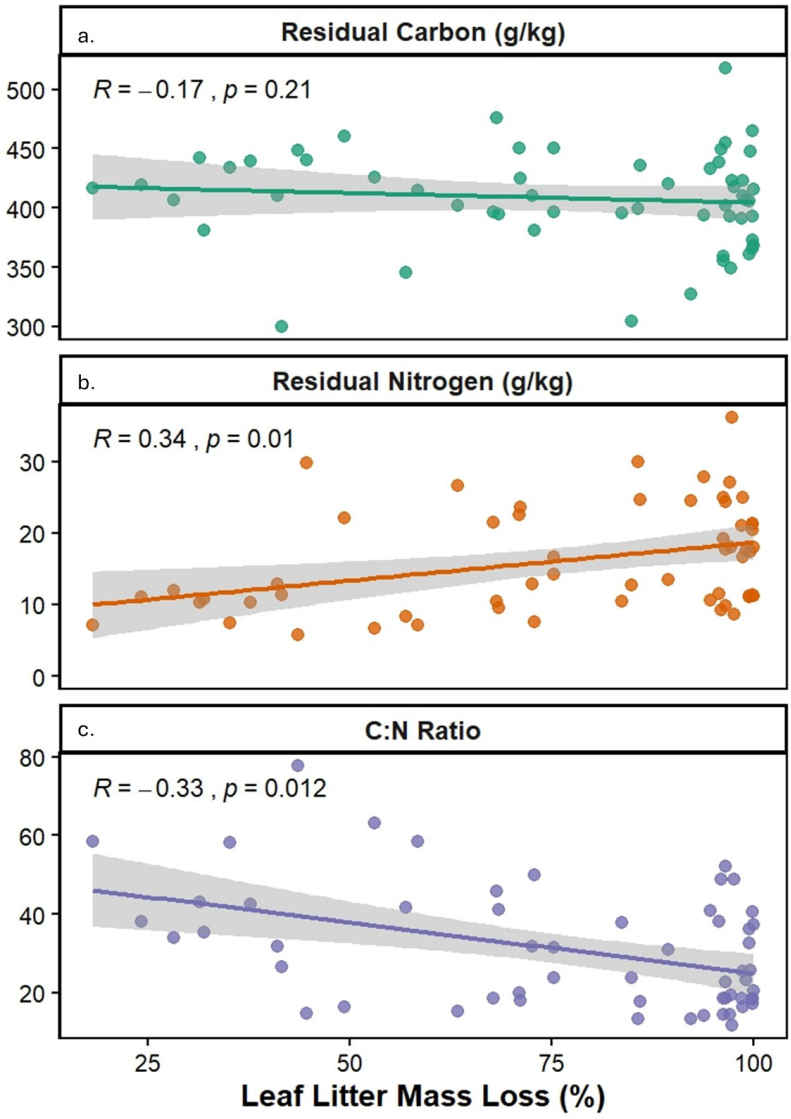


Figure 5.4 Relationship between leaf litter mass loss (%) for both species and their residual nutrient content - (a) C (g/kg), (b) N (g/kg) and (c) C:N ratio

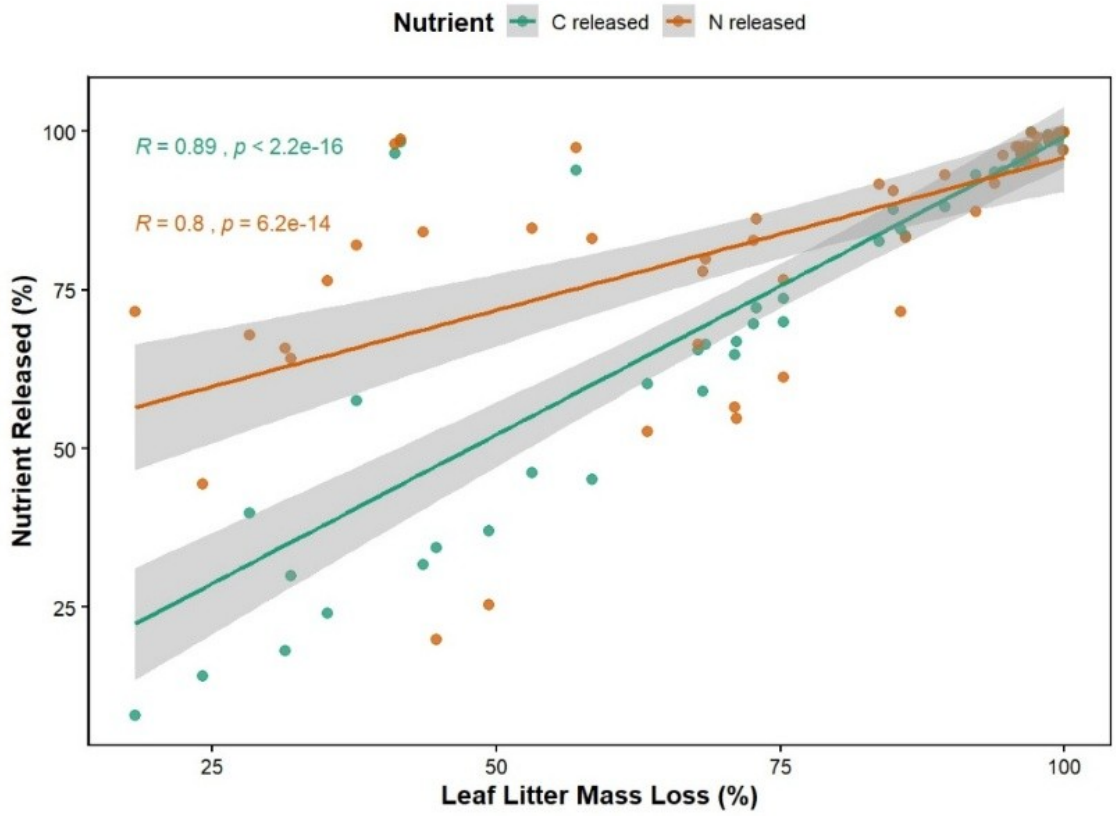


Figure 5.5 Relationship between leaf litter mass loss (%) of both species and their nutrient release (%).

Carbon and N release were both strongly and positively correlated with leaf litter mass loss (C: $\rho = 0.89$, $p < 0.001$; N: $\rho = 0.80$, $p < 0.001$; Fig. 5.5). These relationships indicate that nutrient release increased as decomposition progressed and litter mass declined. The stronger correlation observed for C suggests that C release closely tracked the loss of litter mass throughout decomposition. In contrast, the weaker relationship between N release and litter mass throughout the decomposition indicated greater variability in N dynamics, likely reflecting concurrent processes of mineralisation, immobilisation, and microbial uptake.

Overall, these results demonstrate that litter quality, particularly N concentration and C:N ratio, played an important role in influencing decomposition dynamics, whereas residual C concentration showed little direct association with litter mass loss. The strong relationships between nutrient release and litter decomposition further highlight the close coupling between organic matter turnover and nutrient cycling within the hedgerow ecosystem.

5.4. Discussion

5.4.1. Factors controlling leaf litter decomposition and carbon cycling

Leaf litter decomposition showed a characteristic biphasic pattern of rapid initial mass loss followed by a slower decomposition phase (Fig. 5.2; Table 5.1 and 5.2). This trend is commonly attributed to the loss of labile compounds during the initial phase of decay, followed by the persistence of more recalcitrant materials such as lignin and other structural components (Berg & McClaugherty, 2020; Suseela et al., 2013). Generally, buried litterbags exhibited greater mass loss during the initial phase of decomposition, suggesting that increased litter–soil contact and more favourable moisture conditions enhanced degradation between 0-223 days (Fig 5.2; Chen et al., 2021; Su et al., 2021).

In the later phase, by day 366, both species and litterbag positions had achieved very high mass loss (98.19–99.73%) (Fig. 5.2; Table 5.5). These values were higher than those in other temperate European studies summarised in Table 5.5. For example, Cornelissen (1996) reported a 15.2–48.4% reduction in litter mass by 244 days, while Remy et al. (2018) observed a 52.7–92.7% mass loss after 548 days. Although climatic conditions at the present study site were broadly comparable to those reported by Remy et al. (2018), litter decomposition beneath hedgerows was substantially greater.

This enhanced mass loss may reflect the influence of increased soil biotic activity, more complex habitat structure, and favourable microclimatic conditions associated with hedgerow environments, all of which can promote litter degradation (Cornelissen, 1996; Holden et al., 2019). These findings were consistent with Remy et al. (2018), who observed differences in litter decomposition between forest edges and forest-interior environments.

Differences in decomposition rates between species may also be linked to variation in litter quality (Table 5.1). Previous studies have shown that litter from oak and conifer species often contain higher concentrations of lignin, tannins, and other structural compounds that can alter decomposition, whereas HW and HZ litter may provide more readily available substrates for decomposer communities (Table 5.5). Furthermore, while small differences in decay constants were observed between HW and HZ litter,

these differences were not statistically significant, suggesting that species identity was not a major determinant of decomposition rate. Rather, litterbag position and the resulting microenvironmental conditions were the primary factors influencing decomposition dynamics. Consequently, any apparent differences between species should be interpreted with caution (Tables 5.1 and 5.2; Li et al., 2026).

Moreover, burial of litterbags appeared to modify decomposition pathways by increasing litter–soil contact and exposing litter to more stable moisture and temperature conditions, factors that generally promote microbial activity (Swift et al., 1979; Berg & McClaugherty, 2020). The observed effects of litter burial suggest that decomposition dynamics were influenced primarily by changes in the surrounding microenvironment rather than by species identity (Table 5.2). Burial may also have facilitated interactions among litter, soil microorganisms, and soil fauna, thereby enhancing litter fragmentation and microbial processing, and contributing to the formation of soil organic C (Wei et al., 2022; Sofia et al., 2023). These findings indicate that litter incorporation into soil, whether through earthworm activity or agricultural tillage, can influence decomposition processes and may play an important role in C cycling within hedgerow systems (García-Palacios et al., 2013).

Burial of litterbags appeared to increase decomposition by improving litter–soil contact and exposing litter to more stable moisture and temperature conditions, both of which generally promote microbial activity (Swift et al., 1979; Berg & McClaugherty, 2020). However, as no significant differences in the decay constant were observed between the buried HW and HZ litter, the results suggest that species identity had a limited influence on decomposition rates under burial conditions (Table 5.2)

Table 5.5. Comparative analysis of leaf litter decomposition from litterbag experiments in the temperate zone

Species	Bag position	Veg. type	Country	Days (study period)	Mean annual temp (°C)	Mean annual precipitation (mm)	Leaf litter mass loss %	References
Hawthorn (*Crataegus monogyna*)	Surface	Hedgerow	UK	366	10.7	786.8	99.16	this study
Hawthorn	Buried	Hedgerow	UK	366	10.7	786.8	99.73	this study
Hazel (*Corylus avellana*)	Surface	Hedgerow	UK	366	10.7	786.8	99.45	this study
Hazel	Buried	Hedgerow	UK	366	10.7	786.8	98.19	this study
Aesculus hippocastanum	6m * 5m decomposition bed	Park/woodland	UK	244	8.71	578.7	15.2	(Cornelissen, 1996)
Fraxinus excelsior	6m * 5m decomposition bed	Park/woodland	UK	244	8.71	578.7	32.9	(Cornelissen, 1996)
Laburnum anagyroides	6m * 5m decomposition bed	Park/woodland	UK	244	8.71	578.7	43.4	(Cornelissen, 1996)
Rosa arvensis	6m * 5m decomposition bed	Park/woodland	UK	244	8.71	578.7	22.6	(Cornelissen, 1996)
Rosa canina	6m * 5m decomposition bed	Park/woodland	UK	244	8.71	578.7	25.3	(Cornelissen, 1996)
Rubus fruticosus	6m * 5m decomposition bed	Park/woodland	UK	244	8.71	578.7	17.8	(Cornelissen, 1996)
Rubus idaeus	6m * 5m decomposition bed	Park/woodland	UK	244	8.71	578.7	29.9	(Cornelissen, 1996)
Sambucus nigra	6m * 5m decomposition bed	Park/woodland	UK	244	8.71	578.7	48.4	(Cornelissen, 1996)
Sorbus aucuparia	6m * 5m decomposition bed	Park/woodland	UK	244	8.71	578.7	25.9	(Cornelissen, 1996)
Quercus robur L. (Pedunculate oak)	Forest edge (depth not defined)	Forest	Belgium	548	10.5	800	92.7	(Remy et al., 2018)
Quercus robur L.	Forest interior	Forest	Belgium	548	10.5	800	71.5	(Remy et al., 2018)
Pinus nigra ssp. laricio Maire (Corsican pine)	Forest interior	Forest	Belgium	548	10.5	800	52.7	(Remy et al., 2018)
Pinus nigra ssp. nigra Arnold (Austrian pine)	Forest edge	Forest	Belgium	548	10.5	800	66.4	(Remy et al., 2018)

Nevertheless, decomposition dynamics during the later stages of decay may have been influenced by additional factors beyond initial litter quality. The deposition of surrounding organic residues and the ingress of soil fauna and microfauna into litterbags after placement at a depth of 5 cm may have altered microbial activity and litter fragmentation, thereby may affected overall decomposition and contributing to the formation of soil organic C (Wei et al., 2022; Sofia et al., 2023). These findings indicate that litter incorporation into soil, whether through earthworm activity or agricultural tillage, can substantially influence decomposition processes and consequently play an important role in C cycling within hedgerow systems (García-Palacios et al., 2013)

The 2 mm mesh size used in this study allowed access by small soil fauna, possibly aiding litter fragmentation and facilitating microbial colonisation (Bradford et al., 2002). Soil fauna including snails, earthworms, and spiders were observed within buried litterbags during spring-autumn and winter retrievals, indicating active faunal interactions with litter throughout much of the year, suggesting that decomposition continued under colder seasonal conditions is consistent with studies demonstrating microbial activity can persist beneath snow cover and contribute to litter mass loss (Uchida et al., 2005). These observations emphasise the critical importance of litterbag mesh size in determining the extent to which soil fauna contribute to decomposition processes. Overall, decomposition dynamics in hedgerows adjacent to agricultural systems varied with litter placement; however, the consistently high rates of mass loss observed in this study suggest that UK hedgerows provide favourable conditions for litter decomposition and play a potentially important role in local C cycling (Carreiro et al., 2000).

5.4.2. Influence of litter quality, nutrient dynamics and environmental conditions on leaf litter decomposition

The decomposition of leaf litter is strongly affected by litter chemistry, particularly N availability and C: N stoichiometry, which regulate microbial activity and nutrient cycling (Aerts, 1997; Berg & McClaugherty, 2020; Manzoni et al., 2021; Yu et al., 2025). In the present study, HZ litter had a higher initial N concentration and a lower C:N ratio than HW litter, indicating higher litter quality (Table 5.3; Fig. 5.3). These features are traditionally associated with faster decomposition because they reduce

nutrient limitation for decomposer organisms (Cornelissen et al., 1999). However, litter chemistry alone could not fully account for the decomposition patterns (Dent et al., 2006). Despite its lower initial N concentrations and a higher C: N ratio, HW decomposed more quickly than HZ in surface litterbags and also showed more rapid decomposition when buried (Table 5.3; Fig. 5.3). This implies that decomposition rates were affected not only by litter quality but also by environmental conditions associated with litterbag position, as discussed in section 5.1.

Nitrogen content of the litter varied considerably throughout the decomposition period and did not diminish consistently over time (Table 5.3). This variability was probably driven by microbial immobilisation and mineralisation processes, together with nutrient exchange between the litter and the surrounding hedgerow environment (Prescott & Vesterdal, 2021). The positive relationship between litter mass loss and N concentration, and the negative correlation between litter mass loss and C:N ratio, indicate that the availability of N in the litter is important in regulating decomposition (Fig. 5.4; Zheng et al., 2022). In contrast. residual C content was not significantly associated with litter mass loss. This finding suggests that decomposition was influenced more strongly by litter quality traits and species composition than by total C content alone. Such results are consistent with previous studies showing that plant functional traits and litter chemistry are primary drivers of decomposition processes (Cornwell et al., 2008; Cornelissen et al., 1999).

Although N concentration and C: N ratio contributed to decomposition, their effects depended on environmental conditions, suggesting that N alone did not determine decomposition rates in hedgerow litter. Enhanced microbial activity under hedgerows likely promoted nutrient mineralisation and nutrient exchange between litter and soil, thereby influencing the nutrient dynamics observed throughout the experiment (Akinyele & Donald-Amaeshi, 2021; Froufe et al., 2020).

Litter position influenced decomposition patterns, with differences observed between buried and surface litterbags. These differences may reflect variation in environmental conditions experienced by the litter, although factors such as soil moisture and microbial activity were not directly measured in this study. Seasonal variation may also have contributed to the observed patterns, with warmer periods generally associated

with greater nutrient release and colder winter conditions with reduced nutrient turnover (Aerts, 1997). In addition to microbial processes, decomposer fauna might have influenced patterns of nutrient release and retention. Surface litter was more accessible to detritivorous invertebrate species (e.g., earthworms, woodlice, mites and springtails) whose feeding and fragmentation behaviour increase litter surface area and facilitate microbial colonisation (Hättenschwiler et al., 2005). Seasonal variation in faunal abundance and activity may thus have accounted for the differences observed between surface and buried litterbags.

Moreover, HZ produced significantly more litter than HW, potentially increasing organic matter inputs to the soil and stimulating microbial activity, which may partly explain the relatively faster decomposition of buried HZ litter (Pedersen et al., 2017; Erdem and Karavin, 2016). In contrast, the rapid decomposition of HW despite its lower litter production suggests that litter quality, rather than quantity alone, also played an important role in decomposition dynamics. This finding is consistent with previous studies demonstrating species-specific litter effects of litter traits on decomposition processes (Bellingham et al., 2013).

Overall, the results suggest that nutrient dynamics during decomposition were governed by a combination of litter quality, litter position, seasonal environmental conditions and decomposer activities. Although the N and C: N ratio were important predictors of decomposition, their influence was moderated by environmental conditions and biotic interactions. Consequently, N availability alone was insufficient to explain decomposition patterns in hedgerow litter, highlighting the importance of considering both litter traits and environmental context when evaluating decomposition processes in hedgerow ecosystems.

5.5. Conclusion

1. Leaf litter decomposition. Leaf litter decomposition in temperate hedgerows followed a distinct two-phase pattern, characterised by rapid mass loss during the initial stages (0-223 days) of decomposition followed by a slower rate of breakdown thereafter (223-366 days). Hazel produced higher litterfall than HW on a species basis, suggesting a greater potential for the return of organic matter and nutrients. However, because HW is the dominant hedgerow species within the study area, its overall contribution to litter inputs and soil C and nutrient cycling is likely substantial and may exceed that of HZ at the hedgerow scale. Therefore, both species play important roles in regulating organic matter dynamics and ecosystem functioning within hedgerow systems. Decomposition was strongly influence by decomposition environment, with buried litter decomposing more rapidly than surface litter regardless of species. Although leaf litter mass loss and decay constants did not differ significantly among species, the consistently higher decomposition rates observed in buried litterbags suggests that environmental conditions associated with litter placement was a stronger determinant of decomposition than species identity. These findings highlight the importance of the decomposition environment in regulating litter turnover and C cycling within temperate hedgerow systems.

2. Litter quality and nutrient dynamics. Initial litter quality differed significantly between species, with HW exhibiting higher N concentrations and lower C: N ratios than HW. However, decomposition patterns were not determined solely by initial litter chemistry, indicating that litter quality interacts with environmental conditions to regulate decomposition processes. Temporal changes in N concentration and C: N ratio reflected active nutrient cycling throughout the decomposition period, with litter mass loss positively associated with N concentration and negatively associated with the C: N ratio. These relationships highlight the importance of N availability in promoting litter breakdown and nutrient turnover. Carbon and N release remained closely linked to litter mass loss, highlighting the strong interplay between organic matter decomposition and nutrient cycling. Overall, these findings indicate that nutrient dynamics beneath hedgerows are regulated by both litter quality and local microenvironment, which together appear to provide favourable conditions for decomposer activity and nutrient cycling.

In summary, the study demonstrates that environmental conditions associated with litter placement exert a greater influence on decomposition dynamics than species identity. The enhanced decomposition observed in buried litterbags was likely driven by improved soil contact, more stable moisture conditions, and greater access to decomposer communities. Litterbag mesh size and placement may also influence the accessibility of litter to soil fauna and microbial communities, potentially affecting decomposition and nutrient release.

Given ongoing efforts to expand and restore hedgerows across the UK, further research examining a wider range of hedgerow species, litter qualities, and decomposer communities across contrasting environmental conditions. Such research would improve understanding of the mechanism controlling decomposition and nutrient cycling in hedgerow systems and support evidence-based management strategies aimed at maximising both C sequestration and broader ecosystem service delivery.

Chapter 6: Synthesis

6.1. Outline of chapter

This final chapter synthesises the key findings presented in Chapters 2 to 5 and places them within a broader context of agroforestry as a climate mitigation strategy. Section 6.2 summarises the major findings from each chapter. Building on these findings, Section 6.3 integrates the results to synthesise an understanding of carbon (C) and nutrient dynamics in agroforestry systems.

The implications of the research for policy and practice are discussed in Section 6.4, with particular emphasis on the contribution of agroforestry to climate change mitigation. Section 6.5 outlines the study's limitations and the challenges encountered during the research process. Section 6.6 identifies priorities for future research, highlighting the need to improve understanding of the mechanisms governing C and nutrient dynamics and to support the development of more efficient agroforestry systems for climate change mitigation and sustainable land management. Finally, Section 6.7 concludes the thesis by summarising the principal findings, their significance, and their contribution to advancing knowledge of C and nutrient dynamics in agroforestry systems.

6.2. Summary of key findings

The overall aim of this thesis was to evaluate the aboveground and belowground interactions that lead to the quantification of C storage and accumulation in temperate agroforestry systems and to examine the biotic and abiotic factors influencing C and nutrient cycling. Specifically, this thesis evaluated the effects of agroforestry configuration, tree species, time, climate variability, soil nutrient availability, root dynamics, and litter decomposition on aboveground and belowground C pools and nutrient dynamics. I focused on the four research questions (RQ) to understand the C and nutrient dynamics in temperate agroforestry that were outlined in section 1.2.

1. How do aboveground biomass (AGB) C stocks and annual C increments vary among poplar hybrids over a 30-year period, and to what extent are these variations influenced by local climate variability?
2. What are the effects of a 30-year-old poplar-based silvoarable system on soil organic carbon (SOC) storage and sequestration, and how could expansion of such systems contribute to UK net-zero targets through enhanced C sequestration?
3. How do spatial variations in soil nutrient availability across locations and soil depths influence root traits, root dynamics, and belowground C inputs, turnover, and storage within an agroforestry system?
4. How does leaf litter location (surface vs. buried) influence decomposition rates and the dynamics of C and nitrogen (N) content during decomposition?

Chapters 2-5 address each of the RQs in turn.

Chapter 2 evaluated the AGB C sequestration potential of four poplar hybrids in a mature silvoarable agroforestry system. The results highlighted the value of PAIC for assessing long-term C dynamics in mature agroforestry systems, where growth and C accumulation are influenced by changing weather and environmental conditions. Trichobel outperformed the other hybrids, sequestering approximately 50% more C than the least productive hybrid, whereas Beaupre showed high early productivity followed by a decline, suggesting greater sensitivity to changing environmental conditions. Seasonal climatic variables, particularly summer conditions (R^2 = 43-59%, $p \leq 0.04$), had a stronger effect on C accumulation than annual weather conditions. Soil temperature negatively affected PAIC of Beaupre and Gibecq, reducing it by 22% and 38.2%, respectively, whereas rainfall and wind speed positively influenced Trichobel. Cropping history also influenced C accumulation, with continuous fallow (CF) supporting higher AGB C stocks (53.7%) and increment rates than continuous cropping (CC) during the active silvoarable phase, likely reflecting reduced competition for resources. Overall, the findings demonstrate the combined importance of genotype, climate, and management in determining long-term C sequestration.

Building on these findings, Chapter 3 quantified SOC stocks and sequestration in the same mature poplar silvoarable system and assessed the potential contribution of large-

scale expansion to UK net-zero targets. The silvoarable systems contained 27% more SOC than the adjacent conventional arable field (73.7 vs. 58.9 Mg C ha^{-1}) and achieved an average SOC sequestration rate of 0.55 Mg C ha^{-1} yr^{-1}. Scenario modelling showed that AGB was the dominant C pool, accounting for 66% of total sequestration, while belowground biomass (BGB) and SOC each contributed 17%. Trichobel again showed the greatest sequestration potential, storing approximately 52% more C than Gibecq. Depending on adoption scenario and hybrid selection, estimated CO2 removals ranged from 0.87 to 7.16 Tg CO_2 yr^{-1}, highlighting the significant climate change mitigation potential of agroforestry expansion in the UK.

Chapter 4 examined the interactions between root traits, soil nutrients, and SOC storage. Available nitrogen and P were positively associated with root growth, while bulk density showed negative relationships with root traits, indicating improved soil physical conditions under agroforestry. Although SOC stocks did not differ significantly among locations and depths, SOC concentrations increased at 45cm depth. Most live and dead root biomass (86-99%) was concentrated in the topsoil, and agroforestry also promoted deeper rooting than in the adjacent arable field. Tree rows and alleys supported substantially greater root development, with 80-90% higher live and dead root length density, biomass density, and root C stocks than the arable field. These findings emphasise the importance of root production, turnover, and necromass inputs as key drivers of belowground C accumulation and nutrient cycling.

Lastly, Chapter 5 investigated leaf litter decomposition and nutrient dynamics in temperate hedgerows. Decomposition followed a biphasic pattern, characterised by rapid mass loss (40-60%) during the first 223 days, followed by slower decomposition thereafter. Litter placement –was the main driver of decomposition, with buried litter decomposing significantly faster than surface litter ($p < 0.001$), while species effects were limited. Although hazel litter had higher N concentrations and lower C: N ratios than hawthorn, nutrient release was more strongly influenced by decomposition environment than species identity. Strong relationships between mass loss, N concentration and C:N ratio indicated the close coupling of decomposition and nutrient cycling. Overall, the study highlights the important role of hedgerows in organic matter turnover, nutrient cycling and C dynamics within agriculture.

6.3. Integration of key findings

Collectively, the findings demonstrate that C sequestration in temperate agroforestry systems is governed by interconnected biological and environmental processes operating across multiple C pools and temporal scales. While tree biomass represented the largest C pool, long-term sequestration depended on the combined contributions of root-derived C inputs, SOC, and decomposition-driven nutrient cycling.

The overall aim of the thesis is to examine the key biotic and abiotic factors controlling C storage and sequestration in temperate agroforestry systems by examining AGB, soil properties, root characteristics, and their influence on C and nutrient cycling processes. Overall, the thesis shows how species selection, time/age, tree density, cropping treatments, and climatic conditions interact to regulate C sequestration, C storage, and decomposition-driven nutrient cycling in agroforestry systems. The results of this thesis demonstrate that agroforestry contributes to C sequestration through multiple pathways operating both aboveground and belowground. Findings from the mature poplar-based silvoarable agroforestry system (Chapters 2-4) showed that C storage is distributed among AGB, BGB (based on root-shoot ratio), and SOC, highlighting the multifunctional role of trees in capturing and retaining atmospheric C. This is consistent with previous studies that identify agroforestry as an effective land-use strategy because C is stored simultaneously in vegetation and soil pools (Lorenz & Lal, 2014; Nair et al., 2009ab).

Fig. 6.1 summarises the major pathways contributing to C sequestration in agroforestry systems. In the mature silvoarable agroforestry system (Chapters 2-4), C sequestration was primarily driven by biomass accumulation, with AGB representing the largest C pool. Root growth, turnover, and necromass production contributed to BGB storage and SOC formation. Root-soil interactions, nutrient availability, and lower bulk density created favourable conditions for long-term C accumulation. In the hedgerow agroforestry system (Chapter 5), litter decomposition regulated C and nutrient cycling, with litter placement affecting decomposition rates and nutrient release. Together, these processes illustrate how different agroforestry configurations contribute to ecosystem C sequestration through biomass storage, root-mediated inputs, and organic matter turnover.

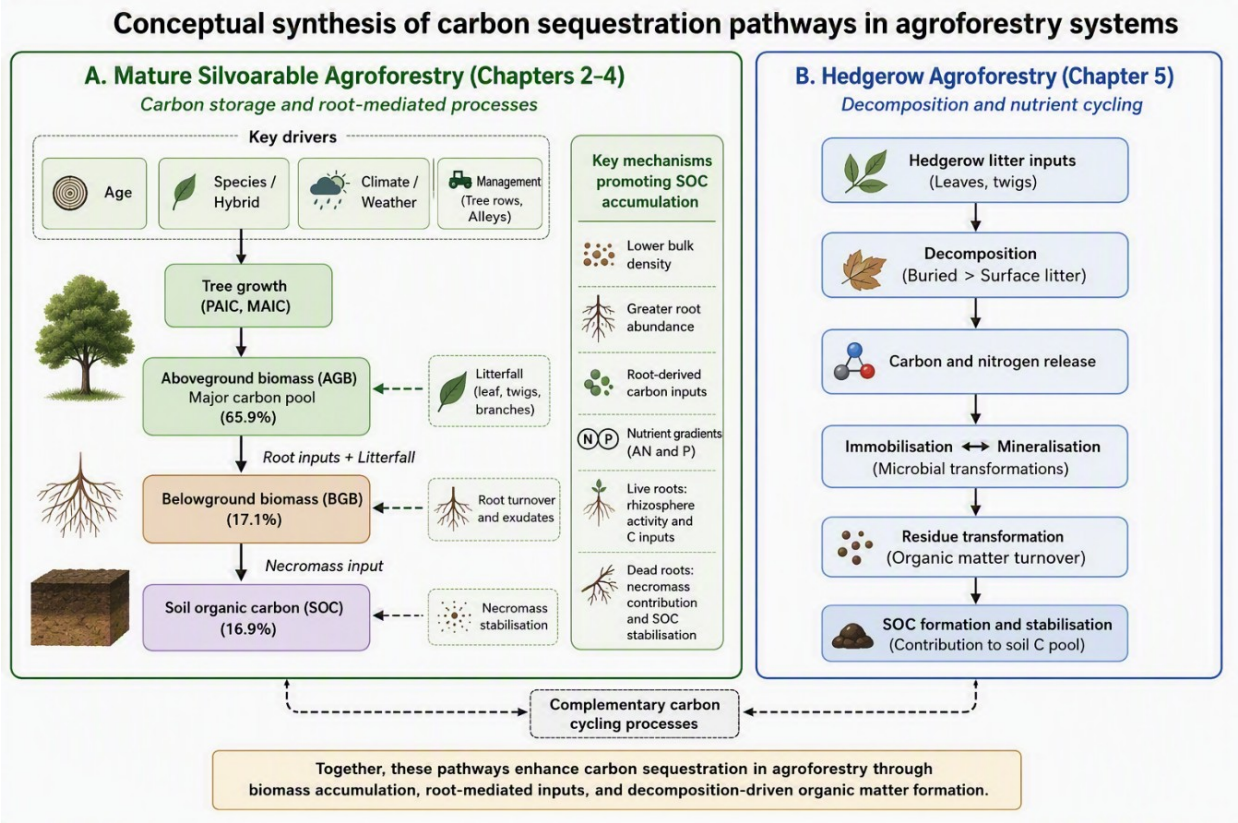


Figure 6.1 Conceptual pathways of carbon sequestration in agroforestry

Overall, Chapters 2-4 demonstrate how mature silvoarable agroforestry systems store C through biomass accumulation and root-mediated soil processes, whereas Chapter 5 highlights the contribution of litter decomposition to C and nutrient cycling within hedgerow agroforestry systems. Together, these findings show that both root turnover (Chapter 4) and litter decomposition (Chapter 5) represent pathways through which plant-derived C enters the soil. Although operating through different mechanisms, both contribute to long-term SOC formation. Hence, agroforestry enhances C sequestration through multiple, interconnected mechanisms operating across different agroforestry designs and C pools.

6.3.1. Role of hybrids and species

Across all chapters, species identity influenced the quantity of C entering agroforestry systems, whereas environmental and management factors more strongly regulated the rate at which C was retained, transformed, or recycled. Species and hybrid selection strongly influenced C sequestration across the agroforestry systems studied,

highlighting that the climate-mitigation potential of agroforestry depends not only on the presence of trees but also on the characteristics of the species selected. Within the mature poplar silvoarable system, substantial variation in long-term C accumulation was observed among hybrids. Trichobel achieved the highest mean AGB C stock (65.77 Mg C ha^{-1}), MAIC (2.13 Mg C ha^{-1} yr^{-1}) and PAIC (2.33 Mg C ha^{-1} yr^{-1}), and sequestered approximately 52% more C than Gibecq in the UK upscaling Scenarios. These findings demonstrate that hybrid selection can significantly influence biomass production and long-term C sequestration potential.

The results further suggest that early growth performance is not necessarily a reliable indicator of long-term C storage. Although Beaupre exhibited significantly greater height growth during the first seven years following establishment, Trichobel maintained superior C accumulation at maturity, indicating that long-term sequestration outcomes depend on interactions between genotype, environmental conditions and management practices. Similar genotype × environment interactions have been widely reported in poplar systems, where hybrid performance varies according to site conditions, competition and resource availability (Burgess et al., 2005).

Species selection also influenced C cycling processes. While Chapter 5 found limited effects of hedgerow species on litter mass loss and decomposition rates, species determined the quantity and quality of litter entering the soil. In contrast, environmental conditions and litter placement exerted a stronger influence on decomposition dynamics and nutrient release. These findings may suggest that species choice affects C inputs to soil, whereas management and environmental conditions largely regulate the rate at which plant-derived C is transformed and incorporated into soil organic matter (Cotrufo et al., 2013; Dong et al., 2026).

The findings also highlight important trade-offs between maximising C sequestration and delivering broader farm benefits. Fast-growing species such as poplar are frequently favoured in silvoarable agroforestry because they rapidly accumulate biomass, sequester large quantities of C and can produce marketable timber within relatively short rotations (Forest Research, 2024; Woodland Trust, 2023). In contrast, productive agroforestry species such as apple, hazel and walnut provide opportunities for income diversification through fruit and nut production while maintaining agricultural output between tree

rows (Agroforestry Research Trust, accessed 2026; Whittick, 2024a, 2024b). Current agri-environment schemes in England, including the Sustainable Farming Incentive (SFI) and Countryside Stewardship Higher Tier (CSHT), increasingly support such systems because of their wider environmental and economic benefits. However, the present study suggests that these systems are unlikely to achieve the same levels of biomass C storage as fast-growing poplar-based systems. Consequently, species selection should balance C sequestration objectives against income diversification, resilience, nutrient cycling, crop compatibility and long-term C permanence.

Overall, the results indicate that species and hybrid selection influence multiple components of agroforestry C dynamics, including biomass accumulation, litter inputs and nutrient cycling. Maximising the climate-change mitigation potential of agroforestry therefore requires selecting species that are appropriate not only for site conditions but also for the primary objectives of the system, whether these are C storage, agricultural productivity, income diversification or a combination of these outcomes.

6.3.2. Management and site design

Beyond species choice and system age, management practices and site design strongly influenced C accumulation and cycling within the agroforestry systems. In the mature poplar silvoarable system, alley management affected AGB C sequestration, particularly during the establishment and active cropping phases when competition between trees and crops was greatest. The CF treatment maintained higher AGB C stocks and greater C accumulation rates than treatments where crops were present, likely because reduced competition for light, water, and nutrients promoted greater tree growth. In contrast, crop production on either side of the tree rows constrained tree growth and C accumulation, highlighting the importance of crop-tree interactions in determining stand productivity (Burgess et al., 2005; Jose, 2009). Following the cessation of cropping, treatment differences may progressively decrease, indicating that the effects of alley management were largely associated with the period of active competition. These findings suggest that minimising crop-tree competition may be advantageous where maximising long-term AGB C sequestration is a primary management objective.

Management effects were also evident belowground. The significant difference between CF and CC treatment effects observed in SOC stock, particularly in Blocks 2 and 3, suggests that historical differences in C accumulation persisted despite the cessation of cropping. However, the long period under unmanaged vegetation likely attenuated treatment contrasts, which may explain the absence of significant differences in SOC concentration and bulk density. Thus, the current SOC stock pattern may appear to reflect both the legacy of past silvoarable management and the subsequent transition toward a vegetation-dominated system. Furthermore, analysis of soil samples from multiple sampling points revealed significantly higher SOC stocks in the agroforestry zones than in the conventional arable field in the 0-15 cm soil layer. However, when SOC stocks were assessed using soil samples collected from only three representative pits and evaluated at a finer spatial scale (5-45 cm depth), no significant differences were detected among the arable field, tree rows, and alley positions. This discrepancy likely reflects the influence of sampling intensity and spatial variability in SOC distribution, whereby a larger number of sampling points provided greater statistical power to detect differences between land-use systems.

Agroforestry zones supported substantially greater root length density, root biomass and root C stocks than conventional arable land, while lower bulk density and deeper rooting patterns created favourable conditions for continued C inputs to the soil profile (Cardinael et al., 2018). Although changes in SOC stocks were not always detectable at fine spatial scales, root turnover, rhizodeposition and root necromass provided a continuous pathway for transferring plant-derived C into the soil, contributing to long-term soil C accumulation and stabilisation (Rasse et al., 2005; Cardinael et al., 2015; Kätterer et al., 2011; Pankaj et al., 2023).

Similarly, findings from the hedgerow system demonstrated the importance of leaf litter decomposition as a complementary pathway linking plant biomass production to soil C and nutrient cycling. The faster decomposition of buried litter highlighted how management can influence the rate at which C and nutrients are transferred into the soil. Changes in C, N, and C: N ratios during decomposition indicated active immobilisation and mineralisation processes that regulate nutrient cycling and contribute to soil organic matter formation (Cotrufo et al., 2013; Dong et al., 2026). Root turnover and litter decomposition represent complementary pathways linking plant production to soil C

formation. Whereas root-derived C enters the soil directly through root mortality, exudation, and rhizodeposition, litter-derived C contributes through aboveground organic matter inputs and subsequent decomposition. Together, these processes form the principal biological pathways through which agroforestry systems transfer atmospheric C into longer-term soil storage. While root turnover (Chapter 4) and litter decomposition (Chapter 5) may work through different mechanisms, both represent important pathways through which plant-derived C enters the soil and contributes to long-term SOC formation (Rasse et al., 2005; Cotrufo et al., 2013).

Together, the findings from Chapters 2 to 5 demonstrate that agroforestry enhances C sequestration through multiple interconnected mechanisms. Aboveground carbon is accumulated through tree biomass growth, while BGB C storage is supported by root production, root turnover, rhizodeposition and litter decomposition. The relative contribution of these processes is influenced by management decisions throughout the lifetime of the system, emphasising that agroforestry performance depends not only on tree establishment but also on how systems are managed over time (Jose, 2009; Cardinael et al., 2018b).

Site design further influenced the magnitude of C storage. Wider alleys supported fewer trees per hectare and therefore lower AGB C at the system scale, whereas narrower alleys and higher tree densities increased biomass C storage by allocating a greater proportion of land to trees. However, increasing tree density may also intensify competition with crops for light, water and nutrients. Tree spacing additionally affects root distribution, litter inputs and rhizosphere processes, influencing both aboveground and belowground C dynamics. Consequently, maximising agroforestry C sequestration requires balancing tree density, spatial arrangement and crop productivity, highlighting the need for site-specific system designs rather than relying on generic C sequestration estimates (Ivezić et al., 2022; Muñoz-Rojas et al., 2023; Sutterlütti et al., 2023).

6.3.3. Tree age and system maturity

The findings of this thesis demonstrate that system age is a fundamental determinant of C sequestration in agroforestry systems. Much of the current evidence base is derived from young agroforestry systems, whereas the present research evaluated a mature

silvoarable system approximately 30 years after establishment. The results showed that C in AGB continued to accumulate over multiple decades, highlighting the importance of long-term monitoring for accurately quantifying agroforestry climate mitigation potential.

The divergence among hybrids became more pronounced over time. Between 2005 and 2023, Trichobel's MAIC increased by 34.8%, compared with increases of only 10.8%–13.8% for the other hybrids. By 2023, Trichobel maintained a MAIC that was 49.0%–77.5% higher than Beaupre, Robusta, and Gibecq. Similar trends were observed for PAIC. Although Beaupre exhibited the highest PAIC during establishment, Trichobel became dominant by 2005, exceeding the other hybrids by 36.8%–44.3%. This advantage continued through stand development, and by 2023 Trichobel's PAIC was 171%–455% greater than the other hybrids. These findings indicate that Trichobel sustained substantially higher C sequestration rates at maturity, whereas the other hybrids showed a stronger age-related decline in productivity.

Importantly, had the assessment been conducted only during the first decade following establishment, Beaupre would have been identified as the best-performing hybrid. However, the 30-year dataset showed that Trichobel ultimately accumulated approximately 65% more C than Beaupre and 95% more than Gibecq. This demonstrates that short-term studies may substantially misrepresent lifetime C sequestration potential and reinforces the need for mature-system datasets when developing C accounting methodologies, sequestration factors and agroforestry policy recommendations.

Rather than directly explaining the SOC patterns observed in the mature silvoarable system, the hedgerow study complements the broader understanding of agroforestry highlighting how litter quality, residue placement, and nutrient dynamics influence the transformation of plant-derived C into soil organic matter. Fluctuations in N concentration and C:N ratio during decomposition reflected periods of immobilisation and mineralisation, suggesting that part of the decomposing organic matter may contribute to the formation of longer-term soil C. Overall, my research in Chapters 2-4 demonstrates how mature silvoarable agroforestry systems store C through biomass accumulation and root-mediated soil processes, whereas Chapter 5 highlights the

contribution of litter decomposition to C and nutrient cycling within hedgerow agroforestry systems. Together, these findings show that agroforestry enhances C sequestration through multiple, interconnected mechanisms operating across different agroforestry designs and C pools.

6.3.4. Weather conditions

Weather was an important driver of tree C accumulation throughout the study period. In Chapter 2, analysis of weather records from 1993-2023 showed that summer climatic conditions, in line with expectations (Kutsokon et al., 2015; Rosso et al., 2023; Gudynaitė-Franckevičienė et al., 2022), exerted the strongest influence on PAIC, particularly for Trichobel and Beaupre. Notably, Trichobel maintained high C accumulation despite climatic variability, indicating a greater capacity to tolerate environmental stress and sustain growth under adverse weather conditions. Genotypes with greater tolerance to weather variability are likely to provide more stable long-term C sequestration under future climate change scenarios (Navarro et al., 2014; Mishra et al., 2025). Overall, the results indicate contrasting climatic sensitivities among the hybrids. Higher soil temperatures reduced PAIC in Beaupre and Gibecq, suggesting greater vulnerability to heat and associated drought stress. In contrast, Trichobel responded positively to summer rainfall and wind speed, highlighting a greater capacity to maintain productivity under variable climatic conditions. Importantly, weather effects were not limited to the active growing season. Positive associations between autumn and winter wind speed and PAIC in Trichobel suggest that favourable conditions during dormancy and the pre-growing season may support carbohydrate storage, bud development, and subsequent cambial activity, thereby contributing to greater annual C accumulation. Conversely, the negative effect of spring wind speed on Beaupre indicates greater sensitivity to early-season climatic stress.

These findings are particularly relevant in the context of UK climate change projections, which indicate continued warming, hotter and drier summers, more frequent droughts, and increased climate variability (Lowe et al., 2018; Kendon et al., 2019). Consequently, future agroforestry deployment strategies should account not only for current productivity but also for climate resilience. The superior long-term performance

of Trichobel suggests that genotypes that maintain growth under increasing heat and moisture stress are likely to provide more reliable long-term C sequestration.

6.3.5. Agroforestry as a climate mitigation strategy

This thesis demonstrates that silvoarable agroforestry can make a significant contribution to climate change mitigation, despite the study system only being actively managed as agroforestry between 1993 and 2004, through C sequestration in biomass, roots, and soils and improvements in ecosystem functioning (Jose, 2009; Cardinael et al., 2015; Cardinael et al., 2018a).

Scenario projections suggest that UK-wide adoption of silvoarable agroforestry could sequester 0.87-7.16 Tg CO_2 yr^{-1}, depending on the scale of implementation and hybrid selected. Carbon sequestration was dominated by tree biomass (AGB + BGB) accumulation (83%), with Trichobel consistently achieving the highest sequestration rates. As a result, agroforestry could offset approximately 3.1-17.3% of projected UK agricultural emissions in 2050, depending on adoption scenario (Department for Energy Security and Net Zero, 2022).

The projected C trajectories indicate that sequestration rates are unlikely to increase indefinitely. As trees mature, growth rates decline and systems move towards a steady state, reducing annual C uptake over time (Truax et al., 2014). This pattern was evident in the present study, where PAIC declined during the final decade for Gibecq, Robusta, and Beaupre, indicating slower biomass growth and C accumulation with stand age.

A key finding is the trade-off between maximum C sequestration and practical implementation. While high-adoption scenarios achieved the greatest C removals, lower-adoption scenarios reduced total sequestration by approximately 70-76% but may be more realistic and achievable within existing farming systems (Westaway et al., 2023). Therefore, although agroforestry represents an effective nature-based climate solution, it is unlikely to deliver net-zero targets alone and will need to be implemented alongside other mitigation measures (Ntawuruhunga et al., 2023; Rosenstock et al., 2023; Westaway et al., 2023).

In summary, the results indicate that agroforestry should be viewed as a multifunctional climate mitigation strategy capable of delivering substantial and sustained C sequestration while simultaneously supporting soil health, nutrient cycling, biodiversity, and agricultural resilience. However, the magnitude and permanence of these benefits depend on species selection, system design, management practices, and long-term stewardship.

6.4. Policy implications

My findings demonstrate that both agroforestry support policies and wider net-zero strategies should move beyond generic C sequestration estimates and recognise that C and nutrient outcomes vary according to species or hybrid, tree age, planting density, alley width, management history, and climatic conditions. Consequently, incentive schemes and C accounting frameworks should support site-specific system design and assessment rather than prescribing a single agroforestry configuration or C value.

The management practices outlined in Table 6.1 are broadly consistent with the findings of this thesis; however, the results suggest that current policy frameworks could place greater emphasis on factors influencing the long-term stability of C sequestration. Existing schemes, including the Sustainable Farming Incentive (SFI), Environmental Land Management (ELM) schemes, and the Woodland Carbon Code, primarily encourage tree establishment, woodland expansion, and C storage (Table 6.1).

In addition, while Table 6.1 highlights practices that enhance biomass production and C storage, the results of Chapters 3 and 4 demonstrate that substantial C benefits are also derived from belowground processes. Current C accounting approaches may therefore underestimate the contribution of root biomass, root-derived C inputs, and SOC. Future policy frameworks could adopt a more comprehensive whole-system accounting approach that explicitly recognises aboveground, belowground, and soil C pools when assessing agroforestry C benefits, improving the accuracy and credibility of C reporting and incentive schemes. Another policy implication is the need for robust monitoring and verification systems. The substantial variation observed among hybrids, cropping management treatments, and C pools suggests that the use of generic sequestration factors may result in over- or under-estimation of agroforestry C benefits.

Table 6.1 Trade-offs, benefits and UK policy instruments

Management Practice	Benefit	Trade-off	Relevant UK Policies references
Higher tree density	Help to enhance C sequestration and biomass storage (Nair et al., 2009; Jose, 2009b)	Potential reduction in crop yields - growth of trees due to competition for resources	Sustainable Farming Incentive (SFI) agroforestry actions can support integration of trees into productive farmland while maintaining agricultural output (UK Government, 2026).
Retention of mature trees	Accumulation of large C stocks and nutrient cycling - biodiversity (Montagnini and Nair, 2004)	Delayed timber income-reduced harvesting opportunities	Woodland Carbon Code and Environmental Land Management (ELM) payments could help compensate landowners for maintaining long-term C stocks (Defra, 2023ab).
Longer rotation periods	Increased long-term C storage and higher-value timber (Proe et al., 2013; Smith et al., 2012)	Reduced short-term cash flow	Carbon market participation and Landscape Recovery projects may incentivise long-term woodland management (Farm Woodland Forum, 2026).
Diversified tree species mixtures	Enhanced biodiversity, resilience and ecosystem services (Jose, 2009; Torralba et al., 2016)	Greater establishment management costs	Countryside Stewardship and Higher Tier Capital Grants provide support for species-diverse planting and agroforestry establishment (Defra, 2023b).
Fast-growing species	Rapid C accumulation and earlier economic returns (Tallis et al., 2013; Arevalo et al., 2011)	Lower biodiversity value and ecological resilience	UK Forestry Standard and species-diversity incentives encourage balanced species selection rather than C maximisation alone (UK Government, 2026).
Conversion of cropland to agroforestry	Increased C storage - soil and climate resilience (Cardinael et al., 2018a; Nair et al., 2010)	Potential reduction in land available for annual crops	SFI agroforestry actions aim to integrate trees while maintaining productive agriculture (Defra, 2024).
Increased canopy cover	Improved - soil, water retention and microclimate regulation (Dupraz and Liagre, 2008; Jose, 2009)	Reduced light availability for understory crops	Agroforestry planning support helps optimise tree placement and canopy management (Defra 2023b).
Reduced harvesting intensity	Increased C storage and habitat quality (Proe et al., 2013; Nair et al., 2009ab)	Lower timber revenues	Sustainable forest management certification and ELM environmental payments can reward ecosystem service provision (Defra, 2023b).
Retention of deadwood and litter	Improved soil C and biodiversity (Bardgett and Wardle, 2010; Montagnini and Nair, 2004)	Operational and management constraints	Biodiversity-focused woodland management supported under Countryside Stewardship and wider nature recovery objectives (UK Government, 2026).
Carbon-focused management	Maximises climate mitigation benefits (Smith et al., 2012; Nair et al., 2010)	May reduce emphasis on food production or farm profitability	ELM schemes seek to balance food production with public goods through multifunctional land-use planning (Defra,

The findings also support several management recommendations identified in Table 6.1, including longer-term tree retention, diversified planting, and maintenance of vegetative cover. However, the observed influence of seasonal climatic conditions on C accumulation suggests that policy mechanisms should move beyond static estimates of C storage and incorporate long-term monitoring requirements. Monitoring of biomass growth, soil C, root dynamics, and climate sensitivity would improve confidence in C accounting while helping to ensure the permanence of C benefits under changing climatic conditions.

Finally, Table 6.1 emphasises management practices that increase C storage, but the findings of this thesis indicate that C sequestration outcomes are strongly influenced by interactions among management, species selection, and climate. Consequently, future agroforestry policies may benefit from a greater focus on adaptive, site-specific management approaches rather than applying uniform prescriptions across systems. Such modifications would better align policy incentives with the processes driving long-term C sequestration identified in this study and strengthen the contribution of agroforestry to the UK's net-zero, climate adaptation, and biodiversity objectives.

6.5. Limitations

While this thesis provides a long-term assessment of C dynamics in a mature temperate agroforestry system, several limitations should be acknowledged. Across Chapters 2 and 3, uncertainties in C stock and sequestration estimates arise from the use of allometric equations, as species-specific models for poplar hybrids are limited and consistent tree height measurements were unavailable over the 30-year study period. Consequently, AGB was estimated using diameter alone, while BGB was modelled using a fixed root-to-shoot ratio (BGB = 0.26 × AGB), introducing additional uncertainty into C sequestration estimates. The upscaling scenarios presented in Chapter 3 should therefore be regarded as indicative assessments of potential C sequestration rather than predictions of future net-zero outcomes, as they do not fully account for uncertainties in farmer adoption, land suitability, management practices, economic constraints, or future climate change. In Chapter 4, root biomass estimates may have

been underestimated due to losses of fine roots during washing and sieving, while distinguishing tree roots from grass and crop roots was challenging, a limitation also reported in agroforestry root studies (Upson and Burgess, 2013). A key limitation of Chapters 2- 4 is that the silvoarable experimental site did not remain a strictly silvoarable system throughout the study period. Following the cessation of cropping, the development of permanent grass cover altered belowground processes and likely contributed to SOC accumulation through additional root inputs and reduced soil disturbance (Ma et al., 2020). Furthermore, because the study was conducted at a single long-term experimental site, caution should be exercised when extrapolating the findings to other temperate agroforestry systems differing in soil type, climate, management intensity, or tree species composition. Although the site provides valuable insights into long-term agroforestry development, broader geographic replication would improve confidence in the generalisability of the results. Therefore, the SOC sequestration rates reported here should be interpreted as reflecting the long-term evolution of a mature agroforestry system rather than a continuously managed silvoarable system.

Finally, Chapter 5 was subject to methodological uncertainties associated with litterbag studies, including the influence of hedgerow management on litter inputs, potential mesh-size effects on decomposition processes, and challenges separating highly decomposed litter from soil material. Despite these limitations, the consistent patterns observed across biomass, soil, root, and litter C pools provide strong evidence that mature agroforestry systems enhance long-term C storage and ecosystem functioning. The study focused on C sequestration within biomass and soils and did not incorporate full life-cycle greenhouse gas accounting, including emissions associated with establishment, management operations, harvesting, processing or product substitution effects.

While these limitations introduce uncertainty into some estimates, they are unlikely to alter the overall conclusions of the thesis because consistent patterns were observed across AGB, soil, root, and litter C pools. Collectively, the findings provide robust evidence that mature agroforestry systems enhance long-term C storage and support key ecosystem functions.

6.6. Future work

6.6.1. Improving carbon accounting and upscaling

Improving Future research should evaluate C sequestration across mature agroforestry systems located on different soil types and along climatic gradients. Multi-site studies would improve understanding of how species composition, vegetation interactions, soil properties, temperature, and rainfall influence long-term C storage and sequestration rates (Jose, 2009; Cardinael et al., 2018b; Cardinael et al., 2025). Further work is also required to strengthen large-scale projections by better quantifying farmer adoption rates, land suitability, species- and site-specific sequestration rates, management impacts, and the long-term stability of C stocks under changing climatic conditions (Graves et al., 2010; Cardinael et al., 2025). Improved understanding of these factors would enable more robust regional and national-scale upscaling of agroforestry and provide stronger evidence for assessing its contribution to net-zero targets and informing land-use policy (Kay et al., 2019; Cardinael et al., 2025).

Future studies should also assess the implications of harvesting trees at their biological rotation age, as extending rotations may increase both C storage and timber value (Fortier et al., 2017). In addition, C-credit valuation and C-pricing mechanisms should be incorporated into cost-benefit analyses to quantify the financial benefits associated with ecosystem service provision and C sequestration (Kumar et al., 2020; Awazi et al., 2025c). Greater integration between scientific projections and policy implementation is also needed to support evidence-based climate mitigation strategies (Kay et al., 2019; Cardinael et al., 2025).

6.6.2. Improving understanding of belowground processes

The findings of Chapter 4 highlighted the importance of root-mediated processes in regulating nutrient availability and soil C inputs. Future work should investigate how management practices such as pruning, thinning, pollarding, and lopping influence root turnover, nutrient cycling, and C sequestration (Jose, 2009; Udawatta and Jose, 2011).

Management strategies should be evaluated across spatial and temporal scales to optimise tree-crop interactions and maximise system productivity.

Particular attention should be given to root exudates, rhizosphere processes, and root-microbial interactions, which play important roles in nutrient cycling and may reduce dependence on synthetic fertilisers (Panchal et al., 2022). The contribution of different root classes to soil C sequestration also requires further investigation (Rasse et al., 2005; Ma et al., 2022). Deep-rooted species, such as poplar, may enhance C storage in deeper soil horizons through coarse-root turnover and long-residence-time residues (Tufekcioglu et al., 1999; Cardinael et al., 2018b). However, these processes may also stimulate C losses through increased microbial activity (Ma et al., 2022). Future research should therefore focus on quantifying the balance between C sequestration and C emissions in deeper soil layers and improving understanding of the mechanisms controlling long-term C persistence and stability within agroforestry systems (Cotrufo et al., 2013; Lavallee et al., 2020).

6.6.2. Improving monitoring and modelling approaches

Improving the accuracy of biomass and C estimates remains a key research priority. Future studies should obtain detailed measurements of tree height, stem density, and crown characteristics, while destructive sampling should be used where feasible to develop and validate species-specific allometric equations for poplar hybrids. Such approaches would improve estimates of both AGB and BGB (Fortier et al., 2017; Cardinael et al., 2025).

Emerging monitoring technologies, including LiDAR, UAV-based photogrammetry, and satellite observations, offer considerable potential to improve long-term assessments of agroforestry C stocks. These tools can provide more accurate and spatially explicit estimates of tree growth, canopy development, biomass accumulation, and C sequestration while reducing the need for repeated field measurements (Cardinael et al., 2025).

In parallel, process-based agroforestry models such as WaNuLCAS, SCUAF, HyPAR, and HyCAS should be further developed and applied to investigate interactions among trees, crops, soils, and microbial communities under different management scenarios

(Matthews et al., 2004). Integrating improved field measurements, remote sensing technologies, and process-based models will strengthen national-scale C projections and support more reliable assessments of agroforestry's contribution to climate-change mitigation (Graves et al., 2010; Cardinael et al., 2025).

Overall, future research should move beyond quantifying C stocks alone and focus on improving C accounting, understanding the biological processes governing C persistence, and developing robust monitoring and modelling approaches. This will be essential for increasing confidence in long-term sequestration estimates and supporting the inclusion of agroforestry within climate mitigation and net-zero strategies (Cardinael et al., 2018b; Cardinael et al., 2025).

6.7. Conclusion

My thesis demonstrates that agroforestry is a multifunctional land-use system capable of delivering substantial climate-change mitigation while enhancing soil health and nutrient cycling, with C sequestration governed by interconnected aboveground and belowground pathways operating across multiple C pools. While AGB represented the largest C pool, long-term C storage was also strongly influenced by root dynamics, soil organic C accumulation, litter decomposition, and nutrient cycling. Together, these processes determine how atmospheric C is captured, transferred, and retained within agroforestry systems.

Across all chapters, species or hybrid selection, management history, climatic conditions, and system maturity emerged as the dominant controls of C sequestration outcomes. The long-term dataset demonstrated that hybrid rankings changed substantially through time. Although Beaupre exhibited superior early growth, Trichobel ultimately accumulated the greatest C stock and maintained the highest sequestration rates at maturity. These findings highlight the importance of long-term monitoring, as short-term studies may underestimate or misrepresent the lifetime C sequestration potential of agroforestry systems.

The results further showed that agroforestry cannot be characterised using generic C sequestration values. Instead, sequestration potential is highly dependent on system

design, species selection, management practices, and site-specific environmental conditions. Consequently, both C accounting and policy development should adopt whole-system approaches that recognise the contributions of AGB, roots, soils, and organic matter dynamics, rather than relying solely on biomass-based estimates.

At larger scales, the upscaling scenarios indicate that agroforestry can make a meaningful contribution to climate-change mitigation and national net-zero ambitions. However, the magnitude of this contribution will depend on rates of adoption, landscape suitability, management decisions, and long-term system performance. Therefore, future agroforestry deployment strategies should consider not only current productivity but also long-term resilience and stability under changing climatic conditions.

Collectively, my thesis provides new evidence from a rare mature temperate agroforestry system and demonstrates that well-designed agroforestry can function as a resilient and multifunctional land-use strategy. By integrating C sequestration, soil health, nutrient cycling, and agricultural production, agroforestry has considerable potential to support climate mitigation, climate adaptation, and sustainable land-use objectives in the UK and beyond.

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Appendix A. Supporting information for Chapter 2

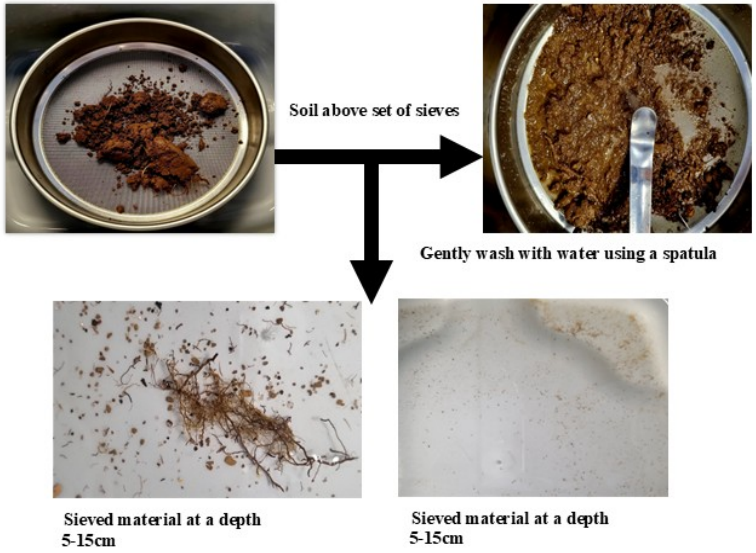
Supplementary Table A.1 Comparison of weather-based models explaining variation in Periodic annual C increment (PAIC) across hybrids.

Model performance is summarized using sample size (n), number of predictors (k), Akaike Information Criterion (AIC), McFadden's pseudo-R^2, and overall model significance. Lower AIC values indicate better model fit, whereas higher McFadden's pseudo-R^2 values indicate greater explanatory power.

Response	Hybrids	Weather variables	n	k	AIC	McFadden R2	Overall p-value
PAIC_B	Beaupre	Summer	23	7	38.36	0.487	0.0079
PAIC_B	Beaupre	Summer half -year	23	8	46.66	0.328	0.2131
PAIC_B	Beaupre	Winter	24	5	47.01	0.205	0.1763
PAIC_B	Beaupre	Autumn	23	7	47.51	0.256	0.3085
PAIC_B	Beaupre	Winter half- year	24	6	52.81	0.113	0.7031
PAIC_G	Gibecq	Summer	23	7	39.59	0.425	0.0357
PAIC_G	Gibecq	Spring	23	6	47.42	0.163	0.5776
PAIC_G	Gibecq	Summer half- year	23	8	47.97	0.255	0.3643
PAIC_G	Gibecq	Winter	24	5	49.9	0.089	0.6093
PAIC_G	Gibecq	Autumn	23	7	49.9	0.151	0.6784
PAIC_G	Gibecq	Winter half -year	24	6	53.62	0.045	0.953
PAIC_R	Robusta	Summer	23	7	30.07	0.585	0.023
PAIC_R	Robusta	Autumn	23	7	35.13	0.411	0.1256
PAIC_R	Robusta	Spring	23	6	39.96	0.177	0.658
PAIC_R	Robusta	Winter	24	5	40.8	0.179	0.3731
PAIC_R	Robusta	Summer half- year	23	8	43.44	0.195	0.8243
PAIC_R	Robusta	Winter half -year	24	6	45.48	0.097	0.8474
PAIC_T	Trichobel	Summer	23	7	46.4	0.47	<0.001
PAIC_T	Trichobel	Autumn	23	7	56.58	0.279	0.0254
PAIC_T	Trichobel	Summer half -year	23	8	61.17	0.231	0.1809
PAIC_T	Trichobel	Winter	24	5	63.28	0.122	0.2583
PAIC_T	Trichobel	Spring	23	6	63.34	0.116	0.3877
PAIC_T	Trichobel	Winter half -year	24	6	64.44	0.137	0.3506

Supplementary Table A.1 shows the performance of weather-based models fitted for each hybrid group across six temporal periods (spring, summer, autumn, winter, summer-half, and winter-half year). Model performance was evaluated using Akaike Information Criterion (AIC), log-likelihood, McFadden's pseudo-R^2, and likelihood-ratio test statistics. Weather predictors included seasonal measures of air temperature, soil temperature, rainfall, sunshine duration, ground frost days, air frost days, growing degree days, wind speed, and windy days. These results indicate that weather variables during the summer weather were most strongly associated with variation in PAIC across all hybrids, whereas climatic conditions outside the summer season explained relatively little additional variations.

Appendix B. Supporting information for Chapter 4



Supplementary Figure B.1 Schematic process of root washing with example images from the samples

Supplementary Table B.1 Mean (and median) of soil nutrients and bulk density (raw data)

Significant differences are indicated by letters: depths — 5 cm (a), 15 cm (b), 25 cm (c), 35 cm (d), and 45 cm (e); locations— Alley (a), Arable (b), and Rows (c) in bold. Asterisks (*) denote significant depth × location interactions. (Bold letters indicate treatments, regular letters represent depths within the same location, and bold letters with stars denote depths across different locations (depth × location)

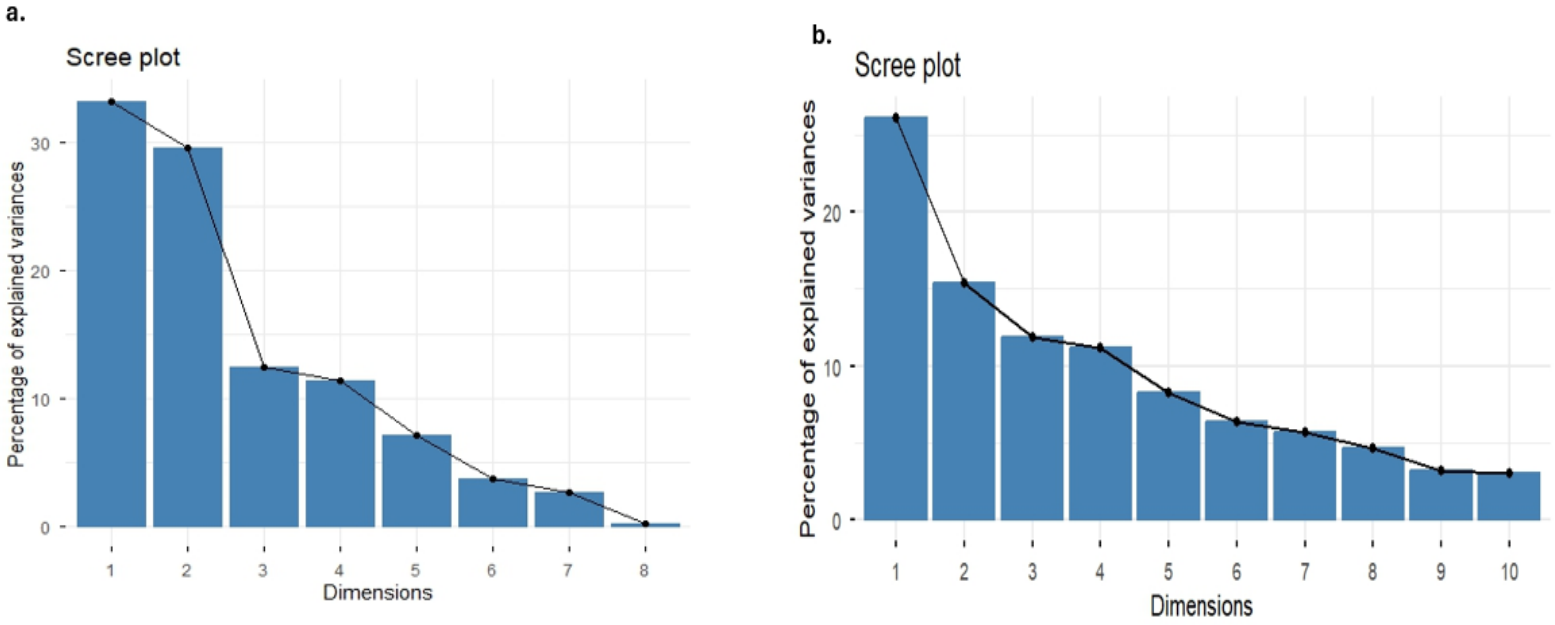
Treatment	Depth	AN (mg kg^{-1})	P (mg kg^{-1})	SOC%	N%	SOC%	SOC stock (Mg ha^{-1})	Bulk density (g cm^{-3})
Alley (n= 19)	5	5.46 (5.19) bcde	23.25 (24.50) b**b**	1.80 (1.74)	0.19 (0.19)	9.52 (9.36)	21.43 (21.30)	1.20 (1.20) **b**
	15	1.70 (1.65) a	10.25 (10.25) a**b**	0.92 (0.91)	0.11 (0.11)	7.74 (7.93)	12.07 (11.57)	1.32 (1.32) **b**
	25	1.09 (1.73) a	8.00 (8.50) a**b**	0.81 (0.78)	0.09 (0.09)	8.37 (8.70)	11.10 (11.35)	1.34 (1.38) **b**
	35	1.54 (1.16) a	6.25 (6.00) a**b**	1.11 (1.28)	0.17 (0.17)	6.81 (8.22)	14.41 (14.76)	1.35 (1.37) **b**
	45	0.55 (0.18) a	4.67 (2.00) a**b**	1.87 (1.69)	0.19 (0.18)	9.60 (9.55)	20.25 (20.92)	1.13 (1.17) **b**
Arable (n=9)	5	2.67 (2.70) bc	39.25 (41.00)bc**ac**	1.43 (1.42)	0.16 (0.16)	9.24 (9.18)	22.64 (22.69)	1.59 (1.59) **ac**
	15	3.37 (2.15) a	25.25 (24.50) a**ac**	1.04 (0.97)	0.13 (0.14)	7.86 (7.56)	15.43 (13.48)	1.50 (1.48) **ac**
	25	0.88 (0.88) a	6.00 (6.00) a**ac**	1.59 (1.59)	0.17 (0.17)	9.33 (9.33)	25.18 (25.18)	1.59 (1.59) **ac**
Row (n=19)	5	5.90 (5.99) bcde	26.75 (27.00) bcde**b**	1.49 (1.28)	0.19 (0.19)	8.35 (9.32)	16.99 (16.23)	1.18 (1.27) **b**
	15	2.23 (1.99) a	13.25 (14.00) a**b**	1.17 (0.96)	0.18 (0.11)	7.23 (7.32)	14.76 (9.02)	1.25 (1.37) **b**
	25	1.19 (1.35) a	12.00 (14.50) a**b**	1.49 (1.71)	0.15 (0.15)	9.88 (9.35)	22.39 (23.45)	1.50 (1.55) **b**
	35	0.85 (0.77) a	3.00 (1.50) a**b**	1.43 (1.03)	0.16 (0.15)	8.17 (8.02)	20.84 (16.40)	1.52 (1.55) **b**
	45	0.46 (0.48) a	2.00 (1.00) a**b**	1.82 (1.72)	0.19 (0.18)	9.55 (9.67)	21.90 (24.48)	1.17 (1.36) **b**

Supplementary Table B.2 Mean (and median) of live root traits

Treatment	Depth	LRD (cm)	LRLD (cm cm^{-3})	LRBD (Mg ha^{-1})	LRN %	LRC %	LRCS (Mg ha^{-1})	LRCN
Alley (n=19)	5	0.28 (0.26) cde**b**	38.12 (0.27) cde**b**	0.89 (0.40) cde**b**	1.01 (1.01) **bb***	44.98 (44.98)	3.99 (2.62) cde**b**	44.53 (44.53) **b**
	15	0.21 (0.20) **b**	8.93 (0.10) **b**	0.08 (0.09) **b**	1.39 (1.03) a**bb***	41.32 (40.39)	0.91 (1.06) **b**	29.73 (39.21) **b**
	25	0.71 (0.24)	5.23 (0.05) a**b**	0.02 (0.02) a**b**	1.36 (1.36) **bb***	41.44 (41.61)	0.24 (0.19) a**b**	30.47 (30.60) **b**
	35	0.24 (0.25)	2.67 (0.02) a**b**	0.01 (0.003) a**b**	1.22 (1.06) **bb***	34.73 (41.80)	0.15 (0.08) a**b**	28.47 (39.43) **b**
	45	0.22 (0.23)	3.51 (0.02) a**b**	0.02 (0.01) a**b**	1.16 (1.16) **bb***	39.89 (39.78)	0.26 (0.09) a**b**	34.39 (34.29) **b**
Arable (n=9)	5	0.21 (0.22)	4.32 (0.03) c**ac**	0.01 (0.01) **ac**	1.77 (1.84)	37.56 (37.50)	0.12 (0.12) c**ac**	21.22 (20.38) **ac**
	15	0.92 (0.29)	1.28 (0.01) **ac**	0.01 (0.01) **ac**	3.31 (3.23) **b*c*d*e*ac**	43.11 (45.83)	0.10 (0.08) **ac**	13.02 (14.19) **ac**
	25	0.21 (0.21)	0.77 (0.01) a**ac**			36.20 (36.20)	0.01 (0.01) a**ac**	
Row (n=19)	5	0.31 (0.25)	18.41 (0.16) cde**b**	0.12 (0.12) cde**b**	1.27 (1.31) **bb***	41.56 (41.72)	1.27 (1.29) cde**b**	32.72 (31.85) **b**
	15	0.69 (0.24)	14.88 (0.16) **b**	0.05 (0.05) **b**	0.71 (0.73) **bb***	42.55 (41.51)	0.51 (0.52) a**b**	59.93 (56.86) **b**
	25	0.28 (0.27)	4.72 (0.03) a**b**	0.01 (0.01) a**b**	0.64 (0.78) **bb***	34.60 (33.93)	0.13 (0.10) a**b**	54.06 (43.50) **b**
	35	0.33 (0.26)	3.66 (0.03) a**b**	0.02 (0.02) a**b**	0.94 (0.94) **bb***	29.36 (32.26)	0.20 (0.20) a**b**	31.23 (34.31) **b**
	45	0.23 (0.23)	1.47 (0.01) a**b**	0.01 (0.01) a**b**	0.12 (0.12) **bb***	27.82 (23.08)	0.07 (0.04) a**b**	23.18 (19.23) **b**

Supplementary Table B.3 Mean (and median) of dead root traits

Treatment	Depth	DRD (cm)	DRLD (cm cm^{-3})	DRBD (Mg ha^{-1})	DRN%	DRC%	DRCS (Mg ha^{-1})	DRCN
Alley	5	0.37 (0.39)	20.75 (0.14) de**b**	0.88 (0.74) **ca***	1.08 (1.20)	41.39 (40.66)	8.44 (7.80) **ca***	38.32 (33.88)
	15	0.46 (0.44)	17.01 (0.14) **b**	0.51 (0.17) **ca***	0.99 (0.89)	56.04 (49.62)	6.49 (3.93) **ca***	56.60 (55.75) e
	25	0.28 (0.32)	5.34 (0.03) **b**	0.16 (0.15) **ca***	0.83 (0.83) e	44.46 (44.01)	1.84 (1.71) **ca***	53.56 (53.02)
	35	0.27 (0.27)	3.44 (0.03) a**b**	0.46 (0.46) **ca***	0.87 (0.87) e	43.60 (43.60)	5.45 (5.45) **ca***	50.11 (50.11) e
	45	0.76 (0.76)	2.28 (0.02) a**b**	0.05 (0.05) **ca***	1.71 (1.72) cd	43.08 (43.08)	0.55 (0.55) **ca***	25.19 (25.04) bd
Arable	5	1.37 (1.01)	2.30 (0.02) **ac**	0.04 (0.04) **ca***	1.04 (1.06)	42.46 (45.50)	0.44 (0.40) **ca***	40.82 (42.92)
	15	0.32 (0.38)	0.49 (0.001) **ac**	0.02 (0.02) **ca***	0.98 (0.99)	45.76 (42.11)	0.29 (0.25) **ca***	46.69 (42.53)
	25	0.47 (0.47)	0.87 (0.001) **ac**	0.04 (0.04) **ca***	1.85 (1.90)	46.49 (46.49)	0.43 (0.43) **ca***	25.12 (24.46)
Row	5	0.43 (0.38)	33.87 (0.36) de**b**	2.29 (2.53) bcde**aba*b*d*e***	1.18 (1.25)	47.14 (47.26)	27.21(30.45) bcde**aba*b*c*d*e***	39.94 (37.80)
	15	0.44 (0.44)	15.68 (0.14) **b**	0.45 (0.49) a	0.93 (1.03)	47.26 (46.11)	5.42 (6.13) a	50.81 (44.76) e
	25	0.36 (0.34)	10.22 (0.05) **b**	0.67 (0.81) a	0.78 (0.82) e	41.02 (43.20)	6.83 (7.83) a	52.58 (52.68)
	35	0.24 (0.24)	2.22 (0.02) a**b**	0.12 (0.10) a	0.61 (0.62) e	39.88 (39.88)	1.03 (1.03) a	65.37 (64.32) e
	45	1.06 (0.25)	3.28 (0.02) a**b**	0.39 (0.27) a	1.30 (1.30) cd	39.18 (39.18)	2.71 (2.71) a	30.13 (30.13) bd



Supplementary Figure B.2 The scree plot displays the variance explained by the principal components for (a) soil nutrients and (b) root characteristics.

The greatest amount of variance is explained by PC1 (≈25%) and PC2 (≈15%). The variance explained decreases gradually from PC3 to PC8, with each component contributing a moderate amount (approximately 5–10%) (Fig. B.2). Beyond this (e.g., PC9 and higher), the components add relatively little additional variance. An "elbow" in the scree plot appears around PC2–PC3, indicating that two to three principal components capture the main structure of the data. Nevertheless, including components up to PC8 could still provide additional detail, depending on the purpose of the analysis.